Artificial Intelligence: Voice Assistants

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Abstract

The increasing amount of data and the need to rapidly retrieve information has prompted individuals to rely on technology. Voice assistants have been developed as an innovative tool that provides a natural interface to retrieving information from the digital world and even instructing simple transactions such as purchasing items off the Web. A voice assistant is a software application driven by AI that operates as a personal information manager, such as Siri, Amazon Alexa and Google Now. However, a deeper understanding of users' perceptions regarding voice assistants is necessary to increase user engagement and ensure users continue using them.

This need motivates the exploratory study presented in this thesis. It aims to investigate the relationship between individuals and their home's voice assistants to identify the main characteristics that facilitate establishing a significant and engaging connection between them, leading to their continuous use of the voice assistant. This research thus aims to enhance the post-adoption stage by providing a valuable contribution to improving the relationship between users and their home voice assistants.

The main findings of this study indicate that both perceived anthropomorphism and perceived intelligence significantly impact the continuous intention to use and engage with the voice assistant. Nevertheless, the importance of these relationships comes from specific crucial mediators such as engagement, rapport, engagement motivations, perceived value and satisfaction. These factors significantly influence the user's intention to use and engage with voice assistant technology.

Declaration

No portion of the work referred to in the thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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Chapter 1. Introduction

This introductory chapter offers an overview of the dissertation project, clarifying the underlying motivation, aim, objectives, methodological approaches, and research questions that shape this investigation. Three academic papers present our findings at the core of this research project, and the chapter also demonstrates how these relate to the overall study.

1.1 Introduction

Information technology's (IT) success is fundamentally associated with the technology's continuous usage (Bhattacherjee, 2001). Following the initial adoption of an IT system, users can only derive advantages from it by continuously engaging with the system. Therefore, it is not unexpected that the concept of IT continuance, which is an important aspect of post-adoption behaviour, has become a significant area of interest in the field of information technology and information systems research (Limayem, Hirt, & Cheung, 2007).

Although the significance of IT continuance is widely recognised, the majority of current continuance research focuses on understanding and predicting the intention of the continuous use intention of IT systems, with a specific emphasis on utilitarian aspects (Mamun, Senn, Peak, Prybutok, & Torres, 2020). While these continuous intention models are valuable, their main focus is on providing insights into the functional dimensions of technology usage. However, the manner in which users engage with specific technologies, such as in-home voice assistants (VAs), goes beyond just functionality. The current theoretical frameworks are believed to fall short of providing a comprehensive understanding of users' adoption and utilisation of VAs (McLean & Osei-Frimpong, 2019), indeed conventional models struggle to capture the dynamic interactions between users and technology. Voice Assistant adoption could depend on user experience, and dynamic human-machine interactions. Theoretical frameworks may not adapt to this field's diverse contexts and changing user expectations. Current frameworks also ignore socio-cultural factors affecting voice assistant adoption. This restricts our understanding of social effects. Moreover, voice Assistants and innovative apps mimic human-to-human interactions. As complex dynamics unfold, current frameworks fail to address user expectations and interactions. To understand VA post-adoption and continued use, theoretical frameworks must include technological, psychological, and societal factors. This implies the necessity for a deeper look into the distinct factors that influence user behaviour

within the context of in-home VAs, exceeding the boundaries of traditional continuance use models.

This study takes into account that users' interactions with in-home VAs frequently show a dispersed and irregular pattern, characterised by irregular usage rather than consistent engagement. According to a report by PwC (2018), the majority of users continue to have a range of concerns, anxieties, and biases when it comes to incorporating VAs into their daily routines. In addition, users avoid and postpone using VAs for transactional purposes (Malodia, Kaur, Ractham, Sakashita, & Dhir, 2022).

The inconsistent user engagement with voice assistants is a major concern in the ever-changing digital technology landscape. This concern necessitates meticulous deliberation given its extensive ramifications for the pervasive incorporation of voice assistants into diverse aspects of everyday existence. The continuous engagement of users is a significant indicator of the efficiency and user satisfaction linked with the technologies (Kim, Kim, & Wachter, 2013). A decrease in regular user interaction may indicate underlying problems associated with usability, user experience, or unfulfilled user expectations. Moreover, the long-lasting success and market sustainability of voice assistants depends on continuous user interest and engagement. In order to fully address this concern and develop effective strategies for improvement, additional research is essential. Thoroughly examining the factors that affect changes in user engagement can provide valuable insights into user behaviour, preferences, and evolving expectations. This research is crucial for customising voice assistant technologies to better match user needs, thereby enhancing functionality and promoting long-lasting and meaningful interactions between users and voice assistants.

In light of this recognition, the goal of this research is to shed light on the complex mechanisms that influence user engagement with these technologies beyond the basic functionality of tools and investigate the domains of emotional connection, anthropomorphic attributions, and perceived cognitive capabilities. This research study, therefore, provides an in-depth examination of the intricate dynamics between users and in-home VA technologies, focusing on the three crucial aspects of engagement (Shneiderman (1998), anthropomorphism (Epley, Waytz, & Cacioppo, 2007), and intelligence (Bartneck, Kulić, Croft, & Zoghbi, 2009).

1.2 Research Background

Conversations with computers have long been the subject of science fiction, but we are now at a point where users can already have basic exchanges through the voice (the most natural way of human-to-human communication) with computers, and that capability will grow as artificial intelligence (AI) models improve. According to AI experts, during the next three decades, AI will improve to the point where it will be able to compete with or surpass humans in tasks such as writing essays, driving trucks, working in retail, translating languages, and even doing surgery (Grace, Salvatier, Dafoe, Zhang, & Evans, 2018). Incorporating artificial intelligence (AI) yields advanced cognitive abilities that mimic human abilities, encompassing manual procedures' automation, decision-making, solving problems, and visual perception (Benbya, Davenport, & Pachidi, 2020). The implementation of advanced capabilities is facilitated by various systems, such as intelligent virtual interfaces, robotic equipment, chatbots, and other digital assistance devices (Fernandes & Oliveira, 2021). Conversational AI combines several services, including natural language processing and dialogue management, to enable interaction via voice entered into an automated user interface. VAs are the advancement in consumer electronics, beginning to permeate everyday life. The term "conversational agents" refers to “software that accepts natural language as input and generates natural language as output, engaging in a conversation with the user” (Griol, Carbó, & Molina, 2013, p.760). These devices demonstrate AI's remarkable growth and potential. However, because of the distinct platform empires of Amazon (Alexa), Apple (Siri), Microsoft (Cortana), and Google (Google Assistant), VAs have become a complicated and fragmented form of computing experience. With every major technology company, including Apple and Amazon, now providing their own virtual assistants, it is highly likely that these devices will develop and become more prevalent in daily life.

Amazon first introduced the concept of a VA to the home, and using Alexa (Echo's virtual assistant) has now become part of daily life for many people. Owners of smart speakers demonstrate a greater diversity of voice usage scenarios and a higher frequency of usage compared to other device owners. This suggests that voice assistants embedded into smart speakers will probably be used daily as part of users' everyday routines, raising users' expectations about voice interactive experiences. Indeed, as Voicebot reported recently (2020), the result of a new user survey in the USA showed that 87.7 million people have started to use smart speakers since January 2020, an increase in the US smart speaker user base of 32% in January 2019 and 85% on January 2018. According to recent statistics, a significant portion of

voice-enabled device users, accounting for 43% of the total, utilise the technology for shopping purposes (Georgiev, 2022).

Furthermore, it is projected that the ownership of smart speakers in US households will reach 55% by the year 2022 (Georgiev, 2022). This reflects the importance of the technology, the rise in popularity of VAs, which are equipped with social agents, has opened up novel avenues for researchers and designers to explore people's perceptions and reactions towards these conversational agents. The versatility of VAs in assuming various roles and functions during multi-user interactions, particularly in personal spaces like the home, makes them highly pertinent for investigation and analysis.

Although AI and robots share technological qualities, they are conceptually distinct (see Appendix A). AI systems are programmed to "learn" from the environment surrounding them, to adapt and act appropriately. As a result, they appear to have humanlike intelligence and can operate independently of human command (Oberoi, 2019). Notably, AI does not have to be physically embodied; it already exists in everyday devices such as mobiles, online chatbots, computers and VAs. Voice assistants are based on AI technology and could be classified as similar to intelligent robots in terms of their interactions. These early-generation AI-powered VAs called Conversational virtual assistants (De Keyser, Köcher, Alkire, Verbeeck, & Kandampully, 2019), can identify and comprehend users' vocal commands, engaging in natural language communication to perform various tasks. These tasks encompass a wide range of activities, such as placing online orders, accessing weather forecasts, making restaurant reservations, or providing news updates (Chattaraman, Kwon, Gilbert, & Ross, 2019; Robinson, Orsingher, Alkire, De Keyser, Giebelhausen, Papamichail, & Temerak, 2020). However, artificial intelligence spans many roles and sectors, and its implications may concern people, particularly if they lack a clear understanding of how AI truly works. Therefore, how humans view AI ontologically may influence their approach to it (Mays, Lei, Giovanetti, & Katz, 2021). Thus, in order to build a strong relationship between users and VAs, users should understand how the VAs' process mechanisms work, and VAs should act intelligently while engaging with the users and provide appropriate responses that fall within the range of user expectations and meet minimum levels of acceptability. The field study on the adoption of AI does not tell about motivations of engagement, but the field study on robot adoption does. The main focus of the study is enhancing the success of continuous usage. However, a literature review on engagement with AI is lacking, but there is a much more mature body of literature review on people engaging with robots and service robots who are driven by AI and about

mobile technology as well. They both have the same properties, so that we can borrow a lot in that field.

The use of home VA technology-mediated services has become common and influential in modern homes, providing various functionalities and conveniences through interactions facilitated by natural language (Smith & Johnson, 2020). The arrival of VAs such as Amazon's Alexa, Google Assistant, and Apple's Siri has significantly transformed how users engage with their smart homes and retrieve information (Brown, 2019). Integrating VAs into a wide range of smart devices could facilitate the ability of users to effortlessly manage their home's surroundings through the use of voice commands. This includes tasks such as modifying lighting settings, regulating thermostats, and even placing orders for groceries. In addition to home automation, VAs offer personalised support by providing up-to-date information on weather, news, and traffic. They also enable hands-free communication through voice-activated calls and messages (Robinson & White, 2022). The importance of home VA technology-mediated services is attributed to their extensive capabilities, which optimise daily routines and enhance users' digital experiences by effectively combining artificial intelligence and humanlike interaction (Smith & Johnson, 2020). These technologies' ongoing development and improvement have significant potential for various aspects of life. This highlights the importance of further investigating their integration and enhancing their role in shaping the future smart home ecosystem (Brown, 2019).

The present study, as documented in this thesis, relates to the domain of information systems and includes aspects derived from human-computer interaction, robotics, and psychology (Garcia & Lee, 2021). This study explores the complex dynamics between users and home VAs, investigating the interactions between individuals and AI-driven technologies within the context of their personal environments. The study investigates the usability and user experience aspects of VAs by using insights from human-computer interaction. Its objective is to improve the interaction between humans and technology by promoting a smooth and natural experience (Chen, Mo, & Zhang, 2020). Furthermore, the incorporation of robotics principles provides an opportunity to examine the societal dimensions of user-agent interactions, clarifying the manner in which users perceive and attribute humanlike characteristics to VAs, thereby influencing their emotional attachments to these virtual entities (Kim & Smith, 2021). Furthermore, incorporating psychology within the research framework facilitates an in-depth investigation of the psychological consequences associated with engaging with VAs, covering aspects such as the development of social abilities and the influence of empathy within

interactions with others (Adams & Baker, 2019). This thesis provides an in-depth examination of home VA technology and its implications by integrating various disciplines. This interdisciplinary approach enhances our understanding of the subject matter and establishes a foundation for future advancements and innovations in the field of information systems research.

Recent studies (Davis, 1989; Venkatesh, Morris, Davis, & Davis, 2003; Moon & Kim, 2001) have explored various aspects of the adoption and influence of technology. Scholars have investigated the determinants that impact consumers' initial adoption and usage intentions, identifying significant factors including perceived usefulness, ease of use, and perceived enjoyment (Davis, 1989; Venkatesh, Morris, Davis, & Davis, 2003; Moon & Kim, 2001). Furthermore, research efforts focused on enhancing the user experience have resulted in advancements in the natural language processing capabilities and conversational design of VAs. These advancements aim to create smoother and more fulfilling user interactions (Rao & Su, 2019). Previous research has also investigated privacy concerns by analysing data collection practises and exploring user attitudes towards sharing data with VAs (Acquisti, Brandimarte, & Loewenstein, 2020).

In addition, there has been a recent focus in academic research on the potential utilisation of home VAs in non-conventional contexts, such as healthcare, education, and home automation (Kim & Ryu, 2022; Kar, Sen, Chakraborty, & Mukherjee, 2023). Previous research has investigated the possibility of utilising VAs to assist elderly individuals in their daily living tasks or deliver specific educational materials to students (Chen, Zhang, & Xu, 2022). The studies mentioned previously emphasise the potential of voice technology to bring about significant changes in different areas of life. However, they also emphasise the significance of ethical considerations and establishing user trust when implementing these creative uses (Gürses, Troncoso, & Diaz, 2021; Koene & Floridi, 2022). Moreover, there has been a recent emphasis on comprehending the social and psychological implications of human engagements with VAs. Previous studies have investigated the manner in which individuals perceive and attribute humanlike characteristics to VAs, as well as the potential for users to develop emotional connections with these artificial intelligence entities (Nass & Moon, 2000; Reeves & Nass, 1996). Furthermore, scholarly research has been conducted to explore the potential psychological consequences associated with engaging with virtual assistants. These investigations have focused on areas such as the getting of social skills and the influence on

empathy within human-human interactions (Nass & Moon, 2000). However, building strong relationships between the VAs and the users to keep the engagement and encourage their continuous intentions to use the technology requires further investigation.

However, notwithstanding the numerous benefits presented by the use of at-home VAs, there continues a lack of broad understanding related to the subsequent adoption phase of this technological innovation. Although there has been considerable research on the factors that influence the initial adoption of home VAs, such as perceived usefulness and ease of use (Venkatesh et al., 2003; Moon and Kim, 2001), there is still a lack of research on the factors that influence continuous intention to use and ongoing engagement with these devices. Furthermore, there is limited knowledge regarding the complexity involved in establishing and sustaining user connections with these artificial intelligence-powered entities. Gaining insight into the underlying motivations and determinants that contribute to users' continuous engagement with VAs, as well as the various factors that may impact their disengagement, is essential for growing and maintaining user satisfaction and loyalty (Davis, 1989; Davis, Bagozzi, & Warshaw, 1992). Investigation in this domain has the potential to provide insights into strategies that developers and designers can utilise to generate more engaging and customised interactions, thereby improving user retention and enhancing the overall user experience. By recognising and addressing these areas of limited understanding, scholars and professionals may develop enhanced methodologies to encourage sustained and substantial interaction with home VAs. This, in turn, will optimise the capacity of such technology to enhance various aspects of users' daily lives.

1.3 Theoretical Background

The Management Information Systems (MIS) success criteria include concepts of user attitudes, actual system usage, and performance impact (Bailey & Pearson, 1983; Ginzberg, 1981; Swanson, 1974). To model the factors leading to adoption and hence success, researchers focus on user attitudes and intentions to use, and develop increasingly sophisticated models (e.g. UTAUT, UTAUT2, TAM3) (Burton-Jones, Stein, & Mishra, 2017) based on the technology acceptance model (TAM) is the most commonly used model to study the adoption of information technology and information systems in the I.S. field. TAM is a model to measure users’ acceptance of any new technology or system. It has been applied to many technologies, such as e-mail and word processors, and in many situations within different cultural and time contexts. It has also been used with various subjects, such as knowledge workers and MBAs,

making TAM a robust model in the opinion of its proponents (Larsen & Lee, 2016). Therefore, TAM is considered the most influential theory describing an individual's acceptance of information systems.

However, after all the success that TAM has achieved, it still has some limitations. For example, according to Larsen and Lee (2016), using self-reported usage rather than measuring actual usage has been identified as a common limitation. Another is the fact that studies have tended to focus on homogeneous groups of subjects performing a single task using the same information system at one point in time, making it more difficult to generalise their findings (Larsen & Lee, 2016). These limitations encouraged researchers to establish more theories and models to improve TAM and include variables that were not previously covered. In order to create a unified theoretical basis, the Unified Theory of Acceptance and Use of Technology (UTAUT) was established by integrating dominant models and theories (Williams, Rana, & Dwivedi, 2011). However, it is important to acknowledge that criticisms have been directed towards these models (McLean & Osei-Frimpong, 2019), for example, regarding their failure to account for the unique attributes of technology (Cho, Lee, & Yang, 2019) and how these attributes influence users' evaluations and responses to the technology (Lee & Chen, 2022).

Going beyond the initial uptake of technology, researchers have focused on models explaining user continuance behaviour using the Expectation-Confirmation Theory (ECT) (Oliver, 1980). This theory explains the pre-behaviour variables, such as expectations, and the post-behaviour variables, such as performance (Lin, Wu, & Tsai, 2005). The Expectation-Confirmation Model (ECM) was formulated by Bhattacherjee (2001b) as an extension of the Expectation-Confirmation Theory (ECT). This model suggests that satisfaction serves as the primary determinant of an individual's intention to continue using a particular technology. Furthermore, the confirmation of post-adoption expectations plays a crucial role in shaping this intention. ECM has gained significant popularity and is frequently employed in research to examine the continuance intentions of IT consumers.

The utilisation of ECT (Expectation-Confirmation Theory) and ECM (Extended Continuance Model) offers a valuable and systematic approach to building a theoretical framework to examine the factors influencing the continuation of IT usage after its initial adoption. However, there is limited attention given by IS research to the mutual cognitive and emotional aspects of IT user satisfaction, as noted by Krasnova, Widjaja, Buxmann, Wenninger, and Benbasat (2015). Furthermore, these theories conceptualise information

technology (IT) as a utilitarian tool and analyse the continued existence of IT usage in relation to cognitive user satisfaction, which has played a crucial role in establishing user satisfaction as the primary determinant of IT continuance (Bhattacherjee, 2001; Bhattacherjee & Lin 2015). Although the utilitarian approach is characterised by its simplicity, it fails to acknowledge the significance of various influential human factors, including emotion (Mamun et al., 2020).

However, these theoretical frameworks have proven to be a valuable lens through which to understand the ways in which individuals interact with technology, mainly viewing it as a means to augment efficiency and productivity. Although these theoretical foundations have gained considerable popularity, they have limited applicability in assessing users' intention to use or adopt technology only during its initial adoption phase (Park & Lee, 2019). Within the limits of this conventional framework, technology is perceived as a non-intelligent and non-anthropomorphic tool that individuals use to achieve particular objectives. Nevertheless, with the rapid advancement of technology, specifically with the development of VAs and artificial intelligence (AI) systems, user interaction with technology has experienced a significant shift in perspective that moves away from perceiving technology simply as a utilitarian tool towards recognising its potential to function as a collaborative entity in various aspects of daily life. The change in viewpoint is motivated by technology's growing intelligence and adaptability, enabling it to effectively address human inquiries, predict requirements, and engage in actual conversations. As McLean and Osei-Frimpong (2019) supported, given the unique attributes of Artificial intelligence technology, conventional adoption and continuance models may fail to capture the underlying motivations behind adopting and utilising advanced technology.

Within this context of advanced technology, it becomes essential to investigate the dynamics of human-to-technology interactions when IT systems show humanlike qualities, including conversational abilities and contextual understanding. As the evolution of technology progresses towards more adaptive and intelligent systems, it is essential that our research approaches adapt accordingly to consider the increasing importance of the human-technology relationship in everyday life. Engaging in thorough research that examines this dynamic association will facilitate a deeper understanding of how technology is transforming our engagements with the world of technology.

1.4 Research Motivation

A primary area of interest for scholars in the discipline of information systems (IS) is understanding how users adopt and maintain their intention to use innovative information system technologies (For example, Kim and Malhotra, 2005). The concept of adoption is commonly understood as an individual's initial decision to utilise a novel information technology innovation, as described by Rogers (2003). The concept of continuance intention refers to a user's determination to continue using a particular information technology innovation following their initial usage (Bhattacherjee, 2001b). Numerous scholarly investigations have examined the users' initial adoption of an information system (IS) (For example, Oliveira, Faria, Thomas, & Popovič, 2014). A key predictor significantly influences the success of the IS. However, it is important to note that the initial adoption of the system does not guarantee the desired managerial outcome. In order to achieve the desired outcome, continued usage of the system must be established (Bhattacherjee, 2001b). The long-term effectiveness of IS depends on the sustained engagement of its users, commonly referred to as users' continuance (Bhattacherjee, 2001b).

The motivation for undertaking this research derives from the notable paradigm shift occurring in our engagement with technology. The appearance of this evolution presents an attractive problem for researchers and practitioners who have traditionally depended on the framework of the Technology Acceptance Model (TAM). Historically, the Technology Acceptance Model (TAM) has played an important role in understanding the process by users who embrace and integrate technology into their lives. However, its primary emphasis has been on examining the interactions between users and the technology as a tool.

Nevertheless, the development of technologically advanced systems possessing human-like attributes, such as the ability to engage in conversation and understand context, has significantly transformed the current scenario. The current paradigm shift requires an in-depth exploration of the interactions related to human-to-human relationships with technology. As technological advancements continue to progress, there is a growing need to modify our research approaches to effectively incorporate the expanding collaboration between humans and technology in everyday activities.

The increasing use of VAs in households is an important driver for this research motivation. The growing usage of this technology highlights the need to understand the factors that impact users' continued intention to use these VAs. The understanding of this information carries

significant implications, not only for individuals who consume technology but also for the individuals who are involved in the development and advancement of technology.

One of the primary motivations behind this research comes from the phenomenon wherein specific users initially adopt technology such as VAs but subsequently discontinue their use. The stoppage of this activity underscores the importance of taking steps to address any potential concerns in order to facilitate sustained adoption in the long term. Solving these issues is considered a fundamental objective of this research, as it is essential to ensuring these technologies' long-term sustainability and success.

Voice assistants, specifically, demonstrate a notable progression in consumer technology, easily integrating into many aspects of our everyday schedules. We were motivated by the dynamic nature of the landscape to start this research project, with the objective of gaining a deeper understanding of how technology is transforming our engagements with the world of technology. The motivation for this research serves as the main reason for our ongoing efforts to investigate, examine, and address the obstacles and possibilities presented by this innovative technological environment.

1.5 Research aim, objectives, questions

This research aims to examine the relationship between users and their home VAs to determine the key factors that contribute to building a meaningful and interactive bond between them. The primary objective of this research is to make a valuable contribution towards improving the relationship between users and their home VAs. This, in turn, will lead to increased engagement and a continued intention among users to continue using the technology.

Therefore, to achieve the study's primary objective, the decision was made to adopt an alternative approach consisting of three separate papers. This approach provides the benefit of conducting thorough and detailed analysis, wherein each paper is devoted to examining a particular aspect or factor that impacts the complex interaction between users and their home VAs. The purposeful use of compartmentalisation facilitates an in-depth understanding of the consequences and influences of these factors. Moreover, the sequential progression of the three-paper approach, whereby the findings of the initial paper influence the direction of the subsequent paper and so on, ensures that the study remains dynamic and up-to-date. The chosen methodology facilitates an in-depth investigation of each factor affecting the relationship

between users and technology, enhancing our understanding of this dynamic and evolving field.

Therefore, the aims of the research and objectives presented in this thesis are as follows:

Paper 1: A bibliometric study (systematic literature review: Voice assistants)

- Research question: What are the current research efforts and the main research directions?
- Aim: Provide a comprehensive picture of the voice assistant technology perspectives that have already been covered in order to identify gaps in existing knowledge.

Towards this aim, one objective is to provide an understanding of how

- Objective 1: Systematically analyse and review the voice assistants' academic progress

As a result of the first paper, two notable concepts emerge regarding our perception of the object of our technology use, wherein it is not regarded simply as a tool but rather as a subject like a person. Therefore, this research intends to provide a deep understanding of the two core factors that influence the technology's continuous usage:

(a) Perceives anthropomorphism.
(b) Perceived intelligence.

These two factors are investigated in the remaining two papers within this thesis:

Paper 2: The power of anthropomorphism on the continuous intention of AI-based technology

- Research question: To what extent could anthropomorphism play a role in building a relationship between voice assistants and users, thus enhancing adoption?
- Aim: Investigate the relationships between users' motivation factors to continue using the technology and their subsequent perceptions and evaluations regarding the role of their home voice assistants.

Towards this aim, two objectives are to provide an understanding of

- Objective 1: The effect of anthropomorphism on building relationships between the home voice assistants and users.

- Objective 2: The impact of engagement and rapport on perceived value and then home voice assistant adoption.

The second study demonstrates the importance of engagement for users continuing to use VA technology. This is reflected in the third research paper, which links perceived intelligence with the underlying motivations for users to engage with technology.

Paper 3: "Are You Talking to Me?": How Perceiving Voice Assistants as Intelligent Impacts the Users' Engagement Intention

- Research question: To what extent does perceived intelligence play a role in motivating users to engage with voice assistants?
- Aim: Examine the influence of perceived intelligence on users' motivations to interact with voice assistants and its effect on their intention to engage with the technology, and investigate the impact of different motivational types on perceived value, satisfaction and users' engagement intention.

Towards this aim, three objectives are to provide an understanding of how

- Objective 1: Examine the effect of perceived intelligence on building relationships between voice assistants and users
- Objective 2: Examine the effect of perceived intelligence on engagement motivations and, subsequently, users' intention to engage with voice assistants.
- Objective 3: Investigate the role of perceived intelligence on perceived value and user satisfaction and, consequently, on users' intention to engage with voice assistants.

The research questions addressed in the three working articles (chapters) that emerge through specific gaps in prior theory revolve around the overarching aims and objectives of the study (presented above).

1.6 Research context

The voice assistant is seen as one of the core mechanisms for people to interact with future artificial intelligence systems using a natural digital channel using voice command and voice interaction. These interactions between the user and the technology are considered as engagement between them. Attfield, Kazai, Lalmas, and Piwowarski (2011) define engagement as an emotional, cognitive, and behavioural connection that exists, at any point in time, between a user and a resource. However, Moriuchi (2021) argued that engagement is often a term used to emphasise the positive aspects of the interaction between humans and technology. On the other hand, for a comprehensive view, O'Brien and Toms (2008) confirmed that engagement is a term used to explain interaction and why and how people develop a relationship with technology and integrate it into their lives. Recent reports show a noticeable increase in the number of VA device users; however, continuous intention usage will lead to this technology's adoption. For instance, those users could use the devices once or more and then stop using them. Therefore, it is essential to consider the continuous use intention to adopt VA devices.

Recent studies (e.g. (Moriuchi, 2021)) clarified that engagement influences the continuous intention to use VA devices. Moreover, Purington, Taft, Sannon, Bazarova, and Taylor (2017) added that in the case of conversational agents, people's perceptions of a machine-generated voice are influenced by its humanlike personality traits. Therefore, the engagement between the users and the VA devices can also be influenced by factors such as anthropomorphism and intelligence. Balakrishnan and Dwivedi (2021) confirmed that factors such as perceived anthropomorphism and intelligence are unique to AI-powered technologies.

With recent technological advancements, embodied virtual agents are becoming more incorporated into everyday life (Purington et al., 2017). Lim, Cheah, Ng, Basha, and Soutar (2021) explained that anthropomorphism is the extreme version of humanisation that acts as a precursor to bridge the gap between humans and technology. Perceived anthropomorphism concerns individuals' perception of how close a device is to a human being (Qiu & Benbasat, 2009). Additionally, Perceived intelligence is associated with perceived anthropomorphism (Moussawi, Koufaris, & Benbunan-Fich, 2021). Individuals tend to assign human-like characteristics to devices that they find to be intelligent. This shows the importance of the relationship between the users and the VA devices and indicates that there is an effect of anthropomorphism and intelligence in the context of the VA devices. Historically, the concept of system intelligence was introduced in technology management research to characterise a system that can aid humans in solving complex tasks (Russell & Norvig, 2002). However,

anthropomorphism is identified as a psychological phenomenon in which people assign human attributes to objects (Fink, 2012). For instance, anthropomorphic language often describes what a computer does (it 'thinks', 'writes', 'reads', 'is nice'). Research indicates that it is natural for humans to anthropomorphise objects (Barrett & Keil, 1996), and one proposed strategy for increasing public acceptance of robots is through the use of anthropomorphic (humanlike) design and "human social" traits (Ingle, Marcus, & Johal, 2021). This would involve creating robots with humanlike characteristics and interactions which are similar to natural human-to-human communication, such as speech. However, a previous study has revealed divergent findings on the effects of anthropomorphic traits and intelligence on A.I. acceptance. Wan and Aggarwal (2015) argue that anthropomorphising a less intelligent object can be an effective strategy for increasing the adoption of an entity, and Pelau, Dabija, and Ene (2021) added that technology with a high level of intelligence may be perceived as a threat to the integrity of human beings.

In contrast, the literature provides empirical support suggesting that perceived intelligence motivates behaviour by forming a positive attitude or stimulating behavioural intention (Balakrishnan & Dwivedi, 2021). In addition, evidence from prior research suggests that the stronger the perception of system intelligence, the stronger the belief that the technology can be more effective and useful in delivering the required services (Moussawi et al., 2021). Duffy (2002) clarified that anthropomorphism in robotics is not intended to create an artificial human but rather to facilitate social interaction. In line with that, Fong, Nourbakhsh, and Dautenhahn (2003) reported that users like to interact with technology in the same way they do with other humans. This could indicate that when the VAs can act more intelligently, the people will prefer them more. Reeves and Nass (1996) concluded that with regard to the Computer as Social Actors Theory, consumers act toward A.I. devices in the same way they act toward humans when they think of them as social actors. This is certainly true in the case of interacting with Alexa (as a conversational agent); the Amazon Echo is inherently a socially interactive device since it requires social interaction to function (Fong, Nourbakhsh, & Dautenhahn, 2003).

1.7 Research design and methodology

This section aims to provide a thorough overview of this study's research design and methodology. The major research technique adopted in this thesis is quantitative research.

However, it is essential to recognise the incorporation of different research designs, particularly qualitative approaches such as thematic analysis. These qualitative methods were applied in the first article and will be briefly discussed. The incorporation of quantitative procedures is driven by the aspiration to improve the capacity to provide theoretical explanations and efficiently tackle research inquiries. This methodological framework guarantees a precise examination of the study's aims, enabling a more profound comprehension of the research area.

1.7.1 Research design

1.7.1.1 Thematic analysis

Thematic analysis is a frequently used qualitative research methodology that entails identifying, investigating, and documenting reoccurring patterns or themes within a specific dataset (Braun & Clarke, 2006). This technique provides researchers with a systematic and comprehensive comprehension of the data, allowing them to examine intricate matters and get important viewpoints. The process of thematic analysis encompasses a set of iterative steps, which include becoming acquainted with the data, creating initial codes, examining themes, evaluating and defining identified themes, and eventually, delivering a full report (Guest, MacQueen, & Namey, 2012). Thematic analysis is a method that aids in the discovery of important themes present in gathered data. This approach allows researchers to discern underlying patterns, common traits, and different aspects within the data's narratives. The methodology outlined in this research demonstrates a notable level of adaptability, as it may be efficiently used to analyse many types of data, including interviews, focus groups, and open-ended survey answers (Braun & Clarke, 2006). Thematic analysis is a fundamental method that plays a substantial role in comprehending human conduct, ideas, and attitudes comprehensively. Consequently, academics from many academic domains, such as the social sciences, psychology, and allied topics, highly regard it. The use of this methodology allows scholars to methodically and meticulously investigate and analyse qualitative data, therefore facilitating a comprehensive comprehension of the topic at hand (Nowell, Norris, White, & Moules, 2017).

1.7.1.2 Quantitative methods

Quantitative approaches include research strategies that use numerical data and statistical analysis to investigate and grasp various phenomena (Bryman, 2016). The

methodologies under discussion are often associated with the positivist and post-positivist research paradigms, which aim to find and measure objective and measurable relationships between variables (Creswell, 2014). Quantitative research is often characterised by using surveys, experiments, or other organised instruments to collect data, enabling the acquisition of comprehensive data from a diverse and wide-ranging sample (Neuman, 2014). Various statistical approaches are used by researchers to analyse the data and formulate hypotheses pertaining to the observed correlations and patterns (Hair, Black, Babin, & Anderson, 2018). The quantitative method is well recognised for its significant contribution to generating insights pertaining to cause-and-effect linkages, patterns, and discoveries that may be generalised. This characteristic makes it especially appropriate for investigating research inquiries that need impartial and numerical evidence (Creswell, 2014).

1.7.2 Conduct of the research

1.7.2.1 Data collection procedures

The data collection procedures used for this research were characterised by a systematic approach that was customised to align with the specific objectives of each paper. The first paper involved the execution of a systematic literature review, employing the PRISMA flow chart as a methodological tool. This flow chart facilitated the systematic identification and selection of relevant studies, establishing a structured review framework. The methodology included the establishment of specific criteria for inclusion and exclusion, which was subsequently followed by a thorough analysis conducted across academic databases in order to retrieve relevant literature. By conducting a methodical examination of academic papers, the researchers were able to identify studies that satisfied the predetermined criteria, thereby influencing the content of the first paper.

Following this, for the subsequent phases of the research, an online survey was utilised as the principal approach for gathering data. The survey was carefully crafted to specifically target users who have experience using at least one home VA device, such as the Amazon Echo or Google Home, and who are 18 years of age or older.

Following this, in the subsequent phases of the investigation, an online survey was used as the principal means of gathering data. The survey was carefully crafted to specifically target users who have owned and used a VA device, such as the Amazon Echo or Google Home, and who are 18 years or older. Using these strategic selection criteria ensured that the participants

had direct experiential knowledge of the technology being investigated. The survey comprised a wide range of carefully designed questions that examined multiple aspects of the research goals. This allowed for collecting valuable information regarding participants' perceptions, behaviours, and interactions with home VA technology. The use of this data collection methodology allowed for the gathering of a varied and broad dataset, which provided an in-depth perspective on the variables being examined. This enabled the studies to derive strong and significant conclusions.

1.7.2.2 The process of obtaining ethical approval

The University of Manchester's ethical approval process for conducting research included a comprehensive and methodical procedure aimed at protecting the ethical integrity as well as the responsibility of the study. The process involved multiple stages, including developing a research proposal defining the research objectives, methodology, participant recruitment strategies, data collection procedures, and potential ethical concerns. Following that, the proposal underwent a thorough evaluation by the university's ethics committee, which consisted of knowledgeable experts in ethical standards and research protocols. This review aimed to evaluate the potential risks, benefits, and implications of the study for its participants, with a focus on protecting their rights, privacy, and well-being.

The ethical dimensions of the research were thoroughly assessed by the university's ethics committee, which took into account various factors, including informed consent, confidentiality, data management, potential harm, and the overall ethical conduct of the study. All potential ethical concerns and considerations were thoroughly examined and resolved before obtaining ethical approval for the research. The procedure for getting ethical clearance from the University of Manchester illustrates a commitment to maintaining the most robust ethical principles in research, emphasising the significance of protecting participants and ensuring the reliability of the study's results. The process of obtaining approval serves the purpose of not only verifying the ethical integrity of the research but also growing a sense of accountability and responsibility within the researcher. This, in turn, establishes a solid basis for conducting research that is both ethically strict.

1.7.2.3 Questionnaire survey

The survey was constructed following the relevant scholarly literature in order to establish content validity. Following this, a pilot study was undertaken to evaluate the questionnaire's initial efficiency. In order to improve the survey's reliability and validity, careful revisions were made to items showing low factor loadings. The survey items and scales were determined to be well-developed through a pre-test, thus establishing a robust and reliable questionnaire. In order to expand the scope and involve a more varied demographic, an online survey was selected as the optimal means of gathering data owing to its related convenience, common availability, and simplified efficiency. The survey was conducted among individuals who use home VAs. The participants were given a letter explaining the questionnaire's purpose to promote openness and obtain informed consent. The implementation of an online survey facilitated the provision of valuable insights regarding users' perspectives and experiences with home VAs.

1.7.2.4 Partial Least Squares Structural Equation Modelling (PLS-SEM)

Structural equation modelling (SEM) is a statistical methodology that facilitates the systematic modelling of relationships between many independent and dependent factors. The PLS-SEM methodology is frequently employed to assess the dependability and accuracy of measures and assess the viability of the proposed research framework. There are several justifications for employing this particular approach. The Partial Least Squares Structural Equation Modelling (PLS-SEM) method has been recognised as particularly advantageous for predictive purposes (Haenlein & Kaplan, 2004). Consequently, when the research objective involves predicting the key factors, PLS-SEM is a suitable statistical approach for evaluating SEMs (Hair, Ringle and Sarstedt, 2011). Given the objective of this study, which aims to identify factors that can predict individuals' intention to continue using home VAs, the PLS-SEM method seems to be the most suitable approach.

Furthermore, it has been noted that Partial Least Squares Structural Equation Modelling (PLS-SEM) demonstrates efficacy in handling complicated structural equation models characterised by many constructs (Henseler, Ringle, & Sinkovics, 2009). The Partial Least Squares Structural Equation Modelling (PLS-SEM) approach was developed as the most appropriate methodology to analyse and interpret the data. Additionally, it is worth noting that PLS-SEM has demonstrated strong performance even when dealing with non-normal

distributions. Furthermore, PLS-SEM requires a smaller sample size than other methods (Hair et al., 2011). Based on the reasons listed above, the selection of PLS-SEM has been made for this research.

1.8 The structure of the thesis

The research approach utilised in this thesis is distinguished by a conscious choice to embrace an alternative approach, consisting of three separate yet interrelated papers. This approach has been selected due to its inherent benefits in enabling comprehensive and meticulous analyses. Each paper in this approach focuses on analysing a particular aspect or factor that affects the complex interaction between users and their home Voice Assistants (VAs). The choice to organise the research in this way enables a concentrated and thorough investigation of each identified factor—Perceived Anthropomorphism and Perceived Intelligence. This sequential progression guarantees a dynamic and current study, as the discoveries from the initial paper inform and direct the subsequent investigations. The research methodology employs a three-paper approach to effectively address the complexity of the subject matter. This approach enables a detailed examination of the various aspects of user engagement with voice assistant technology. The intentional organisation of the research also improves the overall logical flow and impact, resulting in a thorough comprehension of the factors that affect user engagement and ongoing utilisation of virtual assistants in domestic settings.

The thesis is meticulously structured to develop a comprehensive understanding of user engagement with voice assistant technology. This is achieved through a sequential exploration of three interlinked studies. The initial paper, a Systematic Literature Review of Voice Assistants, establishes the basis by thoroughly analysing diverse viewpoints on voice assistant technology. This preliminary study aims to identify deficiencies in current research and crucial aspects that necessitate additional investigation. The results of this review indicate a notable lack of knowledge regarding the factors that influence the acceptance and ongoing use of technology, especially in relation to voice assistants' perceived intelligence. Users view these entities not only as tools but also as cooperative sentient agents. The insight gives rise to two main concepts: Perceived Anthropomorphism and Perceived Intelligence. These concepts then become the main areas of focus for further investigations.

The second paper explores the influence of Perceived Anthropomorphism on the establishment of a connection with voice assistants. This research aims to investigate the impact of users' perception of anthropomorphism on the dynamics of user-voice assistant relationships, building upon the findings of the initial study. This highlights the significant importance of user engagement as a factor in determining ongoing usage and its beneficial effect on the overall relationship between users and voice assistant technology.

The third paper builds upon the previous ones by expanding the investigation into the domain of Perceived Intelligence. This research aims to examine the influence of perceived intelligence on user engagement motivations and the intention to continue using voice assistants, based on the knowledge gained from the previous two studies. This study seeks to establish a comprehensive understanding of the complex relationship between user perceptions, engagement motivations, and the long-term usage of voice assistant technology by examining how perceived intelligence is connected to the underlying motivations that drive user engagement.

This thesis aims to provide detailed insights by conducting a carefully planned series of studies. It not only fills gaps in the existing literature but also offers practical implications for improving the design and development of voice assistant technologies. Ultimately, this will enhance the post-adoption phase and the continued usage of the voice assistant technology.

Chapter 2. Paper One:
Systematic Literature Review of the Voice Assistants

Abstract

In a world characterised by an increasing level of human-computer interactions, AI-powered voice assistants emerge as attractive facilitators of these interactions. Considering the fact that voice assistants are an innovation that is still in the early stages of adoption, this systematic literature review will endeavour to find the extent to which existing research on voice assistants has covered issues related to the way in which these devices are perceived by their users. Building a comprehensive picture of different perspectives on voice assistant technology will help us identify gaps in existing research and important aspects that require further examination. The findings indicate that while academic interest in this novel subject is increasing, there remains a gap in the literature regarding the factors that drive technology acceptance and continuous use because existing research in those areas does not take into account the intelligence of voice assistants, which means that their uses perceive them as collaborative sentient actors and not as tools. Taking these perceptions into consideration can help to increase the engagement of users with the voice assistant technology.

***Keywords**:* Artificial Intelligence (AI); Voice assistant (VA); Usage; Interactions; Satisfaction; Privacy; Security.

2.1 Introduction

The Voice Assistant (VA) can be defined as software that operates as a personal information manager (Almirabi & Chesney, 2018), but they are also "the realisation of the science fiction dream of interacting with our computers by talking to them" (Hoy, 2018; p.82). VAs have become widely used worldwide over the last few years. Systems such as Cortana, Alexa, Siri and Google Assistant have contributed to a shift in the way users complete tasks, interact with companies, consume content, purchase products and search for information (McLean & Osei-Frimpong, 2019). Up to 27% of the global online population now uses voice searches (McCue, 2018), and Juniper (2018) has predicted an increase of up to 1000% in-home VA usage between 2018 and 2023. The rising popularity of VA applications in users' everyday lives is closely linked to convenience (Zhou, Jia, Wang, Dong, Yin, & Lei, 2018).

This technology has found its way into users' living rooms, continuously listening to users' voice inputs and giving them quick responses (Vaidya & Sherr, 2019). According to Chattaraman, Kwon, Gilbert, and Ross (2019), the main objective of the VA is to facilitate the user's daily tasks; however, many questions remain about how VAs can communicate with a user in the most 'natural' way. Due to the increase in VA technology usage, this has become one of the most significant areas that need to be studied. Further studies will also encourage researchers to explore this technology in greater depth to understand its aspects, benefits, the obstacles and risks associated with it, and how it can be developed. This systematic literature review paper aims to provide a comprehensive picture of the VA technology perspectives that have already been covered in order to identify gaps in existing knowledge. Therefore, the research question of this paper is:

What are the current research efforts and the main research directions in relation to voice assistants?

In order to get the best answer to the research question, the paper's objective is to systematically analyse and review the academic literature which foci on VAs.

This paper is organised as follows. After the preliminary scoping and objectives setting in this section, the next section will focus on the principles and method of the systematic literature review. The third section will report on the results of the search, covering two aspects: the first gives an overview of the papers included and provides a general analysis of the data, and the second foci on analysing papers on features and adoption of VAs. The fourth section will present a general research agenda for research into VAs and related topics. The last

concluding section will discuss the following steps to progress research in VAs and propose future studies.

2.2 The method of the systematic literature review

This study's systematic literature review method uses a process of systematic selection of relevant research papers described in the PRISMA systematic review statement (Moher, Liberati, Tetzlaff, & Altman, 2010) and a thematic analysis (Braun & Clarke, 2006) of the papers thus collected.

The PRISMA flow chart in Figure 2.1 below shows the phases of the systematic selection

of papers to be included in the analysis.

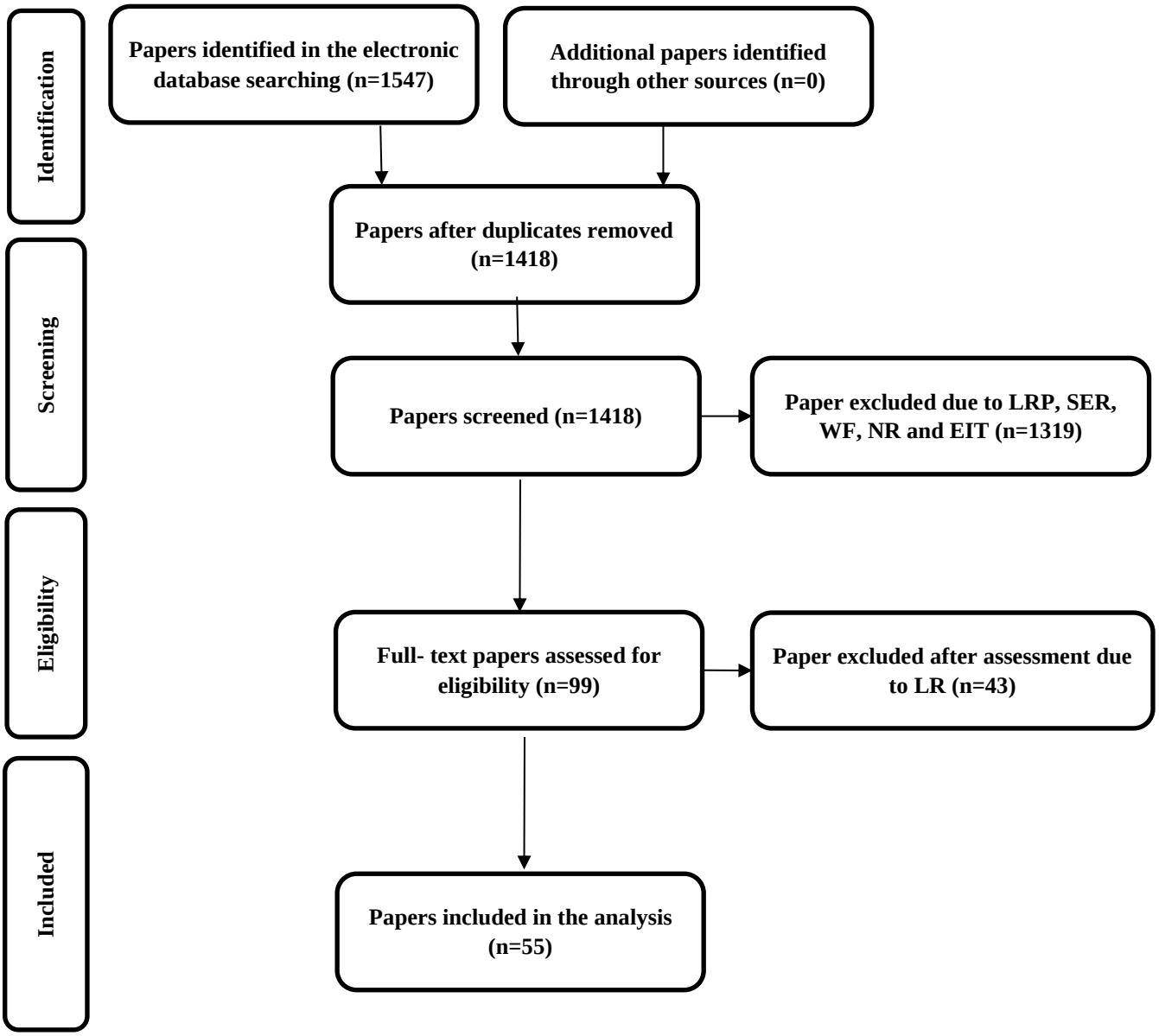


Figure 2.1. The PRISMA flow chart that reports the different phases of the systematic literature review

2.2.1 The principles

Two fundamental principles for review were defined in order to ensure that the papers included in this study could be consistently evaluated from less subjective points of view. The first of these was the inclusion and exclusion criteria, followed by the data collection with evidence criteria.

- **The first principle is the inclusion and exclusion criteria.**

  Table 2.1 below clarifies the criteria applied when collecting papers for this study.

Table 2.1. Inclusion and exclusion criteria

Inclusion or exclusion	Criteria	Principle
Exclusion	Low rank paper (LRP)	LRP: A paper from a conference which ranked B, C or lower or a journal that ranked 2 or lower.
	Search engine reason (SER)	SER: A paper which has only its title, abstract, and keywords in English but not its full text.
	Without full-text (WF)	WF: A paper without full text to be assessed.
	Non-related (NR)	NR: A paper which is not an academic article. For example, editorial materials, conference reviews, contents, or forewords.
	Loosely related (LR)	LR: A paper which does not focus on the review, survey, discussion, or problem-solving of VAs.
	Exclude identified terms (EIT)	EIT: A paper does not include any of the identified terms (used in the search string) in its title, abstract or keywords.
Inclusion	High rank paper (HRP)	HRP: A paper which was published in a conference which ranked A* OR A, or a journal ranked 4*, 4 or 3.
	Partially related (PR)	PR-1: VA is only used to support the description of some challenges, issues, or trends that a paper intends to deal with. PR-2: VA is one of several objects that are to be reviewed, surveyed, or discussed.

	Closely related (CR)	CR: The research efforts of a paper are explicitly and specifically dedicated to VAs.
	Includes identified terms (IIT)	IIT: A paper includes at least one of the identified terms (used in the search string) in its title, abstract or keywords.

- **The second principle is collecting the evidence in the data collection stage.**

To clarify, for particular data, when it requires subjective judgments, this means gathering the original supporting text descriptions of the paper in the database as notes.

2.2.2 Paper collection

Three electronic databases were used for the systematic search to collect academic research papers: SCOPUS, Science Direct and Web of Science. All the collected papers were published online before the end of November 2020 and were collected based on specific characteristics. These were a) that they were published in conference proceedings or journals, b) written in the English language.

In order to collect a comprehensive set of papers, advanced search strings were used by adding the operator 'or' between each of the following terms: 'usage', 'usability', 'using', 'artificial intelligence', 'AI', 'intelligent digital assistant', 'intelligent assistant', 'attitude', 'motivation', 'acceptance', 'accept', 'accepting', 'adoption', 'impact', 'impacting', 'factor', 'Alexa', 'Siri', 'google home', 'google now', 'trust', 'trusting', 'satisfaction', 'satisfy', 'satisfying' in addition to the primary key term 'voice assistant' which connected with the operator 'and'.

The Web of Science had a different string compared to the other two databases. For example, the input used for searching was '(TS=((("voice assistant*") AND (usa* OR usi* OR "artificial intelligence*" OR AI OR "intelligent digital assistant*" OR "intelligent assistant*" OR attitude* OR motivation* OR accept* OR adopti* OR impact* OR factor* OR Alexa OR Siri OR google* OR trust* OR satisf*)))', the search was in all the databases of the web of Science: (WOS, BCI, CCC, DRCI, DIIDW, KJD, MEDLINE, RSCI, SCIELO, ZOOREC). However, in Scopus, the user input for searching was "voice assistant*" AND usa+ OR usi* OR "artificial intelligence*" OR AI OR "intelligent digital assistant*" OR "intelligent assistant*" OR attitude* OR motivation* OR accept* OR adopti* OR

impact* OR factor* OR Alexa OR Siri OR google* OR trust* OR satisf*. However, it found that Science Direct accept just eight terms in one search and does recognise + or *. Therefore, the search consisted of three stages. The first input used for searching was ((“voice assistant”) AND (usage or usability OR using OR “artificial intelligence” OR ai OR “intelligent digital assistant” OR “intelligent assistant” OR attitude OR motivation))), the second one was ((“voice assistant”) AND (acceptance or accept or accepting OR adoption OR impact or impacting OR factor OR Alexa OR Siri OR “google home” or “google now” OR trust or trusting))), and the third one was ((“voice assistant”) AND (satisfaction or satisfy or satisfying))).

After removing duplicates, two steps were applied in the screening process stage (see Figure 2.1). In the first step, the screening process was carried out to exclude papers with a low rank (LRP) based on the ABS 2018 list for the journal papers and the core conference portal website (http://portal.core.edu.au/conf-ranks/). To clarify, only journal papers ranked as 4*, 4 or 3 were included. Meanwhile, only conference papers which ranked as A* or A were included. The papers without access to their full texts (WF) were removed, along with papers not written in English beyond their titles, abstracts and keywords (SER). In addition, we excluded papers which were not academic articles (NR), for example, editorials or book reviews, or did not include any of the identified terms (used in the search string) in its title, abstract or keywords (EIT).

After that, the remaining papers went through the second stage of the screening process, which involved a review of the paper’s title, abstract or even the full text (if required) to exclude papers that did not focus primarily on VAs (LR). After that, all the qualifying papers were reviewed in-depth and then classified into one of the following sub-categories: High-rank paper (HRP), Closely related (CR) and Partially related (PR-1 and PR-2).

2.2.3 Data collection

Two different types of information were gathered regarding each included paper, and all the collected data were entered into a specific database. The first type included the titles, keywords, where they were found (which databases), and their sources, based on categories such as ‘conference’ or ‘journal’ and related information. For conference papers, the names of the conferences and their years were recorded, but for journal papers, their titles and categorisations, considering the impact factors of each paper. This type of information can be described as ‘basic data’.

2.2.4 Paper Categorisation

The included 55 papers were examined in terms of their focus and intent (Vessey, Ramesh, & Glass, 2002). Appendix C shows this analysis done on a sample of the included papers.

Afterwards, the included papers were categorised into four main categories based on a two-step analysis. The first step analysed the papers for keyword co-occurrences based on their titles, abstracts and keywords using the VOSviewer software (see Figure **2.2**). This elucidated the four main clusters, their core concepts, and their connections. The second step was to allocate papers to the clusters thus discovered by developing a concept matrix that records the contribution of the included papers to the relevant core concepts (see Appendix B; a sample of the included papers) and the focus and intent of each paper (Appendix C).

The VOSviewer software was used for analysing all the included papers' titles, abstracts and keywords and clarifying the connections between them. Figure **2.2** presents a graphic representation of the clustering outcomes derived from the implementation of keyword currency analysis using the VOSviewer software. Careful examination of the network identifies four discrete clusters, each providing distinctive perspectives on the domain of voice assistant technology as follows:

1) The yellow cluster, which is anchored on the core concepts of ‘system’, ‘Google Assistant’ and ‘Siri’, is focused on the technology itself, with a specific emphasis on voice assistant brands like Siri and Alexa.
2) The green cluster, anchored on the core concepts of ‘home’, ‘attack’, ‘security’ and ‘accuracy’, focuses on the performance, vulnerabilities and security considerations.
3) The blue cluster, anchored on the core concepts of ‘artificial intelligence’ and the way it changes user perceptions of ‘voice assistants’ such as ‘anthropomorphism’, ‘engagement’ and ‘user experience’, comprises studies focused on the user experience, specifically analysing elements of engagement and overall satisfaction associated with interactions with voice assistants.
4) Finally, the red cluster, anchored on the core concepts of ‘task’, ‘interaction’ and ‘efficiency’, ‘adoption’ and ‘trust’, is focused on investigating the adoption, use and perceived utility of voice assistant systems, based on conventional utility- and ease-of-use- based adoption models such as TAM.

These clusters provide a suitable framework for comprehending the many facets of the ongoing research endeavour that is shaping the landscape of voice assistants.

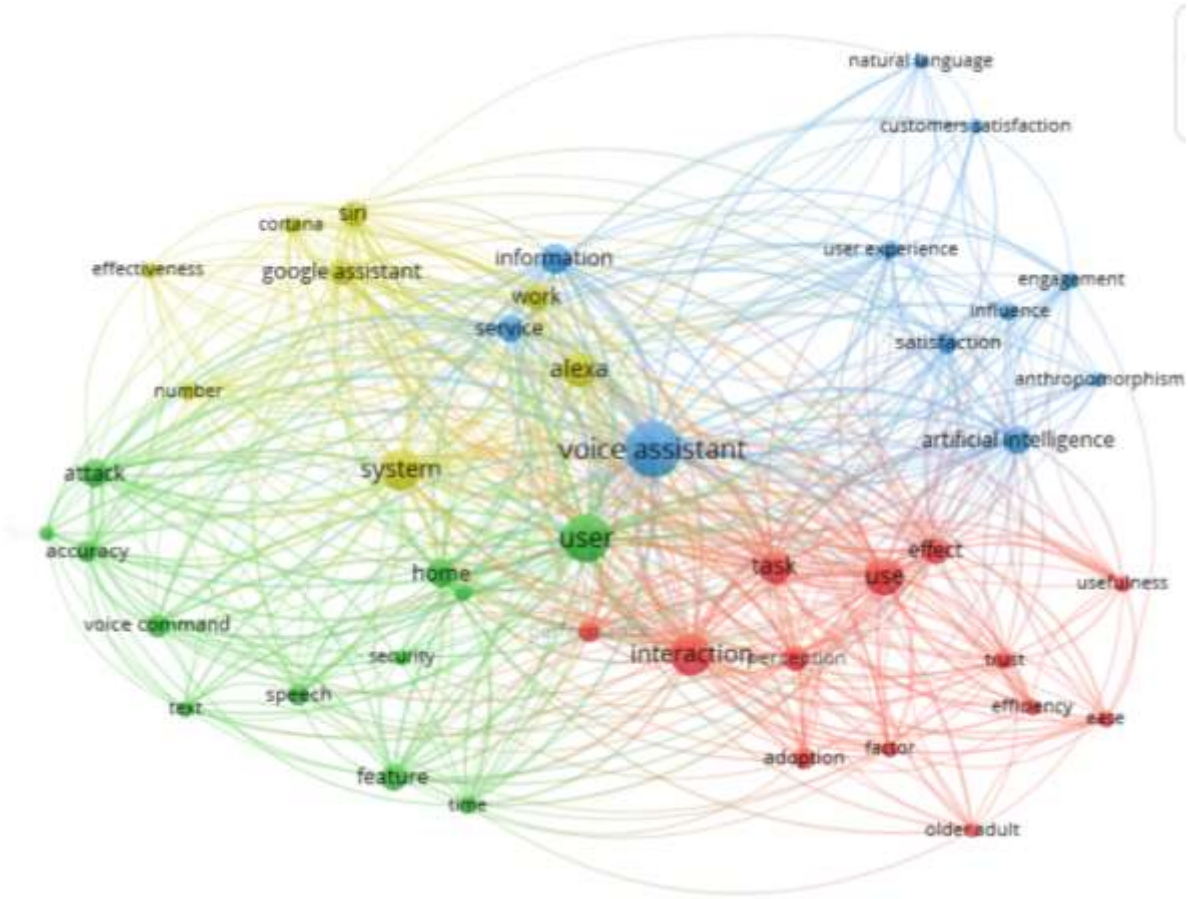


Figure 2.2. VOSviewer map of the included papers

Figure **2.3** shows a secondary analysis that considers the progression of research focus within the realm of voice assistant technology over time. While Figure **2.2** presents a concise overview of the primary clusters, Figure **2.3** showcases the dynamic evolution of research themes. Significantly, there has been a noticeable shift in emphasis in recent years. In recent years, there has been a growing focus in research on the practical implications of technology, user interaction, anthropomorphism, and the enhancement of user engagement. This transition signifies the increasing development and sophistication of voice assistant technology, as well as its integration with practical applications in everyday life. In light of the ongoing advancements in technology, it is crucial to comprehend the dynamic research trends in order to effectively guide future advancements and fully exploit the capabilities of voice assistant systems.

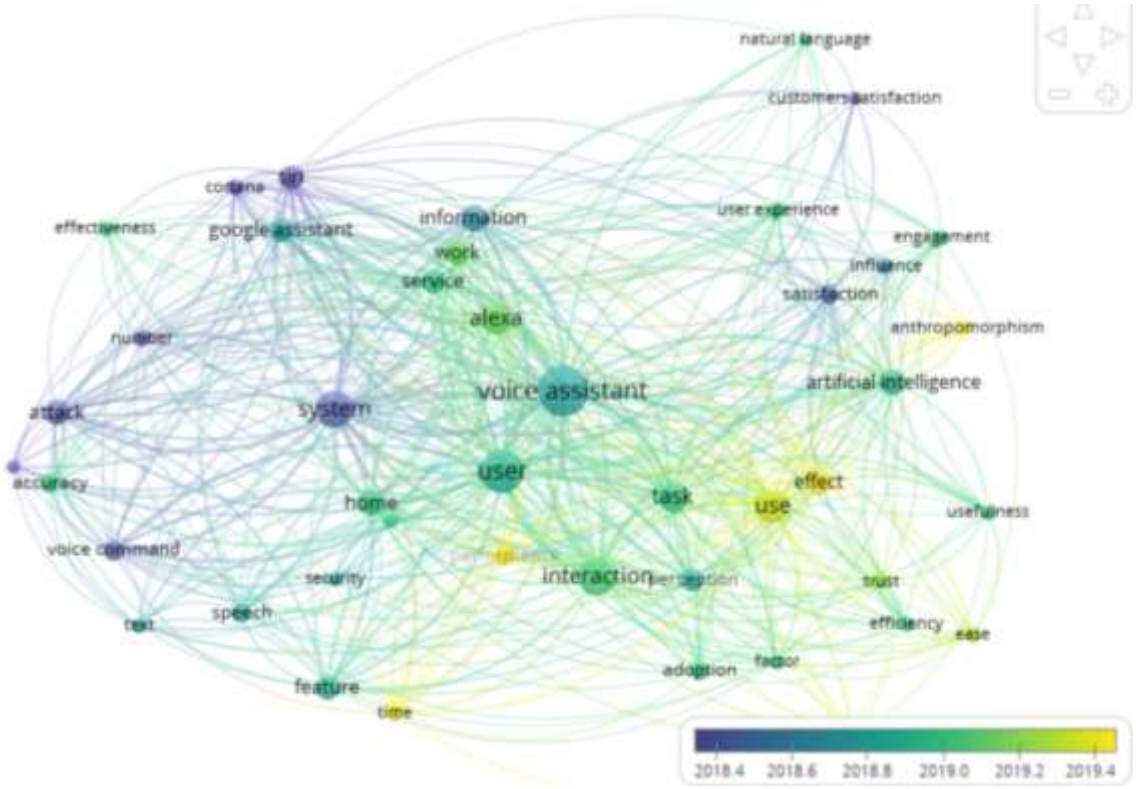


Figure 2.3. VOSviewer map illustrating the shift of focus over time

2.3 Results: The themes

The themes have been categorised according to two primary stages. The initial stage involved examining the titles, abstracts, and keywords of all the papers included in the study and then establishing relationships using the VOSviewer software (see Figure 2.2). The subsequent phase involved the construction of a concept matrix, which outlines the extent to which the selected papers contribute to the relevant concepts. This matrix is presented in Appendix B, which provides a representative sample of the included papers. Figure 2.4 shows the structure of the identified categories.

Table 2.2. The Themes

The themes	Example quotes and references
The use of the voice assistant • **Voice assistant adoption:** This theme focuses on the rising prevalence of voice assistants in different technological ecosystems, indicating the growing trend of users integrating voice assistants, such as Siri, Alexa, or Google	**Voice assistant adoption:** ❖ "In this article, an in-depth comparative analysis is conducted for determining which model best explains the user's acceptance of using Vas". (Pal, Arpnikanondt, Funilkul, & Chutimaskul, 2020). **Adding value to the voice assistant:**

Assistant, into their everyday routines for the sake of convenience and effectiveness. • **Adding value to the voice assistant:** This theme focuses on the techniques and advancements that aim to improve the user's experience by integrating additional features and services into voice assistants. • **Enhancing the voice assistants:** This theme focuses on ongoing endeavours to enhance the functionalities of voice assistants through advancements in natural language processing, machine learning, and AI technologies.	❖ "Here we show that existing Alexa development features such as skill persistence allow for the effective implementation of spaced-repetition and suggest a heuristic adaptation of the spaced-repetition model that is appropriate for use with voice assistants (VAs)". (Skidmore & Moore, 2019). **Enhancing the voice assistants:** ❖ "An open-source library DeepPavlov is tailored for development of conversational agents. The library prioritises efficiency, modularity, and extensibility with the goal to make it easier to develop dialogue systems from scratch and with limited data available. It sup- ports modular as well as end-to-end approaches to implementation of conversational agents". (Burtsev, Seliverstov, Airapetyan, Arkhipov, Baymurzina, Bushkov, ... & Zaynutdinov, 2018).
Interactions with voice assistants	❖ "we investigated the role of non-verbal cues in human-AI interaction as a way to create a more natural and effective interaction with users, which leads to positive emotional, motivational, and cognitive outcomes ". (Kim, Kim, Nam, & Song, 2020). ❖ "The findings demonstrated how families engaged in a diversity of interactions when using the Alexa skills". (Beirl, Rogers, & Yuill, 2019).
Privacy and security considerations	❖ "we explore the feasibility of using users' text-converted voice utterances as classification features to help identify users' genuine commands, and detect suspicious commands". (Kwak, Huh, Han, Kim, & Yoon, 2019). ❖ "we employ low-cost motion sensors, in a novel way, to detect these hidden voice commands." Wang, Anand, Liu, Walker, Chen, & Saxena, 2019).
Increasing user satisfaction	❖ "In the course of an empirical investigation, 26 use cases were evaluated by two samples each of more than 2,000 customers of an online fashion retailer. With the help of the Kano approach and the Segmented Kano perspective, the requirements were categorized to determine the resulting effect on the satisfaction of the customers". (Baier, Rese, Röglinger, Baier, Rese, & Röglinger, 2018).

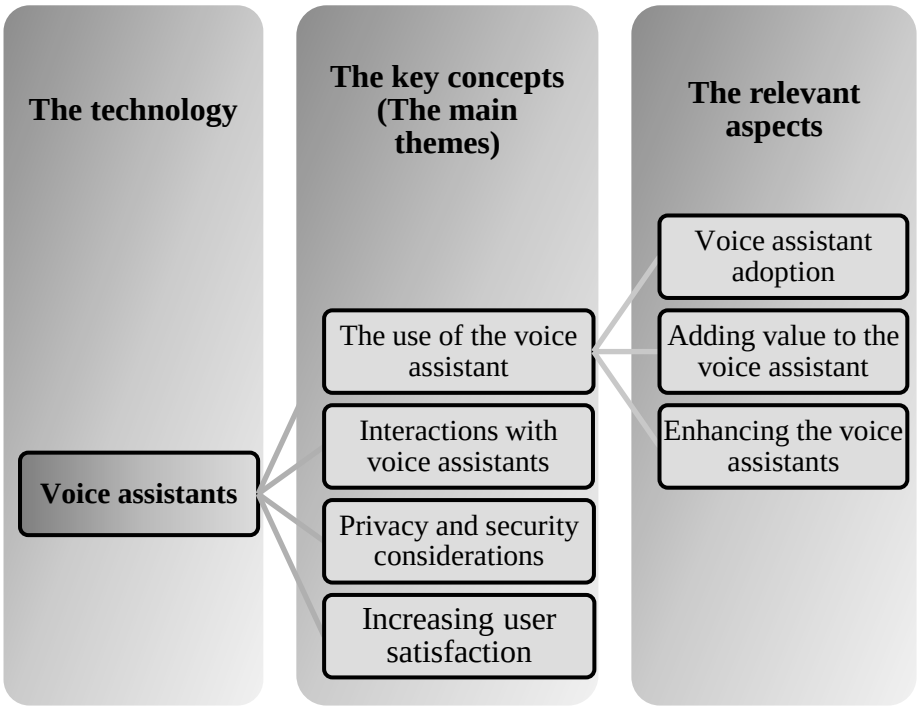


Figure 2.4. The structure of the identified categories

2.3.1 The use of voice assistants

Several studies have focused on using VAs as a significant aspect of new technology. This usage can be measured from different perspectives in order to clearly evaluate the technology and provide recommendations for future development paths. The main perspectives considered so far are the adoption of VAs, the associated user values, and how these can be improved to encourage wider adoption.

2.3.1.1 Voice Assistant Adoption

Technology adoption is an area that has been studied for many years, providing literature to describe general models of technology adoption. However, VAs exhibit human-line characteristics, which makes them different from conventional technologies (Wagner & Schramm-Klein, 2019). This is expected to impact the adoption dynamics and models. Therefore, new customised models are required for emerging technologies based on their features (Nasirian, Ahmadian, & Lee, 2017). McLean and Osei-Frimpong (2019) noted that VA-based Artificial Intelligence has different attributes compared to existing adoption models such as TAM and UTAUT, but these may not be sufficient to explain users' behaviour towards the VA in their own right. As a result, three different conceptual models for adoption have been

developed, focusing on the VA as a new technology. Pal, Arpnikanondt, Funilkul, and Chutimaskul (2020) evaluated the efficacy of four widely used technology acceptance theories, namely TPB, TAM, VAM, and UTAUT, in forecasting the implementation intentions of VAs. Their results indicate that VAM outperforms TAM, UTAUT, and TPB in terms of efficiency. While TAM has been widely used in the past to forecast the adoption of emerging Information and Communication Technologies, its oversimplification renders it the least predictive model and, therefore, unsuitable for VAs (Pal et al., 2020).

Voice assistants generate new types of interactions, which can be described as more implicit and transparent types of exchanges with users; these require high-quality evaluation of Human-Computer Interaction (HCI) in order to assess their quality (Nasirian et al., 2017). In addition, a significant new dimension in the use and adoption of this technology is the attractiveness of the devices involved and the social presence conveyed by the VAs. These have not been captured in previous technology adoption theories that focused on technologies without the humanlike social presence which characterises the VA (McLean & Osei-Frimpong, 2019). As a result of this gap, Nasirian et al. (2017) have developed a new adoption model which combines three theoretical perspectives: the social exchange theory (SET), Human-Computer Interaction (HCI), and the Information systems success model (ISSM). This takes account of the effect of personal innovation on users' trust in their VAs to understand the quality preferences which most affect the adoption of the VA and describe the adoption of new emerging artificial intelligence-based technologies. This adoption model concentrates on a new system quality construct as the main objective to describe the adoption of VA technology, whatever form that technology takes.

Considering the different types of VAs and customising the models according to each specific type (for example, in-home VAs) is likely to increase the adoption of this form of artificial intelligence-based technology. Therefore, McLean and Osei-Frimpong (2019) have developed another conceptual model combining HCI attributes with voice technology and U&GT theory. This considers perceived privacy risks as an attitudinal dimension that could provide insight into the variables that drive using VAs in the home to understand users' motivations for using and adopting them. However, as the theoretical model focuses on the in-home VA, the resulting factors and their effects could not be generalised for VAs in general. In addition, Moriuchi (2019) proposed a model for examining the model of technology acceptance (TAM), a conceptual framework (perceived usefulness and perceived ease of use) and their effect on the engagement and loyalty between consumers and their VAs.

While Nasirian, Ahmadian, and Lee's model consist of three main factors (system quality, service quality and information quality) affecting user satisfaction and usage intention, the second model includes key categories which could motivate in-home VA usage (utilitarian benefits, symbolic benefits, social benefits, and hedonic benefits). The results of both models indicate that, in the generic VA adoption model, interaction quality is the most important factor in adopting new technologies, identifying it as the motivation factor that adds value, creates trust and, as a result, increases usage. Amugongo (2018) confirmed the significance of interaction quality in VA adoption, finding that African users were particularly disappointed with their interactions due to the technology's failure to understand their native languages and their lack of emotional intelligence. He concluded that enhancing the user experience by resolving these interaction issues would lead to the broader adoption of VA technology in Africa.

In addition, in terms of in-home VAs, a further key factor in their success is the social presence created through the use of VA technology. While each model foci on different perspectives to identify the factors which contribute to successful adoption, it is clear that the humanlike social presence associated with VAs is a significant factor in the adoption of this new technology and the reasons which encourage users to adopt VAs in their home may be different to those which motivate them to use other types of VA technology. Fernandes and Oliveira (2021) confirmed that users' motivations play an essential role in adopting the VA in service encounters. Their findings demonstrate that relational, functional, and social factors all contribute to adoption, reveal their crossover impacts, and illuminate the moderating influence of experience and the desire for human interaction.

The practice of applying human characteristics to technological devices and the tendency to make objects 'humanlike' is known as anthropomorphism, and it is a distinctive facet of VAs, along with artificial intelligence (Horstmann, Bock, Linhuber, Szczuka, Straßmann, & Krämer, 2018). As a result, anthropomorphism has received particular attention as a key factor affecting the adoption of VAs. It has been studied from two primary aspects: firstly, its position in relation to digital VAs using a qualitative approach (Wagner & Schramm-Klein, 2019) and, secondly, the impact of an augmented reality visual embodiment of a VA on collective decision-making under three conditions: conducting the role alone, collaborating with a disembodied VA, and collaborating with an embodied VA (Kim, de Melo, Norouzi, Bruder, & Welch, 2020). In addition, Moriuchi (2020) analysed three studies to ascertain the effect of anthropomorphism on users' intention to re-use a VA. The results of these studies indicated that anthropomorphism is viewed as having a positive impact on VA adoption.

However, more recent research by Fernandes and Oliveira (2021) concluded that its effect is not uniformly positive, but it sheds new light on the understudied customer-robot rapport-building role. This suggests that further study may be necessary to validate these findings and contribute to a comprehensive understanding of anthropomorphism's impact on the acceptance of VAs.

2.3.1.2 Adding value to the voice assistant

Another significant aspect in promoting the use of new technology is user value. A value is defined as a "belief that a specific mode of conduct or end-state of existence is personally or socially preferable to an opposite or converse mode of conduct or end-state of existence" (Rokeach 1973, p.5). As Nasirian et al. (2017) found, factors that add value contribute to adopting the VA. Amugongo (2018) argued that advances in the field of machine learning and artificial intelligence offer new ways to create value and improve user experience, notably by enabling people to interact with devices using everyday language. Rzepka (2019) confirmed the importance of adding value through focusing on what users themselves value and providing an in-depth analysis of means-ends value chains. These explain users' decision-making processes regarding human-computer speech interaction by focusing on identifying, defining, and understanding the user values that can be achieved using VAs. Other approaches which have been suggested to increase the VAs' value include merging VAs to increase functionality. Zhao and Rau (2020) designed an all-around merged VA that could perform tasks in both users' personal and professional lives, combining two independent VAs in order to enhance user values by increasing the usefulness of the VA. They also proposed a new interaction between a corporate VA and a personal VA, which they called 'synchronisation' (Zhao & Rau, 2020). This enabled two different VAs (a housekeeper for personal issues and a secretary for work-related issues) to share information and work together in order to complete tasks which could not be done without their collaboration. For instance, the first VA might complete the first part of the task and then wake the second VA to complete it. Taking the idea of collaboration a step further, Hettiachchi, Sarsenbayeva, Allison, van Berkel, Dingler, Marini, and Goncalves (2020) created Crowd Tasker (on Google Assistant), a novel system that enables crowd tasks to be performed via the VA. Using the VA, rather than a traditional online platform, offers crowd workers much greater flexibility, facilitating multitasking and allowing them to access work simply by talking to their devices, making this type of work more attractive.

In addition, Norval and Singh (2019) provided another approach to adding value by exploring the VA to allow users to gather information about the Internet of Things (IoT) system surrounding them. They focused on the Alexa VA platform and created the means for users to ask questions about the system's behaviour. This approach involved functionality within the VA itself, or which could be accessed through it, to enable users to enquire about the automated environment around them. In order to provide meaningful answers, the VA uses recipe, log and provenance information, which enables it to track the sequence of steps involved and their inputs and outputs. This allows a user to ask about what a system should do and what a system has done. Therefore, it established the first stage in considering how VAs could help users better understand, challenge, evaluate, and accept VAs as new technology. Although this study was limited to the Alexa platform and cannot be applied to other VAs, similar implementations could be created for other VAs, data representations, or databases. However, as the Authors have confirmed, this study's approach hinges on data available regarding how each system operates, and such data are not usually collected nor generally accessible. Therefore, this approach may be limited to other VA brands, such as Google.

Another approach to making the VA more widely adopted is to focus on specific countries, societies, or demographic groups and consider the attributes required based on their particular needs. For example, Amugongo (2018) conducted a study focused on six African countries (Namibia, Zimbabwe, South Africa, Nigeria, Zambia and Niger) to understand African users' opinions about VAs. Likewise, Hwang (2019) analysed the conversational styles of some South Korean VAs (Bixby, Nugu, Kakao mini, Clova and Giga Ginie) to examine the effect of female gender stereotypes in assistants with female voices and to understand how these assistants interact with their users. The reason behind this study lies in the fact that while VAs offer gendered voices to obtain natural sound, most users select a female voice as the default setting. Creating VAs which are adapted for use by older users may also increase adoption. A study by Trajkova and Martin-Hammond (2020) found that, over a year, most elderly participants stopped using their in-home VA (Amazon Echo), even though they had been early adopters. This was due to a lack of perceived usefulness, concerns about their own technical abilities, and difficulties using the devices in shared accommodation. This suggests that current interfaces, which use the same style of auditory response in every situation, are unlikely to be widely adopted by older users unless they are adapted to meet their specific needs. Therefore, Kowalski, Jaskulska, Skorupska, Abramczuk, Biele, Kopeć, and Marasek (2019) established a study to explore the potential of VAs to meet the needs of older adults in the context of smart home technology. The four key needs emerged: a) an accessible design

with a low barrier of entry, b) understanding technology and receiving feedback, c) control and assurance of security and d) seamless incorporation into everyday life. These findings were supported by Chattaraman et al. (2019), who demonstrated that a one-size-fits-all agent design strategy failed to meet older users' needs by illustrating the degree to which the level of cognitive, functional and social support The foci and intents of the included papers s could offer with an online shopping task depended on both the style of the agent interaction and the task-competency of the user. This indicates that personalisation would significantly increase the value of VAs and encourage their wider adoption. To illustrate this, Braun, Mainz, Chadowitz, Pfleging, and Alt and (2019) created four different assistant personalities ('Admirer', 'Friend', 'Butler' and 'Aunt') and compared them to a baseline (Default VA) in order to investigate the effect of personalised characters on user experience, acceptance, trust, and workload. Their findings indicate that personalisation increases confidence and likability when the VA character fits the user's personality, but mismatches can increase dissatisfaction.

Another demographic group that has received attention is visually impaired users. Storer, Judge and Branham (2020) confirmed that users with visual impairments value VAs more highly than sighted users, and the technology's potential to aid people with visual impairments has increased usage within this group. As a result, Choi, Kwak, Cho, and Lee (2020) examined the perceptions of people with visual impairments in relation to the speech rate of their VAs and found that they were more satisfied with a default speed rate similar to human speech rather than at higher speeds. This study highlights both the impact of speech rate design on people with vision impairments and the potential of using speech rate as a design feature to promote wider adoption. Although VAs enhance accessibility for visually impaired users, Storer, Judge and Branham (2020) found that challenges can arise in mixed-visual-ability families, and these should be considered before bringing VAs into the home.

Skidmore and Moore (2019) also identified problems when they examined Alexa with a view to enhancing its value in the education field, highlighting elements that hinder language learning, notably the absence of multilingual speech recognition. While their study confirmed that Alexa could support basic language learning, it requires further development to become an efficient language-learning tool. By focusing on specific countries or groups, each study provides different results based on the characteristics of their particular demographic. However, considering their findings together could prove an effective baseline for examining a new area targeting the adoption of VAs. For example, Hwang, Lee, Oh, and Lee's (2019) findings concerning the stereotypical aspects of femininity present in some South Korean VAs and Amugongo's (2018) conclusions that VAs are developed without considering developing

or emerging markets could provide new directions for development, adding user value (by broadening the range of languages and accents available, for example), and thereby contributing to the wider adoption of VA technology.

Moreover, Mayer, Laput and Harrison (2020) offer a different value-adding approach. They presented WorldGaze, a smartphone software tool that engages the front and rear cameras to establish its physical location and follow the user's gaze. This enables the VA to provide context-specific information and respond more naturally. In a similar vein, Shen, Chen, Wei, Yang, and Choudhury (2020) focused on the feasibility of determining the user's position using voice signals obtained from microphones, facilitating the addition of user localisation as a modern feature of VAs within the home. Furthermore, Zhao, Westing, Scully, Nieto, Holenstein, Jeong, and Guestrin (2019) engineered a reliable, low-power detector capable of operating in various scenarios to identify a novel way of interacting with VAs on smartwatches: simply by lifting your hand and talking naturally. Particular features include components to protect consumer privacy and have minimal impact on battery life, memory, and processing power.

These approaches to added value have identified different factors that affect the adoption of the VA. According to Rzepka (2019), five fundamental objectives were defined that maximise users' overall value in using VAs (convenience, efficiency, enjoyment, less cognitive effort and ease of use). The last three objectives are related to existing constructs of "hedonic motivation" and "effort expectancy" in UTAUT2 theory (Venkatesh, 2012) and also "perceived ease of use" in TAM (Davis, Bagozzi, & Warshaw, 1989). However, Zhao and Rau (2020) conclude that merging VAs' performances is the best-preferred type for users in terms of perceived usefulness, satisfaction, mental workload, and performance. The reason could be that merged VAs encompass both work and daily lives, meeting a higher range of user requirements and expectations, thereby enhancing the user experience.

With regards to users' "perceived usefulness" or "performance expectancy", the benefits that users expect most from their VA use are efficiency and convenience; these have been identified as more important than the other constructs (Rzepka, 2019). By looking at the merging of VAs as an example, Zhao and Rau (2020) confirmed that merged VAs are better than other VAs in terms of a) the required instruction (fewer), b) perceived usefulness (higher), c) completed tasks' quality (higher); d) satisfaction (higher) and e) mental workload required (less). In fact, improvements in these factors could indicate an increase in the efficiency of the technology, as noted by Rzepka (2019). Furthermore, as Zhao and Rau (2020) supported, increasing the efficiency and entertainment value offered by VAs can increase users'

satisfaction, thereby enhancing adoption. The outcomes of applying these different approaches show the importance of certain factors (which were common factors in different studies) and their effect on the adoption of the VA from many perspectives.

As Rzepka (2019) concluded, an integrated and comprehensive perspective on the values of using VAs can further understand when and why speech interaction is preferred over other modes of interaction with information systems. As Amugongo (2018) suggests, a key concept in promoting the adoption of the VA in different regions is the need to meet local users' specific needs by resolving issues which negatively affect their user experience, thus enabling users to gain more value from using the devices. In addition, Norval and Singh (2019) argued that value could be added by offering users a new way to understand the IoT ecosystems around them in terms of why specific functionality occurs and how the entities involved interact; this is achieved by providing sources of information, logging previous behaviours and actions and documenting the sequence of steps in a particular process or system. In summary, these approaches to adding value focus on two main concepts: understanding the users' values and helping users to understand the technology (e.g. how the systems operate) in order to contribute to the wider adoption of the VA.

2.3.1.3 Enhancing the voice assistants

Improving voice assistants would help to increase their adoption as an innovative new technology. As a result, the development of VAs has attracted scholarly interest in several areas, including customer queries, interactions, and how to process more complicated utterances. Due to the adoption of message communication via VAs in recent years, there is a need for tools that speed up the prototyping of feature-rich conversation systems. Therefore, Burtsev, Seliverstov, Airapetyan, Arkhipov, Baymurzina, Bushkov and Zaynutdinov (2018) created DeepPavlov, an open-sourced library optimised for conversational agent development. It places a high value on modularity, extensibility, and efficiency in order to facilitate the development of dialogue systems from scratch and when limited data is available, and supports both modular and end-to-end methods of agent development. It also enables the development of multipurpose agents with a variety of skills. This is critical in real-world application contexts since abilities may be added, improved, or deleted separately after a conversation system has been installed, and existing capabilities can be used to accelerate the creation of new services and conversational solutions.

The aim of interaction development is to achieve seamless interactions which better meet user expectations. Generally, voice-activated interactive systems such as VAs operate in two stages: converting the input voice to text using an automatic speech recogniser and extracting information from text using natural language understanding to conduct the required task. However, performance can be affected by factors such as users' accents, pronunciation and use of homophones. As a result, Bhasin, Mathur, Yenigalla, and Natarajan (2020) proposed a multistage CNN architecture system which uses phoneme sequences instead of word sequences to overcome these issues. Capable of processing partially generated phoneme sequences, it attained a level of accuracy comparable to the most advanced text classification systems across various stages and with phoneme sequences of varying lengths.

In addition, VAs frequently rely on a semantic parsing component to determine which actions to perform in response to a user's instruction. Usually, rule-based or statistical slot-filling systems have been used to parse simple inquiries, and shift-reduce parsers have been developed for more complicated utterances. However, while these approaches are effective, they place certain limits on the types of queries that may be parsed. Therefore, Rongali, Soldaini, Monti and Hamza (2020) developed a unified architecture for semantic parsing, which can be used for a variety of query types with high levels of match accuracy. Gamzu, Haikin and Halabi (2020) have also addressed problems that arise in voice shopping, notably mispronounced or misunderstood queries. Voice null queries like this typically fail to elicit the required response and negatively impact users' shopping experience. As a result, Gamzu et al. (2020) developed a new framework for generating possible alternative queries, ranking them for suitability, and then substituting the best alternative for the null query. Testing in commercial VAs and an e-commerce website revealed fundamental differences between null voice queries and web shopping null queries, corroborating the employment of mechanisms designed specifically for the speech domain.

2.3.2 Interactions with voice assistants

The interactions between voice assistants and users significantly affect the user experience and could play a key role in encouraging wider adoption of this technology. Therefore, the interaction perspective has received attention from researchers and has become a significant area for study. For example, Raveh, Siegert, Steiner, Gessinger and Möbius (2019) explored how users interact with their VAs when another speaker is present. They measured the overall accommodation between participants during these interactions and analysed factors

including gender, task performance, and task order to gain insights into the participants' behaviour. Additionally, Parviainen and Søndergaard (2020) examined how users' experiences and interactions with VAs can be influenced by whispering. Research suggests that the manner in which users speak to their devices may reveal more about their personality and emotional condition than they realise. Hildebrand, Efthymiou, Busquet, Hampton, Hoffman and Novak (2020) examined this from a business perspective, creating a conceptual framework for correlating vocal characteristics with human experiences and emotional states depending on user interaction and considering how businesses might use voice analytics in the future.

On the domestic front, Beirl, Rogers and Yuill (2019) demonstrated how VA technology influences family interactions in the home and found that usage patterns changed over time. In addition, Lahoual and Frejus (2019) performed an in-depth investigation in household and driving environments to better understand how VAs are used and to assess the efficacy of vocal interactions in contexts found in nature. Moreover, Kim, Kim, Nam, and Song (2020) examined the role of non-verbal voice cues in human-VA interactions and found that they played a crucial role in promoting positive emotional, motivational, and cognitive outcomes. However, there could be a difference in the interaction results based on users' characteristics, such as age (notably among children or older users). Therefore, Aeschlimann, Bleiker, Wechner and Gampe (2020) examined children's interactions with VAs and the impact these had on their prosocial behaviour and communication. They found that the children enjoyed playing with these technologies but did not expect to cooperate with them like a human.

The results of these studies indicate that the accommodation between users is more evident when the user first interacts with the gadget alone, and the chronological order of interactions affects participants' speech behaviour (Raveh et al., 2019). However, the interpersonal relationship's key variables are connectedness and intimacy, supported by the use of nonverbal empathic signals, and this has implications for human-AI interaction research (Kim et al., 2020). It is also significant to note that in cases where VAs proved to be a barrier to the completion of tasks in household and driving environments (Lahoual & Frejus, 2019), users often forgave and accepted these flaws rather than abandoning their devices, especially among experienced users with higher levels of knowledge and technical competence.

Within families, Beirl et al. (2019) concluded that interactions with the Alexa VAs motivated distinct trends of family engagement, including increased teamwork to scaffold children's relationships with the devices. While children eagerly adopt and interact with VAs, even tiny children are able to distinguish artificial intelligence from human intelligence and

adapt their behaviour accordingly (Aeschlimann et al., 2020). As Aeschlimann et al. (2020) demonstrated, human cooperation principles are distinct from human-computer cooperation standards; however, both the positive and negative sides of family interactions are necessary components of understanding one another and improving empathy skills, and interactions with VAs, such as Alexa, may assist children to learn an appreciation for the art and pleasure of dialogue in the presence of both humans and machines.

2.3.3 Privacy and Security Considerations

Voice assistants are being rapidly improved to address existing privacy and security issues. Nevertheless, security researchers have explained that the voice channel's open nature makes it challenging to secure VAs from cyber-attacks (Feng, Fawaz, & Shin, 2017). Voice assistants continuously monitor conversations, which are usually transported to a cloud, where they are stored, processed, and often transmitted to other service providers. However, users have no chance to control this process, such as by tracking conversation recordings, nor can they control the recording behaviour of surrounding VAs. As a result, Cheng, Bagci, Yan and Roedig (2019) established a prototype acoustic tagging device to signal how VAs and their back-end systems should handle recordings. Based on PocketSphinx, the implementation enables devices to tag conversations and retrieve the tagging signal from conversations saved in Google's back-end system (as their experiment focused on Google Home) (Huggins-Daines, Kumar, Chan, Black, Ravishankar, & Rudnicky, 2006). In addition, Seymour (2019) created a device (the Aretha project) that allows users to engage in conversations about network security and privacy and the sophistication of modern conversational voice interfaces to reduce security and privacy risks in their homes.

However, despite such developments, the open nature of these cloud-based services continues to raise security issues and user concerns about privacy persist (Lee, Lee, & Sheehan, 2020). Security concerns could impede the adoption of VA technology, and existing research is focused on addressing issues such as the leakage of recording content without users' authorisation, defending against adversarial audio attacks, voice squatting attacks and voice masquerading attacks, hidden voice command attacks, and spoofing attacks. As Zhang, Mi, Feng, Wang, Tian, and Qian (2019) reported, new attacks that target the VAs' ecosystems have appeared in their security analyses, including voice squatting attacks and voice masquerading attacks, which seek to steal users' private information through impersonation. The problem, as Mitev, Miettinen and Sadeghi (2019) explain, is that specially developed malicious software

can cause damage by launching ‘man-in-the-middle’ attacks against the VAs’ functionalities, targeting the Skill/App ecosystem and using loopholes in the interface to redirect input (users’ voices) to bogus Skills/Apps, thereby hijacking the conversation between users and their devices to cause the users to harm.

In order to resolve these issues, Feng, Fawaz and Shin (2017) established VAuth, the first system to provide usable and continuous authentication for the VA. Their system guarantees that only commands given in the owner’s own voice will be executed by the VA. The system achieved a high matching accuracy, with a false positive rate of less than 0.1%, and successfully disrupted various practical attacks, including impersonation attacks, mangled voice attacks, and replayed attacks. In addition, Gao, Chandrasekaran, Fawaz and Banerjee (2018) proposed a framework to impose a privacy and security perimeter around the VA. The framework continuously jams the VA’s microphones in order to block unauthorised voice commands and prevent the recording of speech unless the user issues a voice command. This operates through a scheme similar to two-factor authentication, allowing access only when the authorised user is close to the VA.

As Zhang, Zhang, and Zhang (2019) confirmed, relatively little is known in the literature about defending against audio adversarial attacks. As a result, he used audio compression and code modulation to propose several proactive defence mechanisms to target audio adversarial examples in automatic speech recognition systems. Moreover, Kwak, Huh, Han, Kim and Yoon (2019) explored the possibility of using text-converted voice command analysis to support security-critical command processes (e.g. making payments, checking emails, and unlocking devices) by identifying simple user commands, detecting suspicious commands, and maintaining high detection accuracy. They developed a globally trained model, open to all users (including new users) and gradually switched to a user-specific model tailored to a target user’s speech patterns. When an optimal threshold value was used, the model detected 95.7% of attacks and achieved an equal error rate of 3.4%. However, some of the most common attacks on current voice interface systems are “hidden voice commands”. In these cases, the attack is embedded in the user’s voice command through synthetically rendered sounds in order to trick the speech recognition process into executing malicious commands without the user noticing (Wang, Anand, Liu, Walker, Chen, & Saxena, 2019). Therefore, Wang et al. (2019) proposed a system that detects attacks through low-cost motion sensors, focusing on the unique audio signatures in the vibration domain for the issued voice commands.

Additionally, to avoid spoofing attacks launched by adversaries who record users’ voice commands and then replay them, Ahmed, Kwak, Huh, Kim, Oh, and Kim (2020) provided a

"Void" solution that analyses the spectral power patterns of voice signals (between voices replayed through speakers and live human voices) to accurately detect voice replay attacks. Void is based on the use of a single classification model (with just 97 features), and it is resilient against six kinds of attack, with the following reported detection rates: inaudible voice command (100%), hidden voice command (99.7%), voice synthesis (90.2%), combining replay attacks with live-human voices (98.2%) and equalisation manipulation attacks (86.3%) (Ahmed et al., 2020).

Moreover, Vaidya and Sherr (2019) introduced new techniques that seek to prevent malicious technologies from falsifying or misdirecting users' speech and enhance voice recognition capabilities. They sought to safeguard users' privacy without requiring external support from transcription services by sanitising voice inputs and removing distinctive voice characteristics, leaving only the information needed for the cloud-based services to recognise the speech. Zhang et al. (2019) also focused on finding solutions to mitigate security threats. In terms of voice masquerading threats, they designed and implemented a context-sensitive detector with a 95% precision rate. Although new detection software is being developed to identify risks and mitigate threats, notably for Google and Amazon systems, these attacks are likely to increase (Zhang et al., 2019). Given the fact that some users would never use a smart speaker because of privacy concerns and risk perceptions (Rzepka, 2019), it would be highly beneficial to integrate these solutions into mobile VAs and in-home VAs to improve their efficiency and prevent issues that negatively affect their usage.

Furthermore, in terms of users' privacy with regard to health-related information, Cho (2019) observed the effect of the modality (voice or text) and device types (mobile VA or in-home VA) on users' perceptions, specifically in terms of retrieving sensitive health information from their VAs. The study found that voice interaction enhanced the VA's perceived social presence only if the users required less sensitive health information, and fewer privacy concerns were raised in these cases. While generally, only 'low-sensitivity' users felt positive about their voice interactions, the type of device used had no significant effect on users' attitudes toward the VA, regardless of the sensitivity level of the health information being sought or the level of users' privacy concerns.

2.3.4 Increasing user satisfaction

User satisfaction is one of the most important aspects of the VA, and it has been studied from a number of different perspectives over the last few years. For example, Ernst and

Malzahn (2018) developed a model based on the Expectation Confirmation Theory to explain consumers' post-purchase behaviour in relation to the order to evaluate the role of Past Technology Category Satisfaction on people's technology usage behaviour; this included one predecessor of both Perceived Enjoyment and Perceived Usefulness (Past Technology Category Satisfaction), and two predecessors of Behavioural Intention to Use (Perceived Enjoyment, Perceived Usefulness). Meanwhile, Lee et al. (2019) tested the effect of post-adoption use of the Alexa VA (such as habit, continuance of use, etc.) on group dynamics among shared users in order to understand the factors which drive satisfaction, the relationship between satisfaction and continuance usage, and the role of habituation as a mediator between them, as well as investigating how the sharing of VAs contributes to Group Harmony. In addition, Baier, Rese and Röglinger (2018) used the Kano approach and the Segmented Kano perspective to explore conversational user interfaces' use in online shopping and their influence on customer satisfaction. They also investigated the most attractive use cases based on users' opinions and which cases appealed most to particular customer segments.

Another perspective that could affect user satisfaction is that of speech interaction systems, an area which is becoming more important due to their increasing popularity. Ning, Jia, Wu, Li, An, Wang, and Meng (2017) assumed that the primary purpose of speech interaction systems is to produce more satisfactory responses for users regarding their speech utterances and identified this as the most critical problem in analysing user intention. They created a benchmark data set using the Sogou VA (a Chinese brand), which uses a real-world speech interaction system to predict intention prominence. Focusing on modality contribution analysis and multitask learning effects showed a 7% increase in understanding user intention, opening up new ways to increase user satisfaction.

However, Ernst and Malzahn (2018) affirmed that users' current usage behaviour towards specific technologies is indirectly influenced by their satisfaction with technological categories in the past (through Perceived Usefulness and Perceived Enjoyment). For example, if people were satisfied by a technology category in the past, their expectations regarding the usefulness and enjoyment of new technologies in that category will be positively affected today. Furthermore, Baier, Rese and Röglinger (2018) added that while the classification of services showed that there is hardly any influence on user satisfaction due to limited user experience, regular users who are familiar with the VAs seem to have different points of view in terms of the requirements and the evaluations of the scenarios when compared to other users. For instance, they had much greater enthusiasm and showed less rejection when deciding which use cases were most attractive. Furthermore, Lee et al. (2019) compared the relative influence

of three factors, Hedonic Motivation, Compatibility, and Perceived Security, showing the highest beta weight of each of the factors respectively, and finding that, as a whole, they explained over 60% of the variance in satisfaction of post-adoption VA usage.

Although satisfaction is a predictor of continuance intent in most systems, the key drivers of satisfaction differ depending on the system's value propositions and the context in which the system operates (Lee et al., 2019). In terms of investigating the differences in intention understanding, given that different users may have different understandings of the intention conveyed by the same utterance, using intention prominence is more consistent with the real intention understandings of the users (Ning et al., 2017). However, users can be expected to form the intention to use a technology if they believe that it satisfies their expectations with regard to its hedonic benefits and instrumental benefits (Perceived Enjoyment and Perceived Usefulness); thus, the intention drives the degree of actual usage for technology users (Ernst & Malzahn, 2018). While Ernst and Malzahn's study focused only on German-speaking people and on Google Assistant, these empirical findings could be explored in relation to non-German-speaking people or by considering another brand of VA, such as Siri.

To sum up, all these studies aimed to increase user satisfaction; therefore, it would be valuable to assess the percentage increase in satisfaction concerning each approach by examining the findings and applying them in specific case studies using different types of VAs (mobile and in-home VAs) and different brands (Siri, Google and Alexa for example) to ensure that similar results can be generated for all the VAs.

2.4 Discussion and Conclusion

The purpose of this paper was to provide a systematic review and analysis of studies related to VAs in order to provide insights into the main directions of current research and identify gaps in existing knowledge. It began by providing an overview of VAs drawn from papers selected from prominent journals and conference proceedings concerned with voice-assistant-related content, identified via the Chartered Association of Business Schools (CABS) 2018 list and the Core Conference Portal website (http://portal.core.edu.au/conf-ranks/), and illustrating the VAs' significance. Following this, particular consideration was given to four particular aspects which emerged from the included papers: firstly, the use of the VA, including VA adoption, adding value to the VA and enhancing the VAs; secondly, interactions with VAs; thirdly, privacy and security issues; and finally, user satisfaction.

After analysing all the papers, it became clear that while academic interest in this novel subject is increasing, there remains a gap in the literature regarding the factors that drive technology acceptance and continuous use. Research on adopting digital technology, ranging from chatbots to humanoid robots, is sparse and fragmented. Additionally, contemporary research studies from a consumer perspective (e.g. Moussawi, Koufaris and Benbunan-Fich, 2021) often rely on existing acceptance mechanisms such as UTAUT and TAM, and this could be constrained as the usefulness of these frameworks is highly context-dependent. Furthermore, they are mostly concerned with functional characteristics without fully encapsulating automated technologies' extended dimensions and unique, empathic characteristics (Lin, Sun, Chen, & Shi, 2019).

As Chi, Denton and Gursoy (2020) note, while recent research indicates that a growing number of consumers are willing to embrace VA technology, resistance to change remains substantial and forms a fundamental impediment to adoption. One strategy for increasing user acceptance is to incorporate human characteristics into the design of the VA in order to maximise user familiarity and satisfaction by suggesting a humanlike counterpart (Purington, Taft, Sannon, Bazarova, & Taylor, 2017). In respect of this, Araujo (2018) concluded that critical distinctions should be made between anthropomorphic signals provided by embodied agents like robots (such as voice, body movements, facial expressions, dialogue) and disembodied agents like chatbots (e.g. dialogue, personality, voice, name). Fernandes and Oliveira (2021) confirmed that further investigation of the importance of perceived humanness, rapport and interactivity could be a fruitful area for future study.

The effect of anthropomorphism on VAs has become a controversial topic. Based on the analysis presented here, no study has yet explored the role of anthropomorphism in relation to automated VAs and examined the most influential humanlike characteristics using a quantitative approach. More specifically, no one has yet provided an understanding of how anthropomorphism could affect the acceptance of the VA based on quantitative results and which humanlike characteristics could enhance their adoption. Addressing this gap will contribute to solving some of the current issues associated with adopting this technology by understanding what can be done to make users more willing to accept VAs and less resistant to change. This led to the formulation of the following research question: To what extent could anthropomorphism play a role in enhancing VAs' adoption? Two main objectives to answer the research question were then identified: the first to investigate anthropomorphism's significance in relation to VAs, and the second to examine the most powerful humanlike

characteristics. The research will lead to a more comprehensive view of the adoption of VAs and offer managerial advice on integrating that technology successfully.

The limitation of this study is that all the papers were collected from only two multi-disciplinary databases (Web of Science and Science Direct) and one peer-reviewed literature database (Scopus). Another is that the search criteria were restricted to English language publications, which means that papers reporting VA research in other languages were excluded. From a completeness point of view, including more databases and more languages would give a more comprehensive picture; however, these restrictions were required to ensure the systematic literature review conducted here was feasible. The search of the three electronic databases yielded 1547 papers, of which 55 were included. Considering that voice assistance is a new topic in the field, and given the aim of identifying a specific knowledge gap, 55 papers would be a logical and reasonable number of papers whose data and findings could be relied on without doubts about the accuracy of their results or their reliability.

This study contributes to the literature on artificial intelligence-based technology usage, specifically VAs, by explaining the current research efforts and the main research directions to identify some clear potential avenues for future studies. Despite the limitations set out above, this paper has provided an analysis of the current status of VA research and the key approaches adopted. The next stage of this research project will focus on addressing the gap in current knowledge identified here.

Chapter 3. Paper Two

The Role of Anthropomorphism on the Post-Adoption of Voice Assistant Technology

Abstract

The use of Voice Assistants is becoming increasingly common in consumers' everyday lives. An increasing number of voice assistants are being introduced into multi-user settings, including homes, yet it is unclear how to improve users' continuous engagement with voice assistant devices. Conventional models of continuous use are focused on non-anthropomorphic technologies, so the main purpose of this study is to investigate the potential effects of perceived anthropomorphism, engagement, rapport, and perceived value on use continuance. The findings suggest that the user perception of anthropomorphism positively impacts the dynamics of user-voice assistant relationships. This influence appears through an indirect enhancement of the perceived value of voice assistants, consequently encouraging their continuous use. Furthermore, the research highlights the importance of engagement and rapport in shaping users' perceptions of the value of voice assistants, particularly in relation to their intention to continue using the voice assistant.

Keywords: Smart Technology, Artificial Intelligence, Robotics, Voice assistant, perceived value, Rapport, Engagement, Anthropomorphism, Intelligence.

3.1 Introduction

There has been a noticeable level of consumer interest in utilising voice assistants (VAs) powered by artificial intelligence (AI) algorithms. These are commonly referred to as smart speakers (such as Alexa (Amazon's Echo) and OK Google) (McLean, Osei-Frimpong, & Barhorst, 2021). The utilisation of VAs is expected to bring about a significant transformation in the way users interact with technology. This transformation will lead to changes in the manner in which users interact with companies and participate in diverse shopping activities (Hoy, 2018). The potential for this transformation is apparent in the significant rise in the proportion of users who own VAs. As an illustration, the United States experienced a VA ownership rate exceeding 30% in the year 2020, as reported by Kinsella (2020). Similarly, it is worth noting that in Japan, a total of 3.7 million households owned VAs in the year 2018. Estimates indicate that this number will continue to experience a substantial increase of approximately 500%, reaching a startling 22 million households by the year 2024 (Francis, 2019).

However, notwithstanding the increasing prevalence of home VAs, there remains a notable gap in the research surrounding these interactions (Acemoglu & Restrepo, 2018). Specifically, limited attention has been given to understanding the social dimension of home VA interactions and the relationship between constituents of this social dimension and the users' continuous intention to use these VAs. Overall, there is limited knowledge regarding the factors that motivate the continued usage of VAs. Available empirical research is scant and fragmented, focused primarily on utilitarian acceptance models such as the Technology Acceptance Model (TAM) (Davis, 1989), Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh, Morris, Davis, & Davis, 2003), where the technology is considered as simply a tool which may or may not be useful to the user.

Social dimensions are acknowledged in another area of research distinct from VA technology, which is social robotics. Social robot refers to when a robot engages in interactions with humans while considering social norms and emotions (Breazeal, 2003). However, the acceptance of social robots has been predominantly explored in conceptual or controlled settings (Davis, 1989; Rogers, 2003; Venkatesh, Morris, Davis, & Davis, 2003; Venkatesh, Thong, & Xu, 2012), again through the lens of conventional technology acceptance models such as TAM plus some social aspects. These studies have investigated factors such as social influences, anthropomorphism and performance expectancy, drawing upon theories like UTAUT. Significant and interesting outcomes have been discovered within distinct domains,

such as healthcare (Spekman, Konijn, & Hoorn, 2018), gerontology (Chang, Lu, & Yang, 2018), and education (Ponce, Molina, & Grammatikou, 2016). However, these findings only highlight effects observed in these specific sectors and fail to capture the social and relational aspects of users attitudes to social robots, or indeed the impact which robots' empathic, distinctive characteristics have on users continuance intention (Lin, Chi, & Gursoy, 2019).

According to Gunkel (2018), research into Human-Computer Interaction is now entering a new "third wave" where the focus is on the relationship between the users and the technology rather than the technology and its components. This paper contributes to this third wave, which frames the interactions between humans and computers in an innovative fashion by prioritising the conditions and terms of the interactive relationship over the human subject and its computational object. This is not about the operations or capabilities of the two components interacting together (the technology and the human user) but rather about their relationship characteristics. Ascribing human-like features to computing technology has become a fundamental part of its design, use, training, and communications, and it is often regarded as the most frequently used computing language metaphor (Johnson, 1994; Marakas, Johnson, & Palmer, 2000). The literature demonstrates that anthropomorphic characteristics significantly influence the relationship between users and A.I. technologies (Canning, Donahue, & Scheutz, 2014; Lu, Cai, & Gursoy, 2019; Wan & Aggarwal, 2015). The perception of artificial intelligence as a human has the potential to promote acceptance by encouraging the idea that they can be regarded as equal to humans (Pelau, Dabija, & Ene, 2021). This is captured by the concept of 'perceived anthropomorphism', which is the extent to which a user perceives a technology or product as having humanlike characteristics or behaviours (Epley, Waytz, & Cacioppo, 2007). We will focus on the impact of perceived anthropomorphism on the measure of 'perceived value', or the worth that a user places on a product or technology (Zeithaml, 1988).

The study focuses on examining the potential impact of perceived anthropomorphism, engagement, rapport, and perceived value on user perception, specifically in terms of enabling more natural and human-like interactions. Therefore, the research question of this study is: Is perceived anthropomorphism playing a role in building a relationship between VAs and users and thus enhancing the continued use intention? The first objective is to examine the effect of anthropomorphism on building relationships (engagement and rapport) between the home VAs and users, and the second is to examine the impact of engagement and rapport on perceived value and then home VA continued use intention. While engagement is a combination of

psychological states that serves as a driving force behind a user's behavioural expression (Lim, Cheah, Ng, Basha, & Soutar, 2021), rapport refers to as a personal connection that occurs between two interactants engaged in an interaction (Gremler & Gwinner, 2000). These factors are expected to significantly impact the dynamics of the relationship between users and their VAs, thereby influencing users' continued intention to use this technology. A deep understanding of these impacts is crucial for improving users' inclination to sustain their usage of VA technology. This investigation not only improves our understanding of the intricate dynamics between individuals and AI-powered VAs but also provides practical insights for enhancing the design and performance of these digital entities in order to effectively cater to and meet the users' needs.

3.2 Literature Review

The behaviour of attributing social attributes to both individuals and non-living entities, such as computers, is frequently observed in human behaviour and cognition. This behaviour arises due to the inherent cognitive limitations of the human brain, which often struggle to differentiate between actual human individuals and mediated representations of them (Reeves & Nass, 1996). The CASA framework, an abbreviation for "Computers Are Social Actors", provides empirical evidence for this interesting phenomenon, particularly in the context of interactions between humans and computers. It is based on the idea that individuals naturally possess the ability to automatically and unconsciously utilise social norms when interacting with computers, perceiving these machines as authentic partners in communication (Gambino, Fox, & Ratan, 2020; Nass & Moon, 2000). The CASA framework has been widely utilised as a theoretical framework in various research studies, providing valuable insights into user behaviours and interactions with computer systems (Nass & Lee, 2001; Wang, 2017). The existing body of literature emphasises the enduring nature of this effect, as previous research has consistently shown that participants tend to apply social norms commonly observed in human-to-human interactions while engaging with computers (Lee & Nass, 2010; Nass & Lee, 2001). A few recent research studies have provided support for these findings within the domain of VAs (VAs), confirming that users perceive and interact with VAs as genuine social entities (Carolus, Wienrich, Törke, Friedel, Schwietering, & Sperzel, 2021; Whang & Im, 2021). Consistent with the CASA framework, this study takes the position that consumers perceive their interactions with virtual assistants (VAs) as similar to engaging with other individuals.

According to Voorveld and Araujo (2020), the mere existence of social cues, even those that are subtle, has the ability to encourage and inspire social reactions from individuals. Within the domain of VAs (VAs), the voice aspect itself possesses significant social implications. It has been observed that humans have a natural ability to extract social information from auditory signals present in the voices of others (Latinus & Belin, 2011). Furthermore, individuals tend to automatically utilise voice characteristics to make inferences about various human attributes such as gender, age, identity and personality traits(Imhof, 2010). The nature of social perception extends to the domain of human-computer interactions, wherein individuals employ computer-generated voices in a manner similar to forming impressions and determining appropriate responses to these machines (Nass & Moon, 2000). For instance, individuals perceive personality characteristics from artificially generated computerised voices, thereby developing perceptions of "the speaker" and evaluating the credibility or attractiveness of said speaker (Nass & Lee, 2001). The significance of voice design in shaping people's perceptions and intentions towards robots has been emphasised in recent research conducted by Dou, Wu, Lin, Gan, and Tseng (2021) and Niculescu, van Dijk, Nijholt, Li, and See (2013). According to Krenn, Schreitter, and Neubarth (2017), individuals rely on the voice when engaging with AI agents in order to evaluate the AI's social characteristics. The prior proof derived from studies in the fields of human-computer interaction and user research highlights the propensity of humans to interact with non-human entities and machines, such as VAs, in a manner similar to their interactions with other individuals (Whang & Im, 2021). Hence, it is reasonable to consider that the interactions between users and VAs may be comprehended within the social cognition framework (Pitardi & Marriott, 2021).

According to CASA's suggestion, individuals who apply social rules during their interactions with computers may develop a positive evaluation of machines due to the increased similarity between human-computer interaction and relationship building. Based on empirical evidence, it has been observed that specific alterations in the voice characteristics of virtual agents have the potential to create a perception of existence and interaction with other intelligent entities, even if these entities are non-human and only appear to possess intelligence (Lee, & Nass, 2003). Therefore, aligned with the CASA framework, this study aims to enhance the understanding of interactions between users and VAs.

3.3 Research framework and hypotheses development

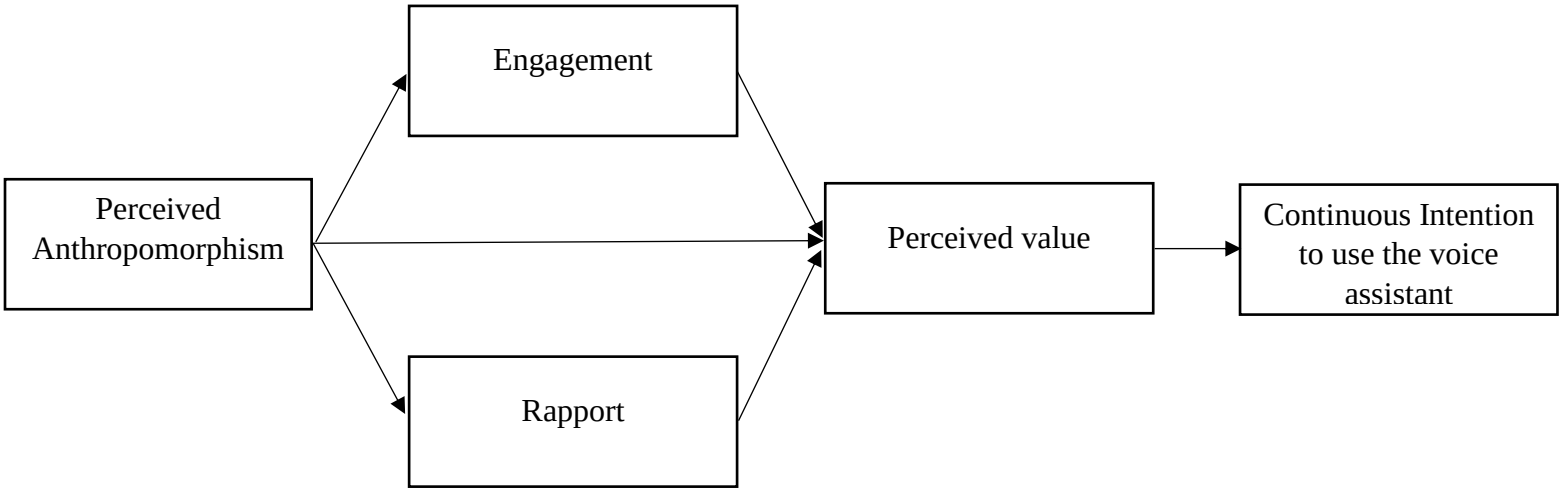


Figure 3.1. Theoretical model

3.3.1 Perceived Anthropomorphism

Anthropomorphism is the process of ascribing humanlike attributes, mental states, behaviours, and characteristics to nonhuman entities such as objects and, more recently, to technological devices (Epley et al., 2007). Whang and Im (2021) stated that anthropomorphism "refers to perceiving a nonhuman as a human or as having human traits" (p.582), while Blut, Wang, Wünderlich, & Brock (2021) defined it as a fundamental psychological process of inductive reasoning that facilitates social relationships (interactions) between humans and nonhumans. In addition, "Personification of a nonhuman entity can also be considered a metaphorical expression of anthropomorphism" (Wang, 2017, p.2). Anthropomorphism, therefore, encompasses a broad variety of characteristics associated with physical appearance or with the mental processes that distinguish human beings, such as moral judgement, intention formation, engaging in reasoning, and emotional experiences (Golossenko, Pillai, & Aroean, 2020; Kim & McGill, 2011).

Once users consider an object to be humanlike, they tend to apply human social rules and expectations to the object (including gender stereotypes and expectations of reciprocity) and treat it like a human partner (Nass & Moon, 2000; Reeves & Nass, 1996). Users readily anthropomorphise inanimate objects if they perceive typical human characteristics (such as a smile) in an entity (Epley et al., 2007), and researchers in human-computer interaction have shown that when machines communicate by voice, users are more likely to anthropomorphise

them (Lee & Nass, 2004). To illustrate this, Nass, Steuer, and Tauber (1994) discovered that when users heard the computer's 'voice' (a pre-recorded human voice), they treated the machine as if it were a human. Earlier studies established that various anthropomorphic cues could be used to remind people of humans. For example, perceived interactivity (Burgoon, Bonito, Bengtsson, & Cederberg, 2000), animacy (Morewedge, Preston, & Wegner, 2007), similarity to the shape of a human (Aggarwal & McGill, 2007; Connell, 2013), and most significantly voice (Qiu & Benbasat, 2009). All of these contribute to the anthropomorphisation of an object. As Whang and Im (2021) noted, the voice is well-recognised as a powerful anthropomorphic indicator, and Moriuchi (2020) confirmed that voice synthesis and voice recognition are among the most common anthropomorphic characteristics, as displayed in systems such as Google Home and Amazon's Alexa.

According to Złotowski, Proudfoot, Yogeeswaran, & Bartneck (2015), anthropomorphic cues are not limited to a robot's appearance; the design of the technical platform must also consider the robot's interactions with people. In this respect, it is essential to understand that anthropomorphism is not limited to visual or audible characteristics that mimic humanness but is also affected by interaction content quality (Moriuchi, 2020). In fact, users have confirmed that it is more significant that a robot performs its function precisely compared to how it looks (Złotowski et al., 2015). This suggests that the importance of anthropomorphism in the context of the VA relates to enhancing users' engagement with their devices, which could lead to more efficient responses, rather than creating devices with a human shape that do not meet users' needs.

The intrinsic human tendency to anthropomorphise entities is particularly pronounced for intelligent objects such as robots (Novak & Hoffman, 2019), and due to the nature of its voice-based conversational interactions, users are more inclined to anthropomorphise VAs. Additionally, the interaction within the conversation between VAs and users has the potential to facilitate anthropomorphism (Whang & Im, 2021). Thanks to artificial intelligence, technology-based A.I., such as VAs, are able to receive users' voice commands and provide responses comparable to what a human would say (Russell & Norvig, 2010). Additionally, VAs personalise the interactions with their users (for example, by modifying their responses based on previous behavioural data and addressing users by name). Hence, VAs can engage more naturally with users, as though they are cognitively capable humans (Whang & Im, 2021).

Creating an anthropomorphic human-computer interface facilitates users' interaction with the system and provides a better user experience (Burgoon et al., 2000). Indeed, by anthropomorphising them, humans want to communicate more effectively with nonhuman agents, improve their ability to explain intricate concepts around them and forecast the future behaviour of these complex entities (Delgosha & Hajiheydari, 2021). Assigning humanlike motivations and attributes to nonhuman agents enhances the user's ability to understand the agent's behaviour, decreases the uncertainty associated with the agent, and increases confidence in future predictions about it (Waytz, Morewedge, Epley, Monteleone, Gao, & Cacioppo, 2010). When users anthropomorphise a nonhuman agent, a sense of familiarity is generated. This helps to decrease uncertainty (Airenti, 2015), facilitates interaction (Damiano & Dumouchel, 2018), and generates shared meaning (Rogers, Ratzan, & Payne, 2001). According to Yang, Aggarwal, and Mcgill (2020), when a user anthropomorphises a nonhuman entity, the object becomes a source of social connection and active belonging. Furthermore, Chen et al. (2018) confirmed that engaging with anthropomorphised agents satisfies users' basic need for relatedness. Therefore, by creating humans from nonhumans, anthropomorphism satisfies two fundamental human needs: the need for environmental control and understanding and the desire for social connection (Epley, Akalis, America, & Cacioppo, 2008).

Delgosha and Hajiheydari (2021) found that anthropomorphism develops a sense of belongingness by shifting the meaning of interactions with objects, such as robots, away from utilitarian, experiential relationships toward more interpersonal and social ones. Moreover, Epley et al. (2007) showed that anthropomorphised agents could develop social ties and that anthropomorphising technical agents enable them to work and interact more effectively. Anthropomorphism also facilitates comprehension of the situation (Yang et al., 2020). For example, anthropomorphising unfamiliar agents (such as consumer robots and VAs) helps users predict their future behaviours by using human schemas to create a sense of understanding. Given this, anthropomorphism is clearly a key element in comprehending how humans react to and interact with robots (Blut et al., 2021).

Furthermore, through anthropomorphism, people tend to perceive a robot to be more life-like and sociable and feel a stronger sense of social connectedness (Fernandes & Oliveira, 2021), making emotional attachment to and bonding with the robot more likely. Thus, anthropomorphism facilitates human-robot rapport, making it easier, more desirable, and more valuable. As a result, we propose that anthropomorphism can be utilised to enhance user

engagement with VAs and their relationship, thereby positively influencing the continuous use of VA technology. We thus expect that:

H1: Perceived Anthropomorphism positively impacts engagement with VAs.

H2: Perceived Anthropomorphism positively impacts the rapport with VAs.

H3: Perceived Anthropomorphism positively impacts the Perceived Value of VAs.

3.3.2 Engagement

Engagement is "a fusion of psychological states that motivates a user's behavioural manifestation" (Lim, Cheah, Ng, Basha, & Soutar, 2021, p.5). While Kim, Kim, and Wachter (2013) concluded that "engagement refers to the state of being involved, occupied, retained, and intrinsically interested in something" (p.361). Additionally, according to Kearsley & Shneiderman (1998), engagement is a sequence of behavioural and emotional activities that include reasoning, cognitive processes, problem-solving, evaluation, and decision-making, which can be described as a complicated procedure that involves behavioural and emotional tasks.

Regarding a medium such as the VA, the term "user experiential engagement" refers to a set of qualitative interactions or experiences with that medium (Malthouse, Calder, & Tamhane, 2007). Experiences can be characterised as the emotions, activities, thoughts, and appraisals resulting from or during an event (Pagani & Mirabello, 2014). Kowatsch and Maass (2010) demonstrated that engagement occurs in the interactions between users and their devices to fulfil the users' needs. These need states could be driven by cognitive motives (Kim et al., 2013). Therefore, engagement with VAs could be valuable due to the facilities they offer to users in terms of immediacy and efficiency of productivity in daily activities. As supported by Venkatesh, Th, & Xu (2012), further engagement can be driven through engagement behaviours that are seen as enjoyable (hedonic), serve a useful purpose (utilitarian), or enable communication with others (social).

As new technologies, such as robots and voice assistants, shape the digital future, it is essential to consider how users engage with these novel technologies (Schmitt, 2019). Engagement could be key to forming strong user relationships with VAs. Moreover, engagement could also contribute to the continuous use of this technology by influencing its perceived value, as the factors that enhance the engagement will add value to the technology.

For example, when individuals discuss their personal values or the values of others, they usually refer to the individual's beliefs about what is desirable or their personal standards. In addition, the perception of value creation can be established through engagement with the technology. As long as a user keeps engaging with the VA devices, as much as the value of the VA devices will increase. This could be because of the increasing level of familiarity, which will influence the users' perceptions of perceived value to evaluate the VA devices (comparing the quality and price).

For instance, Jung, Bapna, Ramaprasad, and Umyarov (2019) concluded that engagement involves users with respect to smartphones. This engagement creates value for users who therefore engage for a more extended period of time. The same effect could be achieved in the context of VAs. Moreover, as much as using VA devices as more as the user can get the advantage of using the technology, which could also lead to a better understanding of the mechanism of the technology (how it works with the AI). Mann, MacDonald, Kuo, Li, and Broadbent (2015) argued that people find robots more enjoyable than tablets. As supported by Ciechanowski, Przegalinska, Magnuski, and Gloor (2019), people interact with humanoid A.I. devices for longer periods than non-humanoid devices. This could significantly increase the continuous use of VAs.

In addition, as Higgins (2006) stated, engagement strength contributes to the value force experienced. Moreover, O'Brien and Toms (2008) concluded that one of the most valuable ways to define engagement with technology is to consider it as the quality of a user's experience, as determined by factors such as aesthetic and sensory appeal, interactivity and perceived user control. Furthermore, O'Brien, Cairns, and Hall (2018) clarified that this encompasses various factors related to usability, including the extent to which the system can hold the user's attention and its aesthetic aspects. Furthermore, Pagani and Mirabello (2011) concluded that, in comparison to discrete individual experiences, user engagement generally provides a higher-level measure of users' relationships with technology. Meanwhile, O'Brien and Toms (2010) confirmed that all the aspects could be evaluated separately or generally as a whole system holistic approach. As Donnermann et al. (2021) noted, the two approaches are popular in the area of human-computer interactions. However, a virtue of the holistic approach is that it provides the ability to evaluate the engagement of totally various systems using the same scale (Nguyen & Meixner, 2019). We thus expect that:

H4: Engagement has a positive impact on Perceived Value.

3.3.3 Rapport

Rapport is commonly defined as a "personal connection between the two interactants" (Gremler & Gwinner, 2000, p.92). It can be understood in the Service Robot Acceptance Model (sRAM); initially, the customer's perspective is that the encounter with the robot is enjoyable, characterised by a sense of care and empathy. Furthermore, the consumer believes that a personal bond is established between themselves and the robot, facilitated by features such as personalised treatment and voice recognition. (Wirtz, Patterson, Kunz, Gruber, Lu, Paluch, & Martins, 2018). In this context, rapport is the personal connection between a user and a VA. Therefore, the definition of rapport in this study is a user's perception of having an enjoyable interaction with a VA, characterised by a personal connection between the two interactants. This is where rapport (with VA) is an important dimension of user experience. For example, In the context of service robot encounters, customers frequently prioritise the establishment of positive relationships with service employees, which is often referred to as rapport (Wirtz et al., 2018). Therefore, in a VA context, when users humanise a VA, they are more likely to build rapport with it. Thus, rapport building between a user and a VA could add value to the VA and would enhance its continued usage. We thus expect that:

H5: Rapport has a positive impact on perceived value.

3.3.4 Perceived Value

Perceived value is regarded as a way to gain a competitive edge (Zeithaml, 1988; Holbrook, 2006). Zeithaml (1988) used the term "perceived value" to refer to an individual's overall evaluation of a service or product's utility depending on their perceptions of what is given (costs) and what is received (benefits). However, according to Coutelle-Brillet, Riviere, & des Garets (2014), the Zeithaml approach is often regarded as simply a balance between price and quality employing utilitarian and economic factors. Holbrook (2006), by contrast, defines value as a comparative, situational, and personal preference that describes a user's experience when interacting with a service or a product. Other research has differentiated between transaction, acquisition, and total value, notably Grewal, Monroe, and Krishnan (1998). However, Kim, Chan, and Gupta (2007) confirmed that the same criteria and measures had been used in all of the studies for both acquisition value and total value. Therefore, the overall value has been determined without any specific reference to acquisition value. Although the Technology Adoption Model (TAM) did not include the concept of perceived value for

innovations and technologies, Venkatesh, Morris, Davis, and Davis (2003) emphasised the importance of an attitude construct when cognition about a particular technology is considered necessary.

According to de Kervenoael, Hasan, Schwob, and Goh (2020), perceived value is influenced by various factors, including types of technology, tangibles, and promises of service types. This provides directionality for cognitions by producing an overall evaluation of responsiveness cues incorporated in the robot, using a model such as that produced by Kim et al. (2007). Meanwhile, Sheth, Newman, and Gross (1991) explained consumption in terms of emotional, conditional, functional, social, and epistemic value. However, depending on the situation, any or all of these values may affect the consumption experience (Kim et al., 2007). In addition, Jamrozy and Lawonk (2017) provided empirical support for including an emotive factor in shaping perceived value. Therefore, perceived value can be considered a multidimensional conceptualisation containing both affective and cognitive aspects (Sheth et al., 1991; Sweeney & Soutar, 2001). For the VAs, the value will be considered by focusing on the significant characteristics representing the critical trade-offs between benefits and costs.

The motivation of users to conduct a behaviour is mainly determined by their perception of its advantages (Li, Lin, Chan, & Wu, 2021). Li et al. (2021) confirmed that when users are unfamiliar with a new service or product or lack information, their decision-making is based on subjective value judgments. As a result, the perceived value could be a significant factor in explaining users' intentions concerning new technology adoption.

However, it is essential to note that past studies related perceived value to user acceptance, adoption attention, and usage in the case of other technologies, such as the smartphone (Kim et al., 2013; Sweeney & Soutar, 2001). To illustrate, Ryan and Jones (2012) high-lighted that the critical success of a mobile app highly depends on its capability to enhance consumers' level of value or engagement. Furthermore, Kim et al. (2007) added that higher perceived value indicates a greater willingness to adopt the technology. This could explain that higher engagement would lead to higher perceived value and a greater willingness to adopt the VA devices. Researchers have proven that there is strong empirical support that perceived value affects perceptual intention to use the technology (Sweeney, Soutar, & Johnson, 1997). This is consistent with the prior research on perceived value, which has recurrently verified perceived value as a predictor of intention (e.g. (Chang & Wildt, 1994)). Kim et al. (2007) confirmed that perceived value dominantly determines adoption intention. We thus expect that:

H6: Perceived Value positively impacts the Continuous Intention to use the VA.

3.4 Research methodology

A survey was developed with the purpose of evaluating the proposed research model. It was developed with the purpose of assessing participants' perceived intelligence, motivation to engage, satisfaction, perceived value and intention to continue engaging with users' VAs. Data was collected from 250 voice assistant device users using a Prolific website. The survey was conducted online and consisted of the following pages. The opening page outlined the aim of the survey, the estimated time required to complete it, 'the participant information sheet', 'the consent form', and the rights of the study participants, such as the ability to withdraw at any moment or to decline to answer any questions. Continuing to the following screen was interpreted as giving "informed consent." The study targeted only participants aged 18 or over who owned at least one VA device (such as the Amazon Echo, Google Home or Apple Home Pod) and had some experience with it. The artificial intelligence industry is focused on younger generations, as they are more likely to act as early adopters with a stronger predisposition for new technology (Chattaraman, Kwon, Gilbert, & Ross, 2019); therefore, this study's consideration of participants who are 18 and over is significant, as millennials are a technologically adept generation (Bilgihan, 2016), who are extremely sensitive to trends (Schepers & Wetzels, 2007). Thus, users who are 18 and above were deemed a pertinent cohort for this research, given that our objective was to identify a range of users who had experience with VA devices. In addition, examining VA device usage among older people could reveal an interest among this age group in using VAs.

The survey was divided into three sections. The first section contained general questions about users' experiences with their voice assistants, including, 'Which of the following voice assistants do you use?', 'What is your main purpose in using voice assistants?', 'How many times have you used the voice assistant?', 'How frequently do you use voice assistant devices?', 'How was your experience with the voice assistant?' and 'How easy was it to engage with the voice assistant's system?' The second section included five sub-sections, each of which focused on a different aspect of voice assistant usage: perceived anthropomorphism, rapport, engagement, perceived value, and intention to continue using the voice assistant. All the questions were adapted from well-established sources in the literature and were scored on a five-point Likert scale, from "Strongly agree" to "Strongly disagree" (See Table 3.1).

Besides demographic data (gender, employment status, age, and educational level), the third section included indices for the human personality aspect (extraversion).

The Partial Least Squares Structural Equation Modelling (PLS-SEM) technique was used through Smart PLS 3.0 software. According to Hair, Hull, Ringle, & Sarstedt (2014), Partial Least Squares Structural Equation Modelling is an efficient modelling method for evaluating complicated prediction models and for the theory-building phases of exploratory studies. Additionally, PLS-SEM is gaining popularity due to its ability to model latent constructs which are not normally distributed (Hair, Ringle, & Sarstedt, 2011) and with small to medium-sized datasets (Chin, Marcolin, & Newsted, 2003). Additionally, Fornell and Bookstein (1982) confirmed that PLS-SEM is an appropriate tool for testing phenomena in their infancy stages, as it does not require a multivariate-normal distribution (Albert & Merunka, 2013). The two-step procedure advised by Diamantopoulos and Winklhofer (2001) was followed, with the measurement (outer) model evaluated first and then the structural (inner) model. SEM is able to analyse all possible paths in a single analysis (Gefen, Straub, & Boudreau, 2000). PLS enables the analysis of both the structural model to evaluate relationships between the theoretical constructs and the measurement model to evaluate the validity and reliability of the items in the questionnaire. Moreover, Partial Least Squares Structural Equation Modelling is appropriate for assessing categorical moderation across many connections (Hair, Sarstedt, Ringle, & Mena, 2012), which is the purpose of this study.

3.5 Research Results and Findings

3.5.1 Descriptive Analysis

The questionnaire was distributed equally among sex, Males (49.20%) and Females (50%). Most of the respondents are 21 to 30 years old (55.20%) and have a diploma (25.20%), a bachelor's (35.60%) or a master's (16.80%) degree. However, just over a third (41.60%) are employed, and (38.40%) are students. All the respondents regularly engage in interactions or frequent engagements with AI-based VA devices, namely Google Home (60.40%), Amazon Echo with Alexa (24.40%) and Apple Home Pod with Siri (14.80%). Most of them used their VA at home (94.40%), but only (5.20%) at work. Most respondents have used their VA devices more than 20 times (56.00%), and only (12%) of the participants used them 1-5 times. They are using VA devices for three main reasons. Over half of those surveyed (55.60%) indicated

that it is useful when the eyes or hands are busy, a pleasure to use (18.40%), and produces results faster (17.20%).

3.5.2 Measurement Model

As part of the measurement model evaluation, five items (AN7, RA3, EN4, EN5, EN6) were removed from the analysis because of low factor loadings (<0.700). The study used Cronbach's alpha and composite reliability (CR) to test the constructs' reliability. All the C.R.s were higher than the recommended value of 0.700 (Wasko & Faraj, 2005). Cronbach's alpha of each construct exceeded the 0.700 thresholds. Convergent validity was acceptable because the Average Variance Extracted (AVE) was over 0.500. The results for reliability and validity, along with the factor loadings for the items, are presented in table 3.1. The Cornell-locker criterion assessed discriminant validity, and the table shows that the square root of AVE for the construct was greater than the inter-construct correlation (see table 3.2). Discriminant validity was also assessed by the heterotrait-monotrait ratio of correlations (Henseler et al., 2015), with values below the threshold of 0.90. Hence, discriminant validity is established (see table 3.3).

The correlation matrix of latent variables (table 3.4) unveils relationships among Anthropomorphism, Continuous Intention, Engagement, Perceived value, and Rapport. The correlation matrix reveals significant associations among the variables. Anthropomorphism demonstrates a moderate positive correlation with Engagement (0.675) and Rapport (0.598), indicating that as perceptions of anthropomorphism increase, so does the level of engagement and rapport. Additionally, a very low positive correlation with Perceived value (0.283) suggests a positive relationship between anthropomorphism and perceived value. Engagement shows a moderate positive association with anthropomorphism (0.675) and rapport (0.683), suggesting that engagement is closely tied to these constructs. Rapport, in turn, exhibits a moderate positive correlation with Engagement (0.683) and anthropomorphism (0.598) and a low positive correlation with Perceived value (0.442), and Continuous Intention (0.406). These findings highlight the interconnections among the variables, providing valuable insights into how perceptions of anthropomorphism, continuous intention, engagement, perceived value, and rapport are intricately linked in the studied context.

Table 3.1. Loadings, Reliability, and Validity

	Loadings	Cronbach's Alpha	Composite Reliability	AVE
Anthropomorphism		0.942	0.950	0.613
(Items adapted from Wang, 2017; Lu et al., 2019; Golossenko et al., 2020; Moriuchi, 2020; Waytz et al., 2010)				
AN1: My voice assistant has thoughts of its own	0.734			
AN10: My voice assistant experiences guilt when it acts incorrectly in relation to the consumer	0.807			
AN11: My voice assistant can feel compassion	0.806			
AN12: My voice assistant has its own personality	0.780			
AN13: My voice assistant has its own preferences and mood	0.789			
AN2: My voice assistant has intentions	0.757			
AN3: My voice assistant has a free will	0.82			
AN4: My voice assistant has a consciousness	0.826			
AN5: My voice assistant has desires	0.82			
AN6: My voice assistant has its own emotions	0.816			
AN8: I felt there was human contact with my voice assistants	0.712			
AN9: My voice assistant has its own decision-making power	0.712			
Continuous Intention		0.919	0.936	0.675
(Items adapted from Moriuchi, 2020; Bhattacharjee, 2001; Kim, Young, Kim, & Wachter, 2013)				
CI1: I intend to continue using my voice assistant rather than discontinue	0.864			
CI2: My intention would be to continue using my voice assistant than use any alternative means (traditional ways)	0.718			
CI3: If I could, I would like to continue my use of my voice assistant	0.844			
CI4: I think it will be worth it for me to use voice assistants when it's available	0.796			
CI5: I would recommend using a voice assistant to my friends and family	0.831			
CI6: I would like to continue to engage with the voice assistant device	0.865			
CI7: I would like to recommend my engagement with the voice assistant device to someone	0.825			
Engagement		0.864	0.917	0.786
(Moriuchi, 2020)				
EN1: I feel that the engagement I have with my voice assistants is very human-like	0.869			
EN2: I feel that the conversation I have with my voice assistant is very engaging	0.900			
EN3: The engagement I have with my voice assistant is very meaningful	0.890			
Perceived Value		0.899	0.921	0.624
(Items adapted from Sirdeshmukh et al., 2002; Kim, Young, Kim, & Wachter, 2013)				
PV1: Compared to the fee I need to pay, the use of my voice assistant offers value for money	0.757			
PV2: Compared to the effort I need to put in, the use of my voice assistant is beneficial to me	0.808			
PV3: Compared to the time I need to spend, the use of my voice assistant is worthwhile to me	0.798			
PV4: Overall, the use of my voice assistant delivers me good value	0.840			
PV5: Using a voice assistant device is an enjoyable experience	0.728			

PV6: The overall value of my experience using a voice assistant device is outstanding	0.800			
PV7: Voice assistant device represents a good use of my time and money	0.792			
Rapport		0.746	0.855	0.663
(Items adapted from: Fernandes & Oliveira, 2021; Gremler & Gwinner, 2000)				
RA1: My voice assistant device relates well to me	0.808			
RA2: I think there is a "bond" between my voice assistant device and myself	0.837			
RA4: In thinking about my relationship with my voice assistant device, I enjoy interacting with my voice assistant device	0.797			

Table 3.2. Fornell-Larcker Criterion

	Anthropomorphism	Continuous Intention	Engagement	Perceived value	Rapport
Anthropomorphism	***0.783***				
Continuous Intention	0.208	***0.822***			
Engagement	0.675	0.345	***0.886***		
Perceived value	0.283	0.756	0.441	***0.790***	
Rapport	0.598	0.406	0.683	0.442	***0.814***

Note: Value in Italic represents Square-root of AVE.

Table 3.3. HTMT Ratio

	Anthropomorphism	Continuous Intention	Engagement	Perceived value	Rapport
Anthropomorphism					
Continuous Intention	0.221				
Engagement	0.740	0.386			
Perceived value	0.299	0.823	0.495		
Rapport	0.703	0.495	0.848	0.537	

Table 3.4. Latent variables - Correlation

	Anthropomorphism	Continuous Intention	Engagement	Perceived value	Rapport
Anthropomorphism	1	0.208	0.675	0.283	0.598
Continuous Intention	0.208	1	0.345	0.756	0.406
Engagement	0.675	0.345	1	0.441	0.683
Perceived value	0.283	0.756	0.441	1	0.442
Rapport	0.598	0.406	0.683	0.442	1

3.5.3 Structural Model

A structural model is assessed based on R2, Q2, and the significance of paths. The structural model reflects the paths hypothesised in the research framework. The strength of each structural path determines the model's goodness by the R^2 value for the dependent variable (Briones Penalver, Bernal Conesa, & de Nieves Nieto, 2018); the value for R^2 should

be equal to or over 0.1 (Falk &Miller, 1992). The results in table 3.5 show that all R^2 values are over 0.1. Hence, the predictive capability is established.

Further, Q^2 establishes the predictive relevance of the endogenous constructs. A Q^2 above 0 shows that the model has predictive relevance. The results show that there is significance in the prediction of the constructs (see table 3.5). Furthermore, the bootstrapped Standardised Root Mean Square Residual (SRMR) is a commonly used metric for assessing the goodness of fit when using SmartPLS. Therefore, the model fit was assessed using SRMR. The SRMR value was 0.067, below the required value of .10, indicating an acceptable model fit (Hair et al., 2016).

Furthermore, in the assessment of the goodness of fit, hypotheses were tested to ascertain the significance of the relationship. H1 evaluates whether Perceived Anthropomorphism has a significant impact on engagement. The results revealed that Perceived Anthropomorphism significantly impacts engagement (ß = 0.675, t = 19.918, p < .001). Hence, H1 was supported. H2 evaluates whether Perceived Anthropomorphism has a significant impact on rapport. The results revealed that Perceived Anthropomorphism significantly impacts rapport (ß = 0.598, t = 15.099, p < .001). Hence, H2 was supported. H3 evaluates whether Perceived Anthropomorphism has a significant impact on Perceived Value. The results revealed that Perceived Anthropomorphism has an insignificant impact on Perceived Value (ß = -0.100, t = 1.203, p = 0.230). Hence, H3 was not supported. H4 evaluates whether engagement has a significant impact on Perceived Value. The results revealed that engagement significantly impacts Perceived Value (ß = 0.311, t = 3.623, p < .001). Hence, H4 was supported. H5 evaluates whether rapport has a significant impact on Perceived Value. The results revealed that rapport significantly impacts Perceived Value (ß = 0.288, t = 3.237, p = 0.001). Hence, H5 was supported. H6 evaluates whether Perceived Value has a significant impact on Continuous Intention. The results revealed that Perceived value significantly impacts Continuous Intention (ß = 0.756, t = 22.622, p < .001). Hence, H6 was supported. This study of 5000 resamples also generated 95% confidence intervals, as shown in Table 3.5. A confidence interval different from zero indicates a significant relationship. Hypotheses testing results are summarised in Table 3.5.

Table 3.5. Hypotheses Testing

	ß	STDEV	T Statistics	P Values	2.50%	97.50%

Anthropomorphism -> Engagement	0.675	0.034	19.918	0.000	0.605	0.736
Anthropomorphism -> Perceived value	-0.100	0.084	1.203	0.230	0.244	0.068
Anthropomorphism -> Rapport	0.598	0.04	15.099	0.000	0.513	0.67
Engagement -> Perceived value	0.311	0.086	3.623	0.000	0.148	0.467
Perceived value -> Continuous Intention	0.756	0.033	22.622	0.000	0.669	0.811
Rapport -> Perceived value	0.288	0.089	3.237	0.001	0.108	0.463
	R^2	Q^2				
Continuous Intention	0.572	0.379				
Engagement	0.455	0.352				
Perceived Value	0.235	0.141				
Rapport	0.357	0.230				

3.5.4 Mediation analysis

Mediation analysis assessed the mediating role of Engagement, Rapport and Perceived Value on the linkage between Perceived Anthropomorphism and Continuous Intention. The results (see Tables 3.6 and 3.7) revealed that the total effect of Anthropomorphism on Continuous Intention was significant (ß = 0.213, t = 4.496, P < 0.10). Including the mediating variables Engagement, Rapport and Perceived Value, the impact of Anthropomorphism on Continuous Intention became insignificant (ß = -0.005, P = 0.894). The indirect effect of Anthropomorphism on Continuous Intention through Engagement and Perceived Value was found to be significant (ß = 0.158, t = 4.126, P < 0.10). Moreover, the indirect effect of Anthropomorphism on Continuous Intention through Rapport and Perceived Value was found to be significant (ß = 0.130, t = 2.801, P < 0.10). Surprisingly, by considering the mediation role of perceived value only on the linkage between Anthropomorphism and Continuous Intention, the indirect effect of Anthropomorphism on Continuous Intention through Perceived Value was found insignificant (ß = -0.076, t = 1.187, P = 0.236). This shows that Rapport, Engagement and Perceived Value fully mediate the relationship between Anthropomorphism and Continuous Intention. For the control variables, we did not find an effect of gender and age.

Table 3.6. Mediation Analysis: the total and the direct effect

	Total Effect			Direct Effect	
	Coefficient	T	P Value	Coefficient	P Value
Anthropomorphism -> Continuous Intention	0.213	4.496	0.000	-0.005	0.894
Anthropomorphism -> Perceived Value	0.282	4.648	0.000	-0.100	0.230

Table 3.7. Mediation Analysis: Overview of indirect effects

	Indirect Paths	Indirect Paths		Path coefficient	T Value	Sig.
Anthropomorphism →	Engagement →	Perceived Value →	Continuous Intention	0.158	3.527	0.000
Anthropomorphism →	Rapport →	Perceived Value →	Continuous Intention	0.130	2.801	0.005
Anthropomorphism →		Perceived Value →	Continuous Intention	-0.076	2.801	0.236

Note: *non-significant path coefficient (p > .05).

3.6 Discussion

Recent technological sector developments have included a rise in VA deployment. However, research in user interaction with VAs is still in its infancy and has largely been conceptual, especially considering the differences between VA and conventional IT, with VA being considerably more human-like when interacting with users. This study investigates the role of such perceived anthropomorphism in building a relationship between VAs and users and the effect this has on users' continuous intentions. In particular, it examines the impact of engagement and rapport on perceived value and subsequent continuous intention to use VAs.

3.6.1 Theoretical Implications

This study conceptualises voice assistants (VAs) as anthropomorphic entities that individuals engage with as if they were engaging with a social interaction partner. Based on the CASA framework, which emphasises social perceptions and interactions, a theoretical model has been developed whereby users use the VAs to draw assumptions about personality characteristics, mirroring the manner in which such inferences are made during interpersonal human interactions. Moreover, through the integration of anthropomorphism, the theoretical framework supports the notion that virtual assistant users activate specific stereotypes related to judgements of warmth and competence. Although the primary hypotheses were not supported, the validation of the serial mediation of engagement and rapport provides support for the efficacy of the Computers Are Social Actors framework and the social cognition application to VAs. This research offers further evidence that supports the extensive body of research in CASA, which has consistently shown the effectiveness of incorporating social cognition principles into machine perception (Belanche, Casaló, Schepers, & Flavián, 2021; Gambino, Fox, & Ratan, 2020).

By considering perceived anthropomorphism as a valuable factor that leads to the continuous intention to use VAs, surprisingly, the results show that the direct relationship between perceived anthropomorphism and the perceived value is insignificant (H3) unless critical factors are used as mediators to make the relationship significant. In other words, just because a user perceives a technology as more humanlike does not necessarily mean they will place a higher value on that technology. There are several potential reasons for this lack of a direct relationship. The reason could be, as demonstrated by Mende et al. (2019), that as anthropomorphism is perceived to increase, consumers are likely to encounter a sense of discomfort, characterised by feelings of unease and a perceived challenge to their human identity. In addition, Blut et al. (2021) confirmed that although anthropomorphic traits have been associated with reported advantages, relying only on anthropomorphism as a replacement for human interaction is not desirable. Moreover, users may value a technology or product based on its functionality or performance rather than its perceived humanlike qualities. Additionally, users may have different personal preferences and attitudes towards anthropomorphism that influence how they perceive and value technology, or they may be unaware that an anthropomorphic VA could benefit them more than the standard version. However, based on the results of this study, perceived anthropomorphism still contributes to perceived value when two factors, namely engagement and rapport, mediate it.

On the other hand, the results confirmed that perceived anthropomorphism influenced engagement and rapport positively (H1, H2) by making users feel more comfortable and familiar with the VA. Engagement refers to the degree to which users are motivated and interested in using technology, while rapport refers to a user's perception of an enjoyable interaction. In other words, rapport refers to the degree to which users feel a positive connection or relationship with the technology. When a technology exhibits humanlike characteristics or behaviour, users may feel more inclined to interact with it as they would with another human being. This feeling can result in a stronger sense of engagement and rapport between the user and the technology. In particular, research has suggested that perceived anthropomorphism can increase feelings of empathy (Pelau et al., 2021) and trust and likability (Blut et al., 2021) towards the technology. These positive feelings can, in turn, lead to a greater rapport between the user and the technology; for example, chatbots or virtual assistants with humanlike voices or personalities can make users feel more comfortable with the technology and enhance their engagement.

Social psychology theories can explain the positive influence of perceived anthropomorphism on engagement and rapport. According to social presence theory, the degree to which a technology or medium allows for social presence, or the feeling of being in the presence of others, can influence users' attitudes and behaviours towards the technology (Short, Williams, & Christie, 1976). Anthropomorphic technology can enhance social presence by simulating humanlike interactions and behaviours. Likewise, social exchange theory proposes that relationships between two parties, including those between humans and technology, rely on a mechanism of social exchange in which both receive and give rewards (Blau, 2017). Anthropomorphic technology can enhance the rewards that users receive from the technology by providing a more engaging and interactive experience.

However, it is worth noting that the relationship between perceived anthropomorphism and engagement/rapport may not always be positive or consistent across all users and contexts. Some users may find anthropomorphic technology unsettling or creepy, while others may not value humanlike characteristics as much. In addition, as Pelau et al. (2021) note, unless an anthropomorphic artificial intelligence device is able to empathise with the user, which shifts the user's perspective, it may be seen as a threat to human identity. By contrast, research indicates that if an artificial intelligence device is perceived to have its own principles, free will, and reasoning, it is thought to be more caring and focused on the needs of its users (Pelau et al., 2021). This suggests that perceived anthropomorphism can build continuing engagement with the VA device, which can then enable it to create and deliver higher value, and this has a positive effect on the users' intention to use the VA again. Therefore, as supported by Moriuchi (2021), perceived anthropomorphism positively influences engagement and thereby contributes to the continuous intention to use the device. However, the effectiveness of anthropomorphism may depend on the technology's specific application or use case.

The findings show that engagement and rapport are two of the most critical factors that add value to the technology and lead to the continuous intention to use that technology (H4, H5). To demonstrate, when users feel engaged and have a positive rapport with technology, they are more likely to use it frequently and continue to use it over time. For example, the ability of VAs and robots to process language and comprehend users is critical in expanding their application in the service industry (Pelau et al., 2021). Moreover, when technology engages, users are more likely to devote their time and attention to it, leading to greater satisfaction and enjoyment. For example, engaging video games can keep users entertained and motivated to continue playing. In addition, when users have a positive rapport with technology,

they feel comfortable and familiar with it, which can lead to a greater sense of trust and loyalty. For instance, developing a positive rapport with a VA can lead to increased reliance on and continued use of the assistant. In addition, Fernandes and Oliveira (2021) confirmed that rapport (such as warmth) is identified as one of the significant direct drivers of digital VA acceptance, and the rapport between humans and robots is seen as essential to services such as elder care (Wirtz et al., 2018). Engagement and rapport are two essential factors that can add value to technology and enhance the continuous intention to use it. Therefore, rapport and engagement strongly predict the intention to use VAs continuously. By understanding and designing for these factors, developers can create more appealing and valuable technology for users.

The value of engagement and rapport in technology is supported by theories such as the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT). These theories suggest that users are more likely to accept and use technology when it is perceived to be useful, easy to use, and enjoyable. Engagement and rapport contribute to technology's perceived usefulness and enjoyment, making users more likely to continue to use it. In addition, research has shown that engagement can influence user behaviour beyond just the continuous use of technology (O'Brien & Toms, 2008). For example, engaged users who feel a rapport with their devices may be more likely to recommend the technology to others or provide feedback to help developers improve it. The importance of engagement and rapport in technology highlights the need for developers to prioritise the user experience in designs that foster these factors. By doing so, developers can create more desirable, useful, and valuable technology for users.

At the same time, these two factors made the relationship between perceived anthropomorphism and perceived value significant. The indirect relationships (through engagement and rapport) are significant, and this indicates that the more the user engages with the VA, the more likely they are to understand the importance and the benefits of anthropomorphic assistants. In addition, when users perceive a technology to be more humanlike, they are more likely to engage with it and develop a positive rapport with it. This, in turn, leads to users perceiving the technology as more valuable and useful. In other words, engagement and rapport act as a bridge between the perception of technology as humanlike and the perceived value of that technology. When people feel engaged and have a positive rapport with the VA, regardless of its anthropomorphism, they are more likely to perceive it as valuable and beneficial. The findings of this study indicate that users who perceived a VA to

be more humanlike also perceived it to be more engaging and to have a stronger rapport with them. This positive perception of engagement and rapport made users perceive the VA as more beneficial and valuable. Thus, the study emphasises the significance of designing a VA that fosters engagement and rapport, as these factors can play a significant role in influencing users' perceptions of the value and utility of the VA and, in particular, the significance of anthropomorphic design characteristics within the context of artificial intelligence technology.

3.6.2 Managerial Implication

The managerial implications of this study are substantial for businesses and organisations that use or intend to use VAs as part of their service offerings. By comprehending the role of anthropomorphism and the significance of establishing relationships and rapport with users, organisations can design and implement VAs that are more effective at engaging and gratifying users. This study suggests that organisations should consider the social and relational aspects of VAs in addition to their functional characteristics. They should also invest in creating VAs to establish rapport and social presence with users. By performing so, companies can boost the continuous use of VAs and improve the overall user experience. This study also emphasises the significance of understanding users' preferences and experiences with VAs, which could guide the design and development of future automated service technologies. The findings of this study have significant implications for administrators and practitioners who wish to increase the continuous use and efficacy of VAs in various contexts. By prioritising the factors under consideration, service providers can increase the perceived value of the technology and the user's intent to continue using it.

3.7 Conclusion

In conclusion, there is a dearth of knowledge regarding users' attitudes and behavioural outcomes subsequent to their interaction with voice-based artificial intelligence assistants, including but not limited to their intentions to persist in using such devices. This study sheds light on the role of anthropomorphism in building relationships between VAs and users and enhancing their continuous use. The research objectives were to investigate the effect of anthropomorphism on building relationships and to examine the impact of engagement and rapport on perceived value and continuous intention to use the VAs.

This study contributes to four-fold. First, it contributes to the literature on automated technologies by providing a deep understanding of the post-adoption phase of these technologies. Second, it adds to the third wave of human-computer interaction (HCI) research, prioritising the study of relationship phenomena over analysing the technical capabilities and operations of the interacting components. In other words, the focus is on exploring how humans and computers relate to each other in interactive contexts rather than on the specifics of their individual functions or features. Third, this study contributes to the existing body of research in the field of Computers Are Social Actors (CASA). Lastly, it contributes to the information systems field by expanding the literature on "perceived value," which has been identified as a critical driver of technology adoption (Kim et al., 2007).

The study examined indirect, mediating effects between the dimensions and elements, considering the factor enhancing the continuous intention to use the VAs. Before our research, no studies had explored the role of anthropomorphism, engagement and rapport on perceived value and continuous use intention in the context of VAs. By filling this gap in the literature, our findings have theoretical implications for technology post-adoption models and highlight the importance of considering engagement and rapport in evaluating perceived value. The results suggest that anthropomorphism is not generally favourable and that technologically advanced users who prefer technology on the front lines could be more sensitive to VAs' relational, high-touch features. Overall, this study provides important insights into post-adopting VAs and contributes to developing theories and frameworks in automated service technologies.

The results indicate that anthropomorphism can positively influence relationships between users and VAs, indirectly increasing their perceived value and continued use. It was discovered that anthropomorphism could increase the perceived value of VAs by nurturing user engagement and rapport with the technology. This study contributes to the existing literature on perceived value by investigating the relationship between engagement, rapport, and intention to use VAs continuously. In addition, the study emphasises the significance of engagement and rapport in moulding users' perceptions of the value of VAs, especially regarding their intention to continue using them. In addition, the study contributes to the existing research on perceived value and technology continuous use and provides novel insights into the factors that may promote VA continuous use. This research can inform the design and development of artificial intelligence technologies that benefit both users and companies by shedding light on how VAs can increase their continuous use.

Despite the insights provided by this study, it is necessary to acknowledge several limitations. Firstly, the sample was restricted to a few nations, which may limit the applicability of our findings to other contexts. This study could be expanded by conducting experiments in work environments to assess the generalizability of the results in the future. In addition, the study was conducted using a home VA (i.e., Amazon Alexa) and no other types of VAs, which limits the findings' applicability to other VAs. Future research could examine the relationship between anthropomorphism, engagement, rapport, perceived value, and VA continuous use using a larger sample size and various varieties of VAs to provide a broader perspective. In addition, this study paves the way for future research to investigate the effect of multiple types of anthropomorphic cues, such as using a humanlike name, on building relationships between VAs and their users. Future studies could also examine how the effect of anthropomorphism on relationship building and continuous use may differ depending on the context of use (e.g., personal versus professional settings). Overall, this study contributes to our understanding of the role of anthropomorphism in building relationships between VAs and their users and emphasises the significance of engagement, rapport, and perceived value in driving VA continuous use.

Chapter 4. Paper Three

"Are You Talking to Me?": How Perceiving Voice Assistants as Intelligent Impacts the Users' Engagement Intention

Abstract

The popularity of voice assistants (VAs) continues to rise, using Artificial Intelligence (AI) to transform the manner in which we engage with technological devices. The motivation for users to engage with VAs is a key to their continuous uptake, and this paper examines if the intelligence of VAs, as perceived by their users, will impact this motivation. We distinguish three types of motivation: hedonic, utilitarian, and social, and we investigate how they mediate the impact of perceived intelligence on continuance intention factors such as engagement intention, perceived value and user satisfaction. The results highlight the significant impact of perceived intelligence on user intention to engage with voice assistant technology. Furthermore, the study's findings indicate that utilitarian motivation plays a crucial role in mediating the relationship between perceived intelligence and intention to engage with the voice assistants.

Keywords: Artificial Intelligence (AI); Voice assistant (VA); Engagement Intention; Perceived Value; Engagement; Hedonic, utilitarian and social engagement motivation; Satisfaction.

4.1 Introduction

Voice recognition technology has revolutionised how individuals interact with their IT devices. Embedded in Voice Assistant (VA) devices, it provides a seamless and hands-free interface to desired functionality and has thus become a crucial part of our digital environment. However, people often stop using VA after a brief period of enthusiasm, and the study of VA adoption has been the focus of increasing numbers of information systems scholars. For example, Davis (1989) and Venkatesh, Morris, Davis, and Davis (2003) have focused on the initial adoption and utilisation of IS systems. However, the frequently observed discontinuance of using VAs has not been fully explored by existing literature despite the recent focus on use continuance (Balakrishnan, Abed, & Jones, 2022).

One of the limitations in the existing literature on adoption and use continuance is its focus on conventional information technology, where the systems to be adopted are perceived as tools helping to achieve a task. Users, however, perceive VAs as intelligent and anthropomorphic, and this perception has the potential to impact how users are motivated to engage with technology.

Different types of motivations impact the users' intention to engage (Kim, Kim, & Wachter, 2013; Lim, Cheah, Ng, Basha, and Soutar, 2021): social, utilitarian and hedonic. This has so far not been applied to devices perceived as intelligent, such as VAs, and this paper foci on the interaction between the different types of motivation to engage and the perceived intelligence of the VAs.

The User Engagement with Technology (MoEN) model (Kim et al., 2013) serves as our starting point, to which we add the concept of perceived intelligence. By exploring the notion of perceived intelligence, this study aims to facilitate the development of VA technologies that are better aligned with users' needs and expectations, reducing the occurrences of discontinuance.

The perceived intelligence of voice-based virtual assistants is an essential aspect that may significantly impact users' motivation to continue engaging with the technology. The concept of perceived intelligence refers to how users perceive an assistant's cognitive abilities, level of responsiveness, and capacity to comprehend and satisfy their requirements (Moussawi, Koufaris, & Benbunan-Fich, 2021). Research has indicated that the perceived intelligence level exhibited by VAs can significantly impact both user satisfaction and usage intention (Marikyan, Papagiannidis, Rana, Ranjan, & Morgan, 2022). The advanced capabilities of VAs

powered by AI, including natural language processing and machine learning algorithms, are instrumental in shaping users' perceptions of the VAs' intelligence and efficacy (Moussawi et al., 2021). Therefore, this study aims to examine the influence of perceived intelligence on users' motivations to interact with VAs and its effect on their intention to engage with the technology. In addition, it foci on the impact of different motivational types; hedonic, utilitarian, and social, on perceived value, satisfaction and users' engagement.

According to Moriuchi (2021), engagement subsequently establishes continuous relationships with the technology. Engagement refers to being actively retained, occupied, involved, and intrinsically interested in a particular subject matter (Pagani & Mirabello, 2011). The term also refers to the attention, level of involvement, and emotional connection that users encounter while interacting with technological systems (Van Doorn, Lemon, Mittal, Nass, Pick, Pirner, & Verhoef, 2010). The extant literature suggests that engagement is a psychological construct that facilitates the establishment of a strong bond between users and technology, resulting in positive outcomes in terms of loyalty for both new and existing users in the short and long term (Pansari & Kumar, 2017). In the case of VAs, developing a strong bond between user and device through motivations to engage with the technology may enhance the intention to engage with the technology. Engagement motivation can then be defined as the desire to engage in activities that require using these devices (Kearsley & Shneiderman, 1998).

In addition, users are more willing to develop a positive attitude and emotional attachment towards a technology when they perceive it to have significant value and when they feel satisfied with its usage. This could enhance the relationship between the user and the technology and encourage a sense of motivation that has the potential to result in continued engagement with the technology. As a result, the study also considers the importance of perceived value and satisfaction as potential mediators in the correlation between engagement motivations and continued engagement intention is also. Perceived value refers to a person's overall assessment of the usefulness of a service or product based on their views of what is provided (costs) and what is gained (benefits) (Zeithaml, 1988). Therefore, the concept of perceived value involves the evaluation of perceived usefulness, relevance, and overall worthiness in meeting the users' needs and objectives. Numerous studies have provided evidence of the significant impact of perceived value (Li, Lin, Chan, & Wu, 2021; Kim, Chan, & Gupta, 2007) and satisfaction (Xu, Peak, & Prybutok, 2015; Zhao, Lu, Zhang, & Chau, 2012; Bhattacherjee, 2001) on users' continued intention to use a given technology, and the likelihood of future technology usage positively correlates with users' perception of its high value.

In contrast, satisfaction reflects the accumulation of feelings and experiences developed through multiple interactions with a service provider (Oliver, 1980). This pertains to the evaluation of technology based on its ability to satisfy user expectations, meet requirements, and provide a favourable user experience. Elevated satisfaction levels could positively correlate with the probability of users continuing to engage with the technology in the long term.

The employment of digital assistants can be attributed to both technology use factors and factors that pertain to the characteristics and capabilities of the assistants themselves (Marikyan et al., 2022). However, knowledge regarding the utilisation of AI-based assistants beyond private contexts remains limited (Marikyan et al., 2022), and further investigation is necessary to ascertain how interactions with these devices transpire within authentic social environments and the specific roles and functions they serve for their users (Purington, Taft, Sannon, Bazarova, and Taylor, 2017). Building on this contextual foundation, our research question asks:

To what extent does perceived intelligence play a role in motivating users to engage with voice assistants?

To address this question, we have defined three primary objectives: (1) to investigate the effect of perceived intelligence on building relationships between VAs and users; (2) to examine the effect of perceived intelligence on engagement motivations and, subsequently, users' intention to engage with VAs; and (3) to investigate the role of perceived intelligence on perceived value and user satisfaction, and, consequently, on users' intention to engage with VAs. By achieving these objectives, this research aims to provide valuable insights into the factors contributing to users' intention to engage with VAs. Understanding the role of engagement motivations, mediated by perceived value and satisfaction, along with the influence of perceived intelligence, will offer valuable guidance for designing and enhancing VA technologies to meet users' needs and preferences.

4.2 Literature review and hypothesis development

The proliferation of home VAs has revolutionised the way users interact with technology in their homes, and voice-controlled devices, such as Amazon's Alexa, Google Assistant, and Apple's Siri, have witnessed a remarkable increase in adoption and usage. Indeed, the results of a recent “new users” survey in the USA showed that 87.7 million people had started to use

intelligent speakers since January 2020, a 32% increase in the US smart speaker user base from January 2019 and a rise of 85% on January 2018 (Kinsella, 2020). According to recent statistics, a significant portion of voice-enabled device users, accounting for 43% of the total, utilise the technology for shopping purposes (Georgiev, 2022), and this growing user base represents a significant market for home VAs. Research indicates that engagement with these devices has contributed to their popularity (Eslami, Ghasemaghaei, & Hassanein, 2022), and this can be attributed to the convenience and accessibility they provide, which align with users' needs. This suggests that users' engagement experience with VAs may be driven by their perception that the system captures and rewards their interests.

Studies by de Kervenoael, Hasan, Schwob, and Goh (2020) and Moriuchi (2021) emphasise that engagement with VAs plays a crucial role in their success. Users' attitudes and behaviour towards technology have shifted from initial adoption choices to daily engagement and involvement, and they now have the freedom to choose when, where, and how they engage with these devices, empowering them to control their user-driven activities. This user-driven engagement behaviour can be fuelled by the attributes, value, and satisfaction derived from interacting with home VAs. As Venkatesh, Th, & Xu (2012) explain, increased engagement can be achieved through behaviours that are perceived as pleasurable (hedonic), serve a practical purpose (utilitarian), or facilitate interpersonal communication (social), so VAs can offer value and satisfaction through their ability to provide information, control smart home devices, and offer entertainment.

Examining engagement behaviours, therefore, plays a crucial role in extending the post-technology acceptance model and understanding users' interaction with home VA technology (de Kervenoael et al., 2020; Moriuchi, 2021).

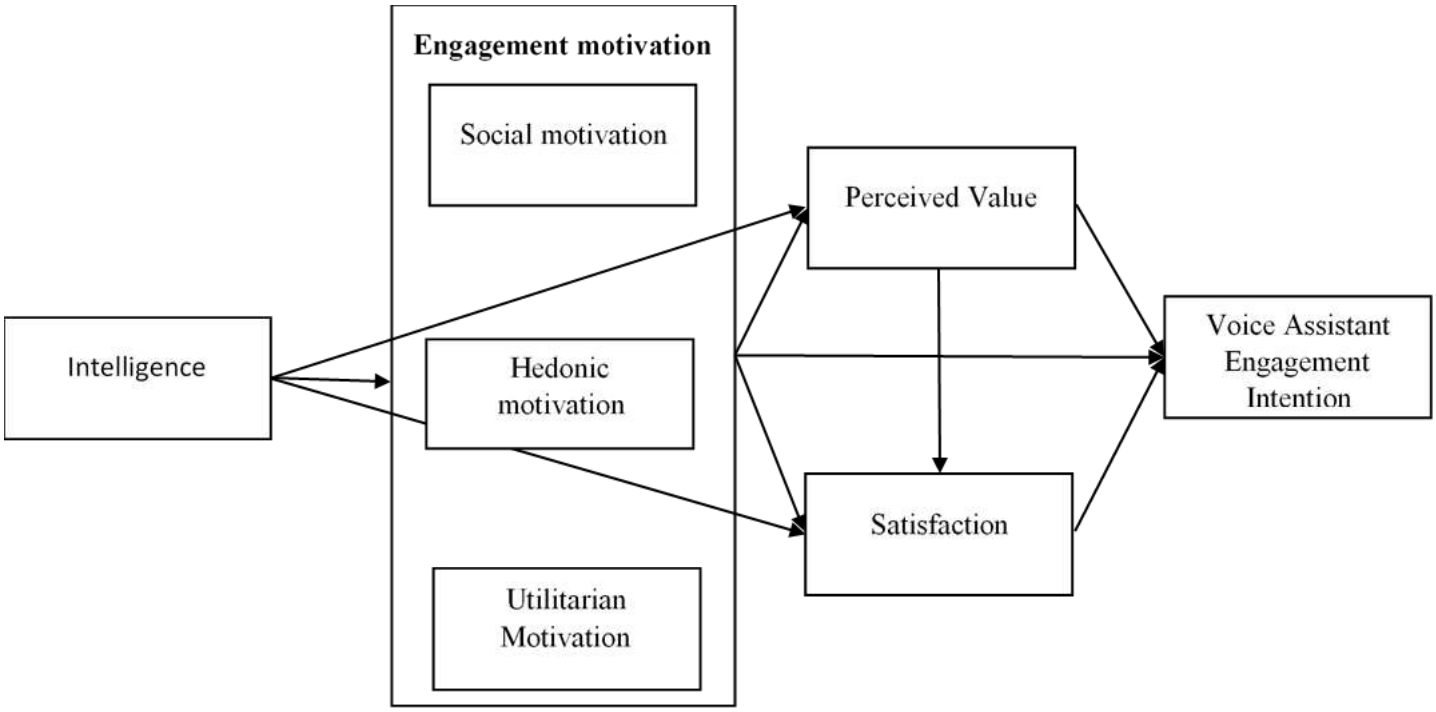


Figure 4.1. Theoretical model

4.2.1 Perceived Intelligence

Humans have always been fascinated by the prospect of developing technologies that can simulate the human brain. Computing technology was long considered to be inert, devoid of intelligence, engagement, initiative or interactivity, and dependent on users to complete and execute tasks successfully. However, a more complex perspective is emerging in which users see computing technology as another component in a complicated social context, "an external, autonomous entity with whom they are forced to interact" (Johnson, Marakas, & Palmer, 2008, p. 172). As such, users regard computing technology as an active instrument that possesses initiative, interactivity, intelligence, and even the ability to control users' behaviour and interactions. The rationale for this stems from the field of Artificial Intelligence (AI) (Bartneck, Kulić, Croft, & Zoghbi, 2009). While interactive robots face significant obstacles when it comes to responding intelligently, AI develops the conceptual tools necessary to comprehend how data is processed (Nilsson, 1982), and robots' behaviours are increasingly predicated on AI-developed methodologies and expertise (Bartneck et al., 2009).

AI has been defined as "a subpart of computer science, concerned with how to give computers the sophistication to act intelligently, and to do so in increasingly wider realms. It participates thoroughly in computer science's passion for abstraction, programming and logical formalisms, and detail - for algorithms over behavioural data, synthesis over analysis, and

engineering (how to do) over science (what to know)" (Nilsson, 1980, p. 1). AI fundamentally differs from other contemporary technologies in several ways, most notably its capacity for autonomy and the resulting increased interactivity (Mays, Lei, Giovanetti, & Katz, 2021). The operation of AI production systems has been described as a search process in which rules are tested until some sequence of them is discovered that produces a database satisfying the termination condition (Nilsson, 1982), and this promotes perceptions of intelligence. Johnson et al. (2008) describe perceptions of intelligence in the context of computing technology that exhibit human-like characteristics regarding users' impressions of the technology's intelligence, purpose and knowledge. According to Bartneck et al. (2009), intelligence is the extent to which a robot appears to be capable of learning, reasoning, and problem-solving, and this typically occurs while engaging with decision-supporting systems or interacting with similar types of technology that give insight into or responses to particular scenarios of decision-making.

Huang and Rust (2018) differentiate four degrees of intelligence: empathetic, analytical, intuitive and mechanical. Empathetic intelligence refers to the capability to read, comprehend, and respond to emotions. Intuitive intelligence is the capacity to process complicated information and function well in unexpected settings; this requires comprehension skills and the ability to think creatively and holistically (this is described as 'strong AI' in the literature). Legal counsel and medical diagnostics are two examples of services that demand intuitive intelligence. Analytical intelligence is concerned with data processing for problem-solving, analytics, and machine learning and involves performing complex processes systematically and predictably (this is described as 'weak AI' in the literature). It typically uses algorithms which derive knowledge from data and is often seen in robot-advisory services. Mechanical intelligence concerns routine and repetitive tasks such as those in call centres. For the purposes of this paper, intelligence will be considered to encompass all four aspects.

Voice assistants are based on AI technology, but they rely heavily on humans to develop the algorithms and datasets required to mimic human intelligence, solve issues and "learn" how to answer questions. The execution of requests involves not just the user's input "ask" or command but also their personal information and usage history. When first used, the VA's role is simply to aid the users in completing everyday tasks, but over time, they become more user-adaptive and learn from their interactions. New users who are unaware of this may be dissatisfied with the initial results; however, if they continue to use their VAs, their perceptions are likely to evolve from seeing it simply as a tool to regarding it as a collaborator in task completion and decision-making processes (Seeber, Bittner, Briggs, De Vreede, De Vreede,

Elkins, & Söllner, 2020). With a collaborative relationship between the user and VA, it is predicted that the user's perceptions of intelligence will be reshaped over time as they learn from one another.

In this respect, VA technology is similar to human-robot interaction technology in terms of its interactions. Canning, Donahue, & Scheutz (2014) argued that users tend to regard humanlike robots as more intelligent than machinelike robots, and Huang & Rust (2018) found that when users interact with a humanlike robot, they often treat it as if it were human and expect it to show aspects of human intelligence. Thus, users are more likely to attribute human intelligence to a VA that appears humanlike to them in terms of its interactions. Furthermore, Blut et al. (2021) confirmed that the more humanlike the robot appears to be, the more human intelligence is ascribed to it. This suggests that users who humanize their VA tend to have higher expectations of its ability to show intelligence. As VAs become more intelligent, it is expected to motivate users to engage with the technology, leading to improved user satisfaction and a higher overall value of the technology.

Consequently, this could potentially have a favourable impact on users' intentions toward the VAs. However, Bartneck et al. (2009) concluded that perceived intelligence depends on the human robot's interaction competency. As a result, we propose that AI can be utilised to motivate users to engage with VAs and positively influence the intention to engage with VA technology. Building upon these insights, we provide the first proposal, supported by five sub-hypotheses, to examine the relationship between perceived intelligence and engagement motivations (utilitarian, hedonic, and social), users' perceived value and users' satisfaction:

Proposition 1. Perceived intelligence positively influences motivations to engage in the use of a VA, users' perceived engagement value, and users' engaged satisfaction.

H1a. Perceived intelligence positively influences users' utilitarian motivation.

H1b. Perceived intelligence positively influences users' hedonic motivation.

H1c. Perceived intelligence positively influences users' social motivation.

H1d. Perceived intelligence positively influences users' perceived value.

H1e. Perceived intelligence positively influences users' satisfaction.

4.2.2 Engagement motivation and value

Engagement motivation and value are key factors in understanding users' interactions with home VAs and their sustained engagement with this technology. Motivation is the intrinsic factors and incentives that stimulate individuals to interact with a particular technology, whereas value denotes the advantages obtained from utilising said technology. As Kearsley & Shneiderman (1998) argue, engagement is a multifaceted process that involves a range of cognitive and affective activities, including reasoning, problem-solving, evaluation, and decision-making, and the engagement experience of these activities is attributed to a confluence of behavioural and emotional tasks. In the context of home VAs, this entails various cognitive and behavioural activities that encompass decision-making, problem-solving, evaluation, and cognitive processes (Pagani & Mirabello, 2011). Users' motivation to engage with home VAs can be understood as their eagerness to partake in tasks that involve using these devices, commonly known as 'engagement motivation' (Kearsley & Shneiderman, 1998). This motivation arises when users interact with Vs to fulfil specific needs or desires (Kowatsch & Maass, 2010). These needs can be driven by cognitive factors, such as organising schedules and prioritising activities, as well as utilitarian motivations focused on functionality and practical purposes (Venkatesh & Brown, 2001).

Investigating the fundamental drivers and values linked to engagement with home VAs yields significant perspectives on user conduct and the determinants that foster their continued usage. As Kim et al. (2013) explain, motivations to engage with a technology positively impact users' perceived engagement value, and the perceived value associated with that engagement significantly influences users' intention to continue using these devices. The underlying motivations, including utilitarian, hedonic, and social motivations, could therefore, play a significant role in shaping users' perceptions of the value they derive from interacting with these intelligent systems. Engagement behaviours that serve utilitarian purposes by providing usefulness and efficiency, as well as hedonic purposes by offering enjoyment and relaxation, contribute to sustained engagement (Venkatesh et al., 2012). Utilitarian motivation, the desire to achieve practical goals and obtain functional benefits from a home VA, is rooted in the belief that engaging with VAs fulfils specific purposes and needs and strongly influences behavioural motivation (Venkatesh & Brown, 2001). Research indicates that when users perceive a high level of utilitarian value in their interactions with intelligent systems, such as task efficiency, accuracy, and convenience, they tend to perceive the engagement value positively. For instance, a study by Wang and Chen (2017) found a positive association between users'

utilitarian motivation and their perceived value when using intelligent personal assistants, indicating that fulfilling practical needs contributes to users' perception of engagement value. Thus, when users find value in these utilitarian activities, they are more likely to sustain their engagement with home VAs (Kim & Han, 2011).

By contrast, hedonic motivation arises from activities that bring pleasure, excitement, and enjoyment. Thus, users engage in pleasurable pursuits through their technologies to satisfy their intrinsic needs for amusement and personal gratification (Holbrook & Batra, 1987). Empirical evidence demonstrates that hedonic motivation significantly drives system use (Turel, Serenko, & Bontis, 2007), with users becoming more engrossed with their technologies when engaging in captivating activities that intrinsically motivate them (Lee & Jun 2005). Hedonic motivation also positively influences users' perceived value (Kim et al., 2007). Users' engagement with home VAs can be driven by the pleasure and enjoyment derived from the interaction. In the context of mobile devices, studies have demonstrated that when users perceive a high level of hedonic value, such as enjoyment, interactivity, and personalisation, they tend to attribute a greater engagement value to the interaction. For example, a study by van Oosterhout et al. (2018) found a positive relationship between users' hedonic motivation and their perceived value when using intelligent virtual assistants, indicating that enjoyment and entertainment contribute to users' perceptions of engagement value. This suggests that a high degree of hedonic engagement motivation may lead to an increased perception of value among users of VAs.

The concept of social engagement motivation in relation to home VAs pertains to the interactions and experiences that users generate and exchange with others, regardless of whether these are physical or virtual in nature (Edvardsson, Tronvoll, & Gruber, 2011). According to Deighton and Grayson (1995), the concept of social engagement motivations refers to activities that are driven by contextual factors and serve to satisfy users' social needs, thereby shaping their perceptions and attitudes towards other users. These social tendencies affect how users structure and perceive their network activities within the environment of the technology (Giddens, 1986). Hence, the social engagement motivation factor, which concerns the yearning for social interaction and camaraderie, may have a favourable impact on users' perceived value. Voice assistants designed for home use that demonstrate social intelligence could potentially satisfy users' social needs and offer them a sense of companionship. Studies have indicated that users' perception of a heightened level of social value, encompassing traits such as responsiveness, empathy, and social presence, positively correlates with their

perception of greater engagement value in their interactions. Chen, Mo, and Zhang (2020) found a favourable correlation between users' social motivation and perceived value while utilising conversational agents. This underscores the significance of social factors in influencing users' perceptions of engagement value. Building upon these insights, we propose the second proposition, supported by three sub-hypotheses, to examine the relationship between engagement motivations (hedonic, utilitarian, and social) and users' perceived value with home VAs:

Proposition 2. Motivations to engage in the use of a VA influence users' perceived engagement value positively.

H2a. Utilitarian motivation influences users' perceived value positively.

H2b. Hedonic motivation influences users' perceived value positively.

H2c. Social motivation influences users' perceived value positively.

4.2.3 Engagement motivation and satisfaction

The level of satisfaction users experience with their VAs may also have a significant impact on their engagement. User satisfaction levels reflect the accumulation of feelings and experiences developed through a series of interactions with a service provider (Oliver, 1980), and studies suggest that they play a crucial role in determining user continuance behaviour and intention (Xu et al., 2015; Zhao et al., 2012). In simple terms, when users are dissatisfied with the technology, there is a stronger possibility that they could end their usage (Kim, Ferrin, & Rao, 2009). Previous research has emphasised the favourable influence of engagement motivations on satisfaction and, thus, users' continuous intention to use the technology (Xu et al., 2015). As with mobile devices, continued engagement with home VAs is driven by the relevance of activities to users' intended motivations, encompassing factors such as connectivity, content, interactivity, and context (Chae, Kim, Kim, & Ryu, 2002), with activities that require minimal effort and provide seamless experiences having a higher likelihood of encouraging user engagement (Davis, 1989). When user activities successfully achieve the intended outcomes, whether they are utilitarian, hedonic, or social in nature, users are likely to experience satisfaction (Edvardsson et al., 2011). Satisfaction could therefore play a role in promoting engagement with VAs and influencing users' choices in utilizing the available features and services. Building upon these insights, we provide the third proposition, supported

by three sub-hypotheses, to examine the relationship between engagement motivations (hedonic, utilitarian, and social) and users' satisfaction with home VAs:

Proposition 3. Motivations to engage in the use of a VA positively influence users' satisfaction.

H3a. Utilitarian motivation influences VA users' satisfaction positively.

H3b. Hedonic motivation influences VA users' satisfaction positively.

H3c. Social motivation influences VA users' satisfaction positively.

4.2.4 Engagement motivation and engagement intention

McLean and Osei-Frimpong (2019) demonstrate the importance of motivations in influencing users' behaviours and usage patterns in the context of VAs. Indeed, motivations could play a crucial role in shaping users' desire to continue interacting with VAs. Research has indicated that motivations associated with enjoyment and entertainment can impact users' intention to engage (Babin, Darden, & Griffin, 1994).

Voice assistant technology facilitates improved user engagement with digital devices and services by offering an effortless and user-friendly method of engagement through conversation, resulting in personalised and interactive experiences that encourage increased engagement behaviours. The behaviours and situations in which users engage, as well as their level of engagement with their VA devices, have the potential to foster ongoing engagement and facilitate the creation and delivery of VA user experiences that are highly valuable, personalised, and engaging (Kowatsch, & Maass, 2010). These behaviours serve as examples of continuous use that extends beyond the initial adoption phase, thereby moving into a state of continued usage rather than remaining in an intentional state (Kim et al., 2007).

The reasons behind using VAs can enhance the user's willingness to sustain engagement in the future (Tojib & Tsarenko, 2012). The intention of VA users to sustain engagement is influenced by the manner in which they perceive and describe their engagement behaviour. When users find these experience contexts applicable to their circumstances, this is expected to result in additional benefits and encourage their continued engagement (Kim & Han, 2011). The various motivations behind engaging in lifestyle activity, whether driven by social, utilitarian, or hedonic factors, contribute to fulfilling user needs by providing value-added

benefits. Creating sustained engagement with users is crucial for delivering valuable and satisfying engagement relationships (Limayem, Hirt, & Cheung, 2007). This suggestion seeks to encourage continuous engagement with VAs in order to fulfil users' requirements (utilitarian) while also ensuring that it is enjoyable (hedonic) (Lee & Kim, 2010). Building upon these insights, we propose a fourth proposition, supported by three sub-hypotheses, to examine the relationship between engagement motivations (hedonic, utilitarian, and social) and users' intentions to engage with home VAs:

Proposition 4. Motivations to engage in the use of a VA positively influence users' intention to engage in the use of the VA.

H4a. Utilitarian motivation positively influences VA users' engagement intention.

H4b. Hedonic motivation positively influences VA users' engagement intention.

H4c. Social motivation positively influences VA users' engagement intention.

4.2.5 Perceived value, satisfaction, and engagement intention

The association between perceived value and users' engagement intention in the context of home VAs holds substantial significance. Perceived value refers to individuals' subjective assessments of the overall benefits and worth they derive from a product, service, or experience (Zeithaml, 1988; Holbrook, 2006), and, in the case of the VA, it encompasses the assessment of the device's utility, relevance, and the enjoyment derived from its features and functionalities. Perceived value also involves the assessment of the advantages obtained in proportion to the expenses incurred (Zeithaml, 1988; Holbrook, 2006), with individuals more likely to feel satisfied when they believe the benefits outweigh the perceived disadvantages. Previous studies have demonstrated an established and positive correlation between perceived value and satisfaction (Xu et al., 2015), which can be explained by the concept that an individual's perception of a product or service's high value enhances their overall level of satisfaction. When a user perceives a high value in a product or service, it can be concluded that their expectations have been fulfilled; thus, as Zeithaml (1988) explains, they experience a positive affective response and a sense of fulfilment, leading to higher satisfaction levels. This suggests that the perception of a high level of value in a user's interactions with a home VA may positively impact their intention to engage with the device.

Moreover, extant research has consistently demonstrated a strong relationship between satisfaction and continuance behaviour (Kahn, 1992). Similar to mobile technology, satisfaction with home VAs is achieved when users find their interactions enjoyable or effective (Pura, 2005), with the level of satisfaction experienced and the intention to continue engaging with the device dependent upon users' perceived value and their intentions to sustain their chosen activities in the future. Studies have established that user satisfaction is a key factor in influencing their decision to continue using a service (Kim et al., 2009; Kuo, Wu, & Deng, 2009); thus, the value users derive from being satisfied with their home VAs plays a crucial role in fostering continued engagement and reducing the likelihood of discontinuation or disengagement (Kahn, 1990). Moreover, Zhao et al. (2012) confirmed that perceived value significantly influences customer satisfaction and post-purchase intentions. These studies suggest that perceived value significantly influences users' satisfaction and shapes their intention to engage with and continue using the VA actively. Therefore, recognizing and enhancing the perceived value can lead to heightened satisfaction, engagement, prolonged usage, and a more gratifying and enriching user experience with the home VA. Thus, we propose:

H5. Home VA users' perceived value positively influences their satisfaction.

H6. Home VA users' perceived value positively influences their engagement intention.

H7. Home VA users' satisfaction positively influences their engagement intention.

4.3 Research methodology

Data was collected from a sample of 250 individuals who were users of home VA devices. The data collection was carried out through a well-established online platform. The survey was conducted electronically. The initial page provided an overview of the survey's objectives, estimated time commitment, a participant information sheet, a consent form, and the rights of the participants, including the option to withdraw at any point or decline to answer specific questions. Progressing to the subsequent page was considered as an indication of informed consent. The study focused specifically on individuals aged 18 and above who were owners of at least one home VA device, such as Apple Home Pod, Amazon Echo, or Google Home, and had some degree of experience using them. It is worth noting that the artificial intelligence

industry places significant emphasis on younger generations, given their early adoption tendencies for new technologies (Chattaraman, Kwon, Gilbert, & Ross, 2019).

Consequently, including participants aged 18 and above holds substantial significance in this study, as millennials represent a technologically adept group (Bilgihan, 2016) with a strong inclination to follow emerging trends (Schepers & Wetzels, 2007). Thus, including participants aged 18 and above was deemed essential to capture a diverse range of home VA device users with prior experience. Moreover, exploring the usage patterns of home VA devices among older individuals could potentially shed light on their interest in utilizing such technology.

The survey consisted of three distinct sections. The initial section comprised general inquiries regarding participants' experiences with their home VAs. These questions encompassed topics such as the VAs being used, the primary purpose of usage, the frequency and number of times the VA was utilized, the overall user experience, and the ease of engagement with the VA system. The second section consisted of seven sub-sections, each focusing on a different facet of VA usage: perceived intelligence, social motivation, hedonic motivation, utilitarian motivation, perceived value, satisfaction and VA engagement intention. All the questions in this section were derived from established sources in the literature and were evaluated using a five-point Likert scale, ranging from "Strongly agree" to "Strongly disagree" (See Table 4.1). Additionally, the third section encompassed demographic information, encompassing variables such as gender, employment status, age, and educational level.

The Smart PLS 3.0 software implemented the Partial Least Squares Structural Equation Modeling (PLS-SEM) technique. PLS-SEM is recognized as an effective modelling approach for analyzing complex prediction models and facilitating theory development in exploratory studies (Hair, Hull, Ringle, & Sarstedt, 2014). This method has gained popularity due to its capability to handle latent constructs in non-normal conditions (Hair, Ringle, & Sarstedt, 2011) and with small- to medium-sized datasets (Chin, Marcolin, & Newsted, 2003). Furthermore, PLS-SEM does not require a multivariate-normal distribution, making it suitable for examining nascent phenomena (Fornell & Bookstein, 1982; Albert & Merunka, 2013).

The research followed the two-step procedure suggested by Diamantopoulos and Winklhofer (2001), commencing with the evaluation of the measurement (outer) model and subsequently analyzing the structural (inner) model. This approach enables the examination of all potential paths within a single analysis (Gefen, Straub, & Boudreau, 2000). PLS-SEM

facilitates the assessment of both the structural model, which examines the relationships between theoretical constructs, and the measurement model, which evaluates the validity and reliability of questionnaire items. Additionally, Partial Least Squares Structural Equation Modelling is suitable for investigating categorical moderation across several connections, aligning with the objective of this study (Hair, Sarstedt, Ringle, & Mena, 2012).

4.4 Research Results and Findings

4.4.1 Descriptive Analysis

The survey was administered to an equal distribution of males (49.20%) and females (50%). The majority of respondents fell within the age range of 21 to 30 years (55.20%) and held educational qualifications such as a diploma (25.20%), bachelor's degree (35.60%), or master's degree (16.80%). Moreover, approximately 41.60% were employed, while 38.40% were students. All participants reported frequent or regular interaction with AI-based VA devices, namely Google Home (60.40%), Amazon Echo with Alexa (24.40%), and Apple Home Pod with Siri (14.80%). Most of the interactions occurred at home (94.40%), with only a tiny percentage (5.20%) occurring at work. Most respondents had used their VA devices more than 20 times (56.00%), while only 12% had used them 1-5 times. They utilized VA devices for three primary reasons: 55.60% found them useful when their hands or eyes were occupied, 18.40% enjoyed using them, and 17.20% appreciated their faster results.

4.4.2 Measurement Model

In the evaluation of the measurement model, all items were included in the analysis due to their high factor loadings (> 0.700). To assess the reliability of the constructs, Cronbach's alpha and composite reliability (C.R.) were employed in this study. All C.R. values surpassed the recommended threshold of 0.700 (Wasko & Faraj, 2005). Additionally, the Cronbach's alpha for each construct exceeded the threshold of 0.700. Convergent validity was deemed acceptable as the Average Variance Extracted (AVE) exceeded 0.500. Table 4.1 presents the results for reliability and validity, along with the factor loadings of the items. Discriminant validity was evaluated using the Cornell-locker criterion, which indicated that the square root of AVE for each construct was greater than the inter-construct correlation (see table 4.2). Discriminant validity was further confirmed using the heterotrait-monotrait ratio of

correlations (Henseler et al., 2015), with values below the threshold of 0.90. Therefore, discriminant validity has been established (see table 4.3).

The correlation matrix of latent variables in table 4.4 provided reveals the interrelationships among various factors influencing voice assistant engagement intention. Notably, there is a low positive correlation (0.494) between hedonic motivation and voice assistant engagement intention, suggesting that an increase in Hedonic motivation is associated with a rise in engagement intention. Furthermore, a moderate positive correlation (0.683) exists between voice assistant engagement intention and perceived value, indicating that higher perceived value is linked to increased engagement intention. Social motivation demonstrates a weaker positive correlation (0.395) with Voice assistant engagement intention, suggesting a less pronounced link compared to other factors. Lastly, utilitarian motivation exhibits a moderate positive correlation (0.551) with voice assistant engagement intention, indicating a moderately positive relationship between utilitarian motives and the intention to engage with voice assistants. It is important to note that these correlations provide insights into associations but do not imply causation between the variables.

Table 4.1. Loadings, Reliability, and Validity

	Loadings	**Cronbach's Alpha**	**Composite Reliability**	**AVE**
Voice Assistant Engagement Intention		0.822	0.918	0.849
(Cronin Jr, & Taylor, 1992)				
CI6: I would like to continue to engage with the voice assistant device	0.921			
CI7: I would like to recommend my engagement with the voice assistant device to someone	0.922			
Hedonic Motivation		0.821	0.892	0.735
(Kim, Goh, & Yuan, 2010)				
HM1: To get rest and relaxation using a voice assistant device	0.815			
HM2: To enjoy the variety of contents (e.g., email, applications, weather, and scheduling) that voice assistant device offers	0.875			
HM3: To enjoy what I like about using a voice assistant device	0.879			
Perceived Intelligence		0.868	0.905	0.655
(Bartneck, Kulić, Croft, & Zoghbi, 2009)				
INT1: My voice assistant is competent	0.817			
INT2: My voice assistant is knowledgeable	0.813			
INT3: My voice assistant is responsible	0.806			
INT4: My voice assistant is intelligent	0.832			
INT5: My voice assistant is sensible	0.777			
Perceived Value		0.804	0.884	0.719

(Petrick, 2002; Zeithaml, 1988)				
PV5: Using a voice assistant device is an enjoyable experience	0.818			
PV6: The overall value of my experience using a voice assistant device is outstanding	0.878			
PV7: Voice assistant device represents a good use of my time and money	0.846			
Satisfaction		0.925	0.953	0.87
(Oliver, 1980)				
SA1: I am satisfied with my use of the voice assistant device	0.938			
SA2: Based on my total time spent, I am satisfied with my engagement with the voice assistant device	0.918			
SA3: Overall, I am satisfied with using/engaging with the voice assistant device	0.942			
Social Motivation		0.871	0.921	0.795
(Kim, Goh, & Yuan, 2010)				
SM1: To keep in touch/share events (community) with my friends and family	0.898			
SM2: To be connected and meet other people with similar interests	0.905			
SM3: To tell my friends and family about what I learned/heard using my voice assistant	0.871			
Utilitarian Motivation		0.872	0.904	0.612
(Kim, Goh, & Yuan, 2010)				
UM1: To try and find new and different things using a voice assistant device	0.713			
UM2: To keep me informed and updated using a voice assistant device	0.838			
UM3: To increase my skills and knowledge in using a voice assistant device	0.815			
UM4: To keep me organized (e.g., checking email, schedule, and plan)	0.763			
UM5: To save and use my time more efficiently and effectively with what I have to do (e.g., work)	0.761			
UM6: It offers a variety of ways to communicate with others (e.g., voice or text message)	0.796			

Table 4.2. Loadings, Reliability, and Validity

	Hedonic Motivation	Perceived Intelligence	Perceived Value	Satisfaction	Social Motivation	Utilitarian Motivation	Voice Assistant Engagement Intention
Hedonic Motivation	***0.857***						
Perceived Intelligence	0.368	***0.809***					
Perceived Value	0.461	0.519	***0.848***				
Satisfaction	0.463	0.506	0.704	***0.933***			
Social Motivation	0.580	0.351	0.398	0.409	***0.891***		
Utilitarian Motivation	0.667	0.446	0.569	0.558	0.605	***0.782***	
Voice Assistant Engagement Intention	0.494	0.475	0.683	0.785	0.395	0.551	***0.921***

Note: Value in Italic represents Square-root of AVE.

Table 4.3. HTMT Ratio

	Hedonic Motivation	Perceived Intelligence	Perceived Value	Satisfaction	Social Motivation	Utilitarian Motivation	Voice Assistant Engagement Intention
Hedonic Motivation							
Perceived Intelligence	0.432						
Perceived Value	0.555	0.622					
Satisfaction	0.522	0.563	0.814				
Social Motivation	0.692	0.404	0.471	0.452			
Utilitarian Motivation	0.783	0.511	0.678	0.620	0.693		
Voice Assistant Engagement Intention	0.590	0.560	0.839	0.899	0.463	0.650	

Table 4.4. Latent variables - Correlation

	Voice Assistant Engagement Intention	Hedonic motivation	Perceived intelligence	Perceived value	Satisfaction	Social Motivation	Utilitarian Motivation
Voice Assistant Engagement Intention	1	0.494	0.475	0.683	0.785	0.395	0.551
Hedonic motivation	0.494	1	0.368	0.461	0.463	0.58	0.667
Perceived Intelligence	0.475	0.368	1	0.519	0.506	0.351	0.446
Perceived value	0.683	0.461	0.519	1	0.703	0.398	0.569
Satisfaction	0.785	0.463	0.506	0.703	1	0.409	0.559
Social Motivation	0.395	0.58	0.351	0.398	0.409	1	0.605
Utilitarian Motivation	0.551	0.667	0.446	0.569	0.559	0.605	1

4.4.3 Structural Model

Assessing a structural model involves evaluating its performance based on R2, Q2, and the significance of paths. The structural model encompasses the hypothesized paths outlined in the research framework. The quality of the model is determined by the strength of each structural path, as indicated by the R2 value for the dependent variable (Briones Penlver et al., 2018). According to Falk and Miller (1992), an R2 value equal to or greater than 0.1 is considered desirable. Examination of the results presented in table 4.5 reveals that all R2 values exceed 0.1, thereby establishing the model's predictive capability.

Furthermore, the Q2 measure substantiates the predictive relevance of the endogenous constructs. A Q2 value above 0 indicates that the model possesses predictive relevance. The results presented in Table 4.5 demonstrate the significance of the constructs' predictions. Additionally, the widely employed measure for assessing goodness-of-fit in SmartPLS, the

bootstrapped Standardised Root Mean Square Residual (SRMR), was utilized in evaluating the model fit. The calculated SRMR value of 0.056 was below the recommended threshold of .10, indicating an acceptable model fit according to Hair, Sarstedt, Matthews, and Ringle (2016).

Furthermore, in assessing the goodness of fit, hypotheses were tested to ascertain the significance of the relationship. H1 evaluates whether perceived intelligence significantly impacts motivations to engage with the VA, perceived value, and satisfaction. The results revealed that perceived intelligence significantly impacts motivations to engage with the VA (Hedonic motivation: ß = 0.368, t = 6.107, p < .001, Utilitarian motivation: ß = 0.446, t = 7.246, p < .001, Social motivation: ß = 0.351, t = 5.626, p < .001), perceived value (ß = 0.320, t = 5.788, p < .001) and satisfaction (ß = 0.145, t = 2.710, p = 0.007). Hence, H1 was fully supported. H2 evaluates whether motivations to engage with the VA significantly impact perceived value. The results revealed that utilitarian motivation significantly impacts perceived value (Hedonic motivation: ß = 0.102, t = 1.343, p = 0.180, Utilitarian motivation: ß = 0.348, t = 3.723, p < .001, Social motivation: ß = 0.015, t = 0.221, p = 0.825). Hence, H2 was partially supported. H3 evaluates whether motivations to engage with the VA significantly impact satisfaction. The results revealed that utilitarian motivation has a significant impact on satisfaction (Hedonic motivation: ß = 0.060, t = 0.939, p = 0.348, Utilitarian motivation: ß = 0.146, t = 2.001, p = 0.046, Social motivation: ß = 0.034, t = 0.603, p = 0.547). Hence, H3 was partially supported. H4 evaluates whether motivations to engage with the VA significantly impact users' engagement intention. The results revealed that hedonic motivation significantly impacts users' engagement intention (Hedonic motivation: ß = 0.108, t = 2.004, p = 0.046, Utilitarian motivation: ß = 0.053, t = 0.858, p = 0.391, Social motivation: ß = -0.013, t = 0.264, p = 0.792). Hence, H4 was partially supported. H5 evaluates whether perceived value has a significant impact on satisfaction. The results revealed that perceived value significantly impacts satisfaction (ß = 0.505, t = 7.846, p = 0.000). Hence, H5 was supported. H6 evaluates whether perceived value significantly impacts users' engagement intention. The results revealed that perceived value significantly impacts users' engagement intention (ß = 0.215, t = 3.164, p = .002). Hence, H6 was supported. H7 evaluates whether satisfaction has a significant impact on users' engagement intention. The results revealed that satisfaction significantly impacts users' engagement intention (ß = 0.560, t = 9.393, p = 0.000). Hence, H7 was supported. This study of 5000 resamples also generated 95% confidence intervals, as shown in Table 4.5. A confidence interval different from zero indicates a significant relationship.

Hypotheses testing results are summarised in Table 4.5. A summary of the hypotheses results is provided in table 4.6.

Table 4.5. Hypotheses Testing

	ß	STDEV	T Statistics	P Values	2.50%	97.50 %
Hedonic Motivation -> Perceived Value	0.102	0.076	1.343	0.180	-0.041	0.256
Hedonic Motivation -> Satisfaction	0.060	0.063	0.939	0.348	-0.062	0.185
Hedonic Motivation -> Voice Assistant Engagement Intention	0.108	0.054	2.004	0.046	0.004	0.217
Perceived Intelligence -> Hedonic Motivation	0.368	0.060	6.107	0.000	0.24	0.485
Perceived Intelligence -> Perceived Value	0.320	0.055	5.788	0.000	0.213	0.423
Perceived Intelligence -> Satisfaction	0.145	0.053	2.710	0.007	0.05	0.251
Perceived Intelligence -> Social Motivation	0.351	0.062	5.626	0.000	0.218	0.462
Perceived Intelligence -> Utilitarian Motivation	0.446	0.062	7.246	0.000	0.32	0.558
Perceived Value -> Satisfaction	0.505	0.064	7.846	0.000	0.382	0.642
Perceived Value -> Voice Assistant Engagement Intention	0.215	0.068	3.164	0.002	0.09	0.356
Satisfaction -> Voice Assistant Engagement Intention	0.56	0.060	9.393	0.000	0.44	0.667
Social Motivation -> Perceived Value	0.015	0.070	0.221	0.825	-0.117	0.151
Social Motivation -> Satisfaction	0.034	0.056	0.603	0.547	-0.092	0.145
Social Motivation -> Voice Assistant Engagement Intention	-0.013	0.051	0.264	0.792	-0.105	0.087
Utilitarian Motivation -> Perceived Value	0.348	0.093	3.723	0.000	0.167	0.508
Utilitarian Motivation -> Satisfaction	0.146	0.073	2.001	0.046	0.007	0.279
Utilitarian Motivation -> Voice Assistant Engagement Intention	0.053	0.061	0.858	0.391	-0.067	0.169
	R^2	Q^2				
Hedonic Motivation	0.135	0.096				
Perceived Value	0.418	0.291				
Satisfaction	0.552	0.471				
Social Motivation	0.123	0.096				
Utilitarian Motivation	0.199	0.119				
Voice Assistant Engagement Intention	0.663	0.544				

Table 4.6. Hypotheses Testing

Proposition and hypothesis		**Results**
Proposition 1	Perceived Intelligence -> Engagement Motivations, Perceived Value, Satisfaction	Fully supported
H1a	Perceived Intelligence -> Utilitarian Motivation	***
H1b	Perceived Intelligence -> Hedonic Motivation	***
H1c	Perceived Intelligence -> Social Motivation	**
H1d	Perceived Intelligence -> Perceived Value	***

H1e	Perceived Intelligence -> Satisfaction	***
Proposition 2	Engagement Motivations -> Perceived Value	Partially supported
H2a	Hedonic Motivation -> Perceived Value	NS
H2b	Utilitarian Motivation -> Perceived Value	***
H2c	Social Motivation -> Perceived Value	NS
Proposition 3	Engagement Motivations -> Satisfaction	Partially supported
H3a	Utilitarian Motivation -> Satisfaction	**
H3b	Hedonic Motivation -> Satisfaction	NS
H3c	Social Motivation -> Satisfaction	NS
Proposition 4	Engagement Motivation -> Voice Assistant Engagement Intention	Partially supported
H4a	Utilitarian Motivation -> Voice Assistant Engagement Intention	NS
H4b	Hedonic Motivation -> Voice Assistant Engagement Intention	**
H4c	Social Motivation -> Voice Assistant Engagement Intention	NS
H5	Perceived Value -> Satisfaction	***
H6	Perceived Value -> Voice Assistant Engagement Intention	***
H7	Satisfaction -> Voice Assistant Engagement Intention	***

Note: S = Significant, NS = Not Significant.

* Significant at the 0.1 level.

** Significant at the 0.05 level.

*** Significant at the 0.01 level.

4.4.4 Mediation analysis

Mediation analysis assessed the mediating role of motivations to engagement, satisfaction and perceived value on the linkage between perceived intelligence and home VA engagement intention. The results (see Tables 4.7 and 4.8) revealed that the total effect of Perceived intelligence on home VA engagement intention was significant (ß = 0.453, t = 8.991, P < 0.10). Including the mediating variables engagement motivations (hedonic, Social, utilitarian), perceived value and satisfaction, the impact of Perceived intelligence on home VA engagement intention became insignificant (ß = 0.032, P = 0.413). The indirect effect of perceived intelligence on home VA engagement intention through utilitarian motivation, satisfaction and perceived value was found to be significant (ß = 0.044, t = 2.925, P < 0.10). Moreover, the indirect effect of perceived intelligence on home VA engagement intention through utilitarian motivation and perceived value was found to be significant (ß = 0.033, t =

2.424, P < 0.10). However, Surprisingly, by considering the mediation role of utilitarian motivation, satisfaction excluding perceived value on the linkage between perceived intelligence and home VA engagement intention, the indirect effect of perceived intelligence on home VA engagement intention through utilitarian motivation, satisfaction was found insignificant (ß = 0.037, t = 1.911, P = 0.057). Additionally, by considering the mediation role of utilitarian motivation only on the linkage between perceived intelligence and home VA engagement intention, the indirect effect of perceived intelligence on home VA engagement intention through utilitarian motivation was found insignificant (ß = 0.023, t = 0.829, P = 0.408). In contrast, the mediation role of satisfaction and perceived value are significant. The indirect effect of perceived intelligence on home VA engagement intention through perceived value was found to be significant (ß = 0.069, t = 2.522, P < 0.10). In addition, the indirect effect of perceived intelligence on home VA engagement intention through satisfaction was found to be significant (ß = 0.081, t = 2.477, P < 0.10). This shows that utilitarian motivation, perceived value, and satisfaction fully mediate the relationship between perceived intelligence and home VA engagement intention. For the control variables, we did not find an effect of gender and age.

Table 4.7. Mediation Analysis: The total and the direct effect

	Total Effect			Direct Effect	
	Coefficient	T	P Value	Coefficient	P Value
Perceived Intelligence -> Voice Assistant Engagement Intention	0.453	8.991	0.000	0.032	0.413

Table 4.8. Mediation Analysis: Overview of indirect effects

Indirect Paths	Path coefficient	T Value	Sig.
Perceived Intelligence -> Utilitarian Motivation -> Perceived Value -> Satisfaction -> Voice Assistant Engagement Intention	0.044	2.925	0.004
Perceived Intelligence -> Utilitarian Motivation -> Perceived Value -> Voice Assistant Engagement Intention	0.033	2.424	0.016
Perceived Intelligence -> Utilitarian Motivation -> Satisfaction -> Voice Assistant Engagement Intention	0.037	1.911	0.057
Perceived Intelligence -> Utilitarian Motivation -> Voice Assistant Engagement Intention	0.023	0.829	0.408
Perceived Intelligence -> Perceived Value_ -> Voice Assistant Engagement Intention	0.069	2.522	0.012

Perceived Intelligence -> Satisfaction -> Voice Assistant Engagement Intention	0.081	2.447	0.015

Note: *non-significant path coefficient (p > .05).

4.5 Discussion

The current research employs The User Engagement with Technology model (Kim et al., 2013) as the theoretical framework within the context of VA technology. The mentioned model, previously utilised in the mobile technology domain, is currently being adapted and applied to investigate motivation engagement, especially regarding VAs. This study examines the impact of perceived intelligence on the users' intention to engage with VAs, utilising a widely recognised framework. This examination, which considers various aspects, including user engagement motivations, perceived value, satisfaction, and the impact of perceived intelligence, will provide invaluable insights for enhancing and advancing VA technologies to align with evolving user needs and preferences.

4.5.1 The theoretical implication

The study conducted in this research makes a significant theoretical contribution by including perceived intelligence as a crucial factor in the User Engagement with Technology model. The results indicate several notable findings that highlight the significant impact of perceived intelligence on different dimensions of users' intention to engage with the technology. The study demonstrates a positive association between users' perceived intelligence of VAs and their engagement motivations, including hedonic, utilitarian, and social motivations. As VAs' perceived intelligence and capability increase, users become more likely to engage with the technology. As Whang and Im (2021) concluded, VAs can engage in conversations with users in a manner that closely resembles human interaction, including natural language and cognitive-like abilities. These abilities focus on the aspect of the VA as intelligent technology. Chang, Lu, and Yang (2018) added that incorporating human-like attributes in social robots encourages extended continuing user engagement. This observation highlights the idea that the perception of intelligence plays a key role in enhancing user motivation to engage with technology. The motivation behind users' desire to explore the capabilities and functionalities of VAs can be rooted in their interests. This interest could be

driven by the belief that these technologies possess a higher level of intelligence and can provide meaningful assistance.

Additionally, the study shows that perceived intelligence impacts users' perception of the value derived from the VA. Perceived value is recognised as a variable that is influenced by multiple factors, including the specific type of technology, the promises made by the service provider, and the actual attributes associated with the product or service (de Kervenoael et al., 2020). The presence of directionality in cognitive processes contributes to assessing responsiveness cues integrated within the robot (Kim et al., 2007). As a result, the intelligence attribute in a VA positively impacts users' perceptions regarding the perceived value of the VA. Users tend to attribute greater importance to VAs that they perceive to have higher levels of intelligence. This perception of intelligence increases the users' confidence in the technology's ability to provide accurate and relevant information, offer valuable recommendations, and effectively fulfil their requirements. Thus, users recognise enhanced practicality and utility in engaging with the VA, resulting in a more favourable evaluation of its overall value.

Furthermore, the research shows a positive correlation between users' perceived intelligence and their satisfaction. Users' satisfaction with their interactions and experiences is positively influenced by perceiving the VA as more intelligent and capable. As confirmed in the research conducted by Lee, Tang, and Jiang (2023), there is a positive correlation between intelligence and user satisfaction, and intelligence can strengthen user satisfaction. The increased level of satisfaction observed can be attributed to the users' favourable receives of the technology, thereby building their confidence in its efficacy and dependability. Furthermore, intelligence's efficacy enables users to believe that using this technology can effectively address challenges and speed up completing diverse tasks (Lee & Chen, 2022). Therefore, it becomes clear that the perception of intelligence plays an essential part in establishing strong and significant connections between individuals and voice-based virtual assistants. Users' perceptions regarding the technology's intelligence have a consequential impact on their motivations to engage, the perceived value they attribute to the technology, and their overall satisfaction with the VA. Establishing positive relationships between the users and the VA plays a crucial role in building enduring engagement with it, resulting in its ongoing and consistent utilisation.

In addition, this research seeks to examine and contrast User Engagement with the Technology model in VA technology and mobile technology, drawing upon the work of Kim

et al. (2013) as a foundational reference. The empirical evidence indicates that the association between engagement motivations (specifically, Hedonic, Social, and Utilitarian) and perceived value is partially supported within the domain of VA technology. It is worth noting that only utilitarian motivation shows statistical significance. On the other hand, mobile technology showcases a relationship between engagement motivations and perceived value that is only partially supported, with Hedonic and Social Motivations showing statistical significance (Kim et al., 2013). Additionally, the findings indicate differences in the correlation between engagement motivations and satisfaction, depending on the technological context. The relationship between motivations to engage with the VA technology and satisfaction is partially supported by utilitarian motivation.

In contrast, within the mobile technology context, the relationship between engagement motivations and satisfaction is fully supported, indicating a strong association (Kim et al., 2013). In fact, McLean and Osei-Frimpong (2019) conclude that the primary purpose of in-home VAs is to meet utilitarian requirements. As a result, users are motivated to engage with in-home VAs in order to enhance task performance, obtain information, seek assistance, and make purchases. This preference is motivated by the belief that the support provided by the VA enhances the functionality of the technology, which may increase perceived value and user satisfaction.

The study also examines the relationship between motivations for engagement and intentions to engage in both technological contexts. The results indicate that the relationship between engagement motivations and intentions to engage with the VA technology is partially supported by hedonic motivation. As supported by Chuah, Aw, and Yee (2021), in the domain of intelligent robotics, it is observed that hedonic motivation can significantly influence users' behavioural intentions. However, within the context of mobile technology, there exists comprehensive support for the relationship (Kim et al., 2013), thereby indicating a strong and statistically significant correlation between Engagement Motivations and Engagement Intention. The variations observed in relationships can be ascribed to factors such as the emphasis on utilitarian tasks in VA technology, which restricts its hedonic aspects, whereas mobile technology's broad functionalities align with hedonic motivations. Furthermore, mobile technology's enhanced maturity and seamless integration significantly encourage more robust relationships. Furthermore, the relative newness and continuous advancement of VA technology could limit its capacity to adequately respond to users' hedonic needs, whereas the

well-established nature of mobile technology enables a more seamless and comprehensive user experience.

Surprisingly, the direct relationship between perceived intelligence and VA engagement intention is insignificant. However, utilitarian motivation is the only type of engagement motivation that is a significant mediator. Utilitarian motivation is a significant mediator between users' perception of the VA's intelligence and their satisfaction, perceived value and intention to engage with the technology. The belief that the VA is intelligent and capable of effectively fulfilling utilitarian needs enhances users' overall satisfaction and perceived value, leading to a higher likelihood of continued engagement with the VA. As supported by Babin et al. (1994), utilitarian motivation is a significant predictor of users' specific behaviour. For example, when users perceive the VA to have high levels of intelligence and efficacy in fulfilling their requirements, it enhances their perceived value and overall satisfaction, leading to engagement with the technology. The concept of utilitarian motivation plays a significant role in mediating the relationship between perceived intelligence and various outcomes, such as satisfaction, perceived value, and VA engagement intention. These results highlight the significance of prioritising utilitarian capabilities in VAs as a means to increase user engagement and satisfaction.

4.5.2 Managerial Implication

The research results hold considerable managerial implications for developers and providers of VA technologies. In order to build deeper connections with users and promote long-term engagement, it is essential to prioritise the enhancement of perceived intelligence. Highlighting the utilitarian functionalities holds particular significance, as it serves as a substantial mediator between perceived intelligence and various factors that influence user engagement, including satisfaction, perceived value, and intention to engage with the VAs. In order to accomplish this objective, it is necessary that VA technologies are purposefully designed to efficiently respond to users' practical requirements, provide accurate and appropriate information, and give valuable recommendations.

Moreover, it is important for developers to take into account the integration of functionalities that facilitate personalised and customised user experiences. This strategic approach can significantly improve the perceived intelligence of the VA and subsequently raise user satisfaction. In addition, ongoing efforts to enhance both the accuracy and efficiency of

voice recognition and response functionalities will shape users' perceptions of increased intelligence within the technology.

4.6 Conclusion

This research highlights the significant impact of perceived intelligence on user intention to engage with VA technology. The correlation between users' perception of intelligence and their engagement motivations, perceived value, and satisfaction points out the importance of users' beliefs regarding the cognitive capabilities of VAs. The perception of higher intelligence among users is associated with enhanced motivation to engage with the technology, a higher perception of its value, and increased satisfaction levels. Moreover, positive relationships between users and VAs play an important role in building strong connections, ultimately resulting in ongoing and continuing engagement. Furthermore, the study's findings indicate that utilitarian motivation plays a crucial role in mediating the relationship between perceived intelligence and intention to engage with the VAs. The results mentioned above offer significant insights for managers and developers in the process of designing VA technologies that effectively address users' needs and enhance their engagement experience. By prioritising the improvement of perceived intelligence, providers of VAs have the potential to build long-lasting relationships with users and facilitate deeper and more fulfilling interactions with the technology.

Regardless of the valuable contributions made by this study, it is essential to acknowledge the presence of certain limitations that require further investigation. The primary focus of this study applies to home VAs, thereby potentially limiting the scope of the findings to VAs integrated within phones. Potential future research efforts may involve the integration of phone-based VAs in the investigative process. Expanding the scope of analysis would facilitate a more comprehensive comprehension of user engagement, thereby resulting in insights into the potential influence of various contexts on the relationship between the factors. Furthermore, the research primarily centres on the perception of intelligence, excluding other possible variables that could impact user engagement, including personalisation, trust, and privacy apprehensions. Examining the interaction between these variables could lead to a deeper understanding of user engagement with VAs. Moreover, it is important to note that the research is carried out within a particular demographic framework, thereby potentially limiting its applicability to other populations. Subsequent research could include a broader range of

participants to investigate potential cultural differences in building relationships between the users and the VA.

Chapter 5. Conclusion

The research project described in the preceding chapters of this thesis aims to investigate different aspects of the relationship between individuals and voice assistants (VAs) in order to identify the main factors that foster a significant and engaging relationship between them in the context of individuals' homes. Achieving this aim will result in enhanced user engagement and a continued intention among individuals to keep using the technology.

The investigation pursued three interdependent objectives – charting the landscape of existing research, investigating the impact of perceived anthropomorphism on perceived value and continuous intention and researching the role of intelligence in engagement motivation. These were captured in three distinct journal papers presented as Chapters 2, 3 and 4. This approach offers the advantage of conducting comprehensive and in-depth analysis, whereby each document is dedicated to examining a specific aspect or variable that influences the complex relationship between users and their home VAs. This enables an in-depth understanding of the implications and impacts of these factors. Furthermore, the iterative nature of the three-paper approach, in which the outcomes of the first paper shape the direction of the subsequent papers, guarantees the study's continued relevance of focus, reflecting the dynamic nature of the developing domain of AI-powered VAs.

5.1 Paper One

The first paper offered an in-depth review of the various perspectives on VA technology that have already been explored to identify any weaknesses or gaps in the existing body of knowledge. The initial step involved presenting a comprehensive summary of voice assistants (VAs) based on scholarly articles from reliable journals and conference proceedings focusing on VA-related topics. These sources were identified using the quality metrics provided by the Chartered Association of Business Schools (CABS) 2018 list and the Core Conference Portal website (http://portal.core.edu.au/conf-ranks/). The purpose was to discover the landscape of academic research regarding VAs and their adoption. Subsequently, careful attention was given to four distinct aspects that came out from the included literature: firstly, the use of the VA, including VA adoption, adding value to the VA and enhancing the VAs; secondly, interactions with VAs; thirdly, privacy and security issues; and finally, user satisfaction. It has become evident that although there is a growing academic interest in this emerging field, there is still a lack of comprehensive literature on the determinants of technology acceptance and sustained

usage. There is a lack of comprehensive and coherent research on the post-adoption of digital technology, covering a broad spectrum from chatbots to humanoid robots.

Moreover, recent scholarly investigations from consumers' perspectives (e.g., Moussawi et al., 2021) frequently utilise established acceptance models like UTAUT and TAM. However, it is essential to acknowledge that the applicability of these frameworks is heavily influenced by the specific context in which they are employed. Moreover, their primary focus lies on the functional attributes, failing to fully encompass automated technologies' broader scope and distinctive empathetic qualities (Lin, Sun, Chen, & Shi, 2019).

According to Chi, Denton, and Gursoy (2020), current research suggests that many consumers are more likely to accept VA technology. However, it is essential to acknowledge that resistance to change remains a significant obstacle to its widespread adoption. One approach to enhance user acceptance involves integrating human attributes into the design of VAs to optimise user familiarity and satisfaction by suggesting a human-like counterpart (Purington, Taft, Sannon, Bazarova, & Taylor, 2017). Regarding this matter, Araujo (2018) concluded that it is crucial to establish clear differentiations between anthropomorphic cues shown by embodied agents, such as robots, including elements like voice, body movements, facial expressions, and dialogue, and disembodied agents, such as chatbots, which primarily rely on dialogue, personality, voice, and name. Fernandes and Oliveira (2021) stated that there is potential for fruitful exploration in future research regarding the significance of perceived humanness, rapport, and interactivity.

The impact of anthropomorphism on VAs has emerged as a subject of controversy. Based on the analysis provided in this paper, there is currently a lack of research investigating the impact of anthropomorphism on VAs and the quantitative assessment of the most influential human-like attributes. To date, there has been a lack of scholarly examination regarding the potential impact of anthropomorphism on the reception of VAs in terms of quantitative findings. Additionally, there remains a dearth of research on the specific human-like attributes that may contribute to the increased adoption of VAs. By addressing this knowledge gap, it will be possible to make progress in resolving the present challenges related to adopting and post-adoption of this technology. This can be achieved by gaining insights into strategies that can enhance user acceptance of VAs and reduce resistance to change.

This study makes an academic contribution to the existing body of literature on using artificial intelligence-based technology, specifically focusing on VAs. It clarifies the ongoing

research efforts in this domain and outlines the primary research directions, thereby identifying promising avenues for future investigations. This paper analyses the present state of VA research and the primary methodologies employed. The subsequent phase of this research will focus on addressing the gap in existing knowledge as outlined in this study.

5.2 Paper Two

This study presents an understanding of voice assistants (VAs) as anthropomorphic entities that individuals interact with in a manner approaching social interaction with a partner. This research provides insights into the impact of anthropomorphism on the development of relationships between VAs and users, as well as its influence on promoting continuous usage. In accordance with the CASA framework, the primary objective of this research is to enhance understanding regarding the dynamics of user-voice assistant interactions. The primary objective of this study is to investigate the potential effects of perceived anthropomorphism, engagement, rapport, and perceived value on user perception. The specific focus is on how these factors contribute to facilitating more natural and human-like interactions. The findings suggest that attributing human-like qualities to VAs can benefit the user-VA relationship, ultimately enhancing the perceived value of the VAs and promoting their sustained usage. The findings indicate that anthropomorphism can enhance the perceived value of VAs by encouraging user engagement and establishing a sense of rapport with the technology. Furthermore, the research emphasises the importance of engagement and rapport in shaping users' perceptions of the value of VAs, particularly in relation to their intention to sustain usage.

This study makes a four-fold contribution. This study contributes to the existing body of literature on VA technologies by offering an in-depth examination of the post-adoption phase of these technologies. Furthermore, it contributes to the ongoing third wave of research in the field of human-computer interaction (HCI), which emphasises examining relationship phenomena rather than simply focusing on the technical capabilities and operations of the interacting components. Put simply, the primary emphasis lies in investigating the dynamics of human-computer interaction within interactive environments rather than delving into the particularities of their distinct functionalities or characteristics. Furthermore, this study makes a valuable contribution to the extant literature in the field of Computers Are Social Actors (CASA). Finally, it makes a valuable contribution to the field of information systems by

enhancing the existing body of literature on the concept of "perceived value," which has been recognised as a crucial factor influencing technology adoption (Kim, Chan, & Gupta, 2007).

5.3 Paper Three

This research study helps to extend the User Engagement with Technology model by incorporating perceived intelligence as a critical determinant. The study's findings show several noteworthy observations that highlight the substantial influence of perceived intelligence on various aspects of users' inclination to interact with the technology. The research findings indicate a positive correlation between users' perceived intelligence of VAs and their motivations for engagement, including hedonic, utilitarian, and social motivations. As the perceived intelligence and capability of VAs increase, users are more likely to engage with the technology. According to Whang and Im (2021), VAs have the ability to engage in conversations with users in a manner that closely emulates human interaction, including natural language and cognitive-like capabilities. These capabilities centre on the aspect of the VA as an intelligent technological entity.

It is therefore expected that including human-like attributes in social robots would encourage continuous and sustained user engagement (Chang, Lu, and Yang, 2018). Indeed, this expectation is confirmed by the study reported here, which discovers that the perception of intelligence significantly influences respondents' motivation to interact with technology.

The study seeks to explain the significant impact of perceived intelligence on users' intention to engage with VA technology. The significance of users' perceptions regarding the cognitive abilities of VA is highlighted by the significant correlation observed between users' perception of intelligence and their motivations for engagement, perceived value, and satisfaction. A correlation has been found between users' perception of intelligence, increased motivation to interact with the technology, increased perception of its value, and elevated satisfaction levels. Furthermore, establishing positive relationships between users and VAs significantly builds robust relationships, ultimately leading to sustained and continuous engagement. Additionally, the study's results suggest that utilitarian motivation is a significant factor in mediating the connection between perceived intelligence and the intention to engage with VAs. These results provide valuable insights for managers and developers involved in the design of VA technologies, aiming to meet users' requirements and improve their engagement level effectively. By emphasising enhancing perceived intelligence, VA providers can establish

lasting relationships with users and enable more profound and more satisfying engagements with the technology.

5.4 Synthesis of knowledge

The journey started by contrasting the conventional/non-intelligent technology, which is commonly regarded as a mere tool, with the AI-powered technology behind VAs, which users perceive as having intelligence. The study relates to the influence of two distinct attributes associated with this perception, namely technology anthropomorphism and perceived intelligence, on the users' continuous intention to use and engage with VAs. The study demonstrated that both perceived anthropomorphism and perceived intelligence significantly impact the continuous intention to use and engage with the voice assistant. However, the significance of these relationships is due to some important mediators. For anthropomorphism, engagement and rapport mediated the impact of perceived anthropomorphism on perceived value, thus enhancing the user's continued intention to use the voice assistant. On the other hand, perceived intelligence, engagement motivations, perceived value and satisfaction mediate the impact of perceived intelligence on the user's intention to engage with the voice assistant. In addition, perceived anthropomorphism impacts the perceived value of continuous engagement, and perceived intelligence impacts the motivations for the user to engage. These factors play a key role in the user's intention to use and engage with the voice assistant technology and distinguish the use continuance dynamics of AI-powered technology such as VAs in contrast to the models focused on conventional technology such as office applications.

5.5 Reflection

Getting a PhD is a life-changing journey that changes your academic skills and helps you grow as a person. When I started my PhD study in 2019, I did so with a strong desire to learn and a never-give-up attitude. Still, my trip was challenging in its own way because of things like the world COVID-19 outbreak and the responsibilities of being a single mother who travelled with her daughter. These factors have affected and sped up my academic and personal growth over my PhD programme.

5.5.1 Academic Journey

My academic journey has tested the full extent of my determination and flexibility, attributes that have become more important in light of the obstacles presented by the COVID-19 epidemic. The sudden shift to distant learning and research was an immediate challenge, requiring me to adapt my study habits and accept new methods of academic participation. The mentioned changing experience enhanced my ability for independent work and emphasised my proficiency in using digital resources successfully.

Furthermore, the worldwide pandemic has fostered within me a greater sense of global awareness and a feeling of responsibility in my studies. The observation of global reactions to this crisis has highlighted the importance of multidisciplinary approaches and the need for new solutions to complex situations. The recognition of this fact significantly influenced the direction of my study, motivating me to investigate new areas of investigation that are directly related to the changing demands of society, therefore enhancing the influence of my academic contributions.

The pandemic's limits unexpectedly presented me with a chance to improve my data analysis and computational tools skills, which have become very useful assets in my area of study, particularly in research production, and the period of compulsory isolation provided me with a continuous opportunity to engage in deep thought, scholarly writing, and reflection, resulting in successfully written chapters. This accomplishment is significant for me as it marks a noteworthy milestone in my academic journey.

Furthermore, my position as a Teaching Assistant (TA) at the institution for a duration of one year had an essential impact in supporting my academic development. The experience not only enhanced my comprehension of the topics I taught but also improved my teaching abilities, increasing my recognition of the skill of education and its essential role in the diffusion of information. This particular experience has deeply embedded within me a strong feeling of accountability and improved my ability to effectively communicate complex concepts to other learners, consequently enhancing my scholarly ability. In my role as a Teaching Assistant (TA), I had the opportunity to provide support and guidance to students, which significantly enhanced my academic progression as I saw the progress and development of my own pupils.

Overall, achieving the goal of my doctoral degree, which started in 2019, has undoubtedly been an insightful and impactful effort, characterised by significant advancements

in both my scholarly and personal development. The confluence of the obstacles presented by the COVID-19 epidemic, the challenges associated with single parenthood, and my position as a Teaching Assistant have together contributed to my personal growth, encouraging increased strength, resilience, and a more comprehensive skill set. The COVID-19 epidemic has impacted research and my position as a Teaching Assistant, expanding the length and significance of my work and enhancing its depth and importance. Subsequently, this journey has enhanced my life by imparting crucial insights relating to the effective use of time, the practice of self-preservation, and the significance of establishing a beneficial example for my child. As I approach the end of my PhD study, I reflect on the significant gain of knowledge and personal growth that has occurred during my academic odyssey, exceeding the early anticipations I had. This particular event has provided substantiation for my unwavering commitment to education, serving as a testament to my ongoing pursuit of both professional and personal development and my desire to inspire others to overcome challenges and attain their objectives.

5.5.2 Conducting research process

The study of a Doctor of Philosophy (PhD) includes more than just a cognitive endeavour; it involves an emotional journey marked by instances of curiosity, discontent, and accomplishment. Reflecting on my trajectory in starting my doctorate studies, I am tempted to share the complicated emotional expedition and human interactions that have shaped my intellectual growth.

The journey's beginning was characterised by a combination of heightened excitement and prudent nervousness as I confronted the task of developing a research question. The encounter was similar to beginning upon an exciting expedition into unfamiliar terrain. The expedition was marked by a profound feeling of inquisitiveness, particularly the instant of recognition that inside the vast domain of academics was a distinct field that matched with my own interests and fervour. The improvement process showed intermittent instances of doubt, although a profound feeling of purpose accompanied it as I concentrated on constructing a question that would not only challenge my cognitive abilities but also inspire a profound emotional response.

The subsequent phase of my expedition involved traversing the scholarly waters of the literature review. During this phase, I assumed the role of an explorer, mapping out the intellectual landscape that others had previously explored. The experience included exploring

the past documents of human understanding, wherein I encountered prominent figures within my academic discipline and their influential contributions. In certain instances, I experienced a sense of being flooded by the extensive body of literature, feeling overwhelmed by the large number of references. Nevertheless, within the vast expanse of information, I encountered areas of profound understanding that functioned as illuminating markers of direction. During these instances, I experienced a profound connection with my scholarly heritage, indicating my position as a beneficiary of the academic achievements made by my predecessors.

The data collection phase encompassed the search for reliability, an expedition that frequently drove me into unfamiliar domains. Equipped with appropriate research methodologies and tools, I embarked upon a scholarly endeavour to collect data that would explain my research question. A prevailing feeling of expectation and occasional anxiety characterised the affective landscape. There were instances of increased excitement when data flow occurred seamlessly, compared with occasions where challenges appeared overwhelming. During this period, individuals showed qualities of patience and persistence as they perceived each data point as a constituent element of a broader puzzle awaiting integration.

The data analysis phase proved to be my research's most emotionally charged aspect. The endeavour involved the extraction of the underlying narrative hidden within the dataset, constituting an expedition towards the core of knowledge gain. Instances of frustration arose when discerning patterns proved to be unclear, while instances of joy occurred when valuable insights surfaced among the confusion. During that period, I actively participated in data analysis, engaging in a cognitive exchange that necessitated the utilisation of both logical reasoning and intuitive understanding. Every individual data point came with a sense of expectation, while each code possessed the inherent capacity to reveal new insights. The endeavour entailed exploring the cognitive processes and personal encounters of the individuals involved, producing a profound and occasionally humbling encounter.

The expedition encountered various stormy conditions along its course. A combination of anticipated and unanticipated challenges tested my resilience. There were instances during my research where it felt as though my work was comparable to a ship navigating tumultuous waters. The emotional terrain was characterised by uncertainty, exasperation, and despondency. However, these challenges also presented opportunities for personal and professional development. I was compelled to review my methodologies, question my

assumptions, and enhance the accuracy of my research questions. During hardship, I realised the extent of my resolve and the significance of persistence.

A noticeable alteration in emotional dynamics was seen throughout the transition to the writing phase. The act of writing fulfilled a dual role, serving as both a medium of communication and a systematic procedure through which I meticulously constructed a story that adeptly encapsulated the essential elements of my study. The writer encountered episodes of writer's block, characterised by difficulties in accessing appropriate words, in contrast to episodes of creative flow, characterised by the simple transcription of thoughts onto the paper. The experience examined the art of storytelling, where each paragraph functioned as a deliberate addition to a broad scope of intellectual comprehension. Writing entails amalgamating many components, including theoretical constructs, empirical evidence, and analytical perspectives, to generate a cohesive and integrated composition.

In summary, my PhD studies have been a significant emotional experience marked by moments of academic curiosity, frustration, enlightenment, and success. The abovementioned experience includes a deep and thorough examination of intellectual inquiry and substantial growth on an individual level. Throughout my academic path, I have contributed significantly to the scholarly discourse and acquired deep insights into my resilience and the tremendous impact of my passion for knowledge. The spectrum of emotions encountered throughout this expedition has shown a resemblance to the multifaceted and intricate characteristics of the scholarly realm, acting as a poignant reminder that the pursuit of a PhD degree comprises more than the acquisition of specific expertise. Conversely, it involves embracing the fullness of the human experience.

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Appendices

Appendix A: Conceptual Distinction between AI and Robots

Aspect	AI (Artificial Intelligence)	Robots
Definition	AI refers to the simulation of human intelligence processes by machines, encompassing learning, reasoning, problem-solving, and decision-making.	Robots are physical or virtual machines designed to perform tasks autonomously, often with physical interaction with the environment.
Conceptual Distinction	AI is primarily a concept related to the cognitive functions of machines, focusing on replicating humanlike intelligence.	Robots are a broader concept, encompassing physical machines capable of performing various tasks, including those that do not require high-level intelligence.
Emphasis on Intelligence	AI emphasises replicating and augmenting humanlike cognitive abilities, such as natural language understanding, image recognition, and decision-making.	While some robots may incorporate AI for intelligent decision-making, many robots are designed for physical tasks like assembly, transportation, or exploration.
Examples	Examples of AI include virtual assistants (e.g., Siri, Google Assistant), recommendation algorithms (e.g., Netflix recommendations), and autonomous vehicles.	Examples of robots include industrial robots used in manufacturing, delivery robots, surgical robots, and even humanoid robots like ASIMO.
Nature of Interaction	AI often interacts with users through digital interfaces, such as voice commands or text input.	Robots interact with the physical world and can perform actions such as moving objects, assembling components, or navigating obstacles in real-world environments.
Learning and Adaptation	AI systems frequently employ machine learning techniques to improve their performance and adapt to new data.	Some robots may incorporate AI for tasks like autonomous navigation and object recognition, allowing them to adapt to changing environments, while others rely on pre-programmed instructions.

Appendix B: The Concept Matrix of the contribution of the included papers to the relevant concepts

The Concepts / The sources	Use	developed a Model or rely on known theory	adaption	Satisfaction	Trust	Interaction	Intention	Privacy or security	Performance
(McLean & Osei-Frimpong, 2019)	X	X	X						
(Norval & Singh, 2019)	X					X			
(Mitev, Miettinen, & Sadeghi, 2019)		X				X		X	
(Cho, 2019)						X		X	
(Kowalski et al., 2019)						X			
(Hwang, 2019)						X			
(Moriuchi, 2019)	X	X	X						
(Cheng, Bagci, Yan, & Roedig, 2019)								X	X
(Vaidya & Sherr, 2019)		X						X	
(Zhang et al., 2019)		X						X	
(Siegel & Sarma, 2019)					X			X	
(Lee, Lee, & Sheehan, 2020)	X	X	X	X				X	
(Rzepka, 2019)	X	X	X		X	X		X	
(Amugongo, 2018)						X			
(Ernst & Malzahn, 2018)	X	X		X			X		
(Baier, Rese, & Röglinger, 2018)				X			X		
(Zhou et al., 2018)									X
(Nasirian et al., 2017)	X	X	X	X	X	X	X		
(Ning et al., 2017)		X					X		X
(Zhao & Rau, 2020)				X					
(Chattaraman, Kwon, Gilbert, & Ross, 2019)					X	X	X		

Appendix C: The foci and intents of the included papers

This is a sample of the 55 papers considered for the analysis presented in Chapter 2.

Source	The Focus and intent of the study
(McLean & Osei-Frimpong, 2019)	This research combines the theoretical foundations of U&GT with technology theories to gain a clearer understanding on the motivations for adopting and using in-home voice assistants. This research furthers our understanding in this domain through taking a Uses and Gratification theory (U&GT) approach to understanding the use of voice assistants focusing on voice interactions, while also integrating Human-Computer Interaction (HCI) literature on the social attributes of the system and individuals' perceived privacy risks.
(Norval & Singh, 2019)	This paper explores the use of voice assistants (an accessible, intuitive and increasingly common interface) as a means for allowing users to interrogate what is happening in the IoT systems that surround them. In presenting an exploratory Alexa 'Skill', we discuss several considerations for the implementation of such a system. This work represents a starting point for considering how such assistants could help people better understand—and indeed, evaluate, challenge, and accept—technology that is increasingly pervading our world.
(Mitev & Miettinen, 2019)	In this paper, we show that specially crafted malicious Skills can use the seemingly limited Skill interaction model to cause harm. We present to a malicious Skill, thereby hijacking the conversation between Alexa and the victim. To the best of our knowledge this is the first man-in-the-middle attack novel man-in-the-middle attacks against benign Skills and Virtual Assistant functionalities. Our attack uses loopholes in the Skill interface to redirect a victim's voice input targeting the Skill ecosystem. We present the design of our attack and demonstrate its feasibility based on a proof-of-concept implementation attacking the Alexa Skills of a smart lock as well as a home security system.
(Cho, 2019)	This study examined if the differences in modality (i.e., voice vs. text) and device (i.e., smartphone vs. smart home device) affect user perceptions when users attempt to retrieve sensitive health information from voice assistants.
(Kowalski et al., 2019)	an exploratory study examining the potential of voice assistants (VA) for some groups of older adults in the context of Smart Home Technology (SHT)
(Hwang, 2019)	Our study explored how gender stereotypes of women are reflected in voice assistants with female voices through analysing five South Korean Vas
(Moriuchi, 2019)	We propose a model that investigates the technology acceptance model constructs (perceived ease of use and perceived usefulness) and its effect on the engagement and loyalty between VA and consumers.
(Cheng, Bagci, Yan, & Roedig, 2019)	We explore techniques for embedding additional information into acoustic signals processed by PVAs.
(Vaidya & Sherr, 2019)	The observation that the increasing popularity of voice interfaces that use cloud-backed speech recognition (e.g., Siri, Google Assistant, and Amazon Alexa) increases the public's vulnerability to voice synthesis attacks.
(Zhang et al., 2019)	We report a study that concludes such remote, large- scale attacks are indeed realistic.
(Siegel & Sarma, 2019)	We describe bridging IoT's air gap using device speakers to communicate with voice assistants. We recommend protecting voice channels using two- factor authentication, contextual monitoring, and voice source embedding. These techniques are necessary when connecting critical applications to open channels, and may apply to other exploits resulting from IoT's growing capabilities.

Source	The Focus and intent of the study
(Lee, Lee, & Sheehan, 2019)	We hypothesize that the social cues emanated from the continued, and habituated, use of SVAS develop the "illusion of intimacy" which, in turn, ripples through the entire group, and help fulfil the need for social integration.
(Rzepka, 2019)	To better understand when and why users choose speech interaction over traditional user interfaces.
(Amugongo, 2018)	Survey across 6 African countries to establish what African users think of existing voice assistants, such as Siri, Google assistant and alike. Additionally, provides recommendations
(Ernst, 2018)	Drew from the Expectation Confirmation Theory to evaluate the role of Past Technology Category Satisfaction on people's technology usage behaviour.
(Baier, Rese, & Röglinger, 2018)	How do conversational user interfaces for online shops via messaging services and voice assistants influence customers' satisfaction? Which use cases are attractive from a customer's view point? Which use cases are must-be and for which customer segments?
(Zhou et al., 2018)	To better infer emotion from conversational voice data, we propose a semi- supervised multi-path generative neural network
(Nasirian et al., 2017)	Having more customized models for emerging technologies upon their features seems necessary. In this study, we developed a conceptual model involving a new system quality construct, i.e., interaction quality, which we believe can better describe adoption of AI-based technologies.
(Ning et al., 2017)	How to incorporate these heterogeneous attributes to infer user intention remains an open problem
(Zhao & Patrick Rau, 2020)	To explore potential trends in the use of VAs, we designed a merged VA and two either familiar (interconnected) or strange (completely independent) VAs, of which one acted as a secretary for work-related issues and the other acted as a housekeeper for personal issues
(Chattaraman, Kwon, Gilbert, & Ross, 2019)	This study suggests that agent design cannot be done with a one-size-fits-all strategy by empirically demonstrating that the extent to which an intelligent digital assistant can provide social, cognitive, and functional assistance to older users in an online shopping task depends on the agent's interaction style in combination with users' task-competency